Photon Voice

v2.64

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Chapter 1

Photon Voice

Photon Voice 2 has three key classes:

- Photon.Voice.Unity.VoiceConnection (extends `Photon.Realtime.ConnectionHandler`)
- Photon.Voice.Unity.Recorder
- Photon.Voice.Unity.Speaker

If you also use the integration with PUN 2, we added two components for ease-of-use and more convenience:

- Photon.Voice.PUN.PhotonVoiceNetwork
- Photon.Voice.PUN.PhotonVoiceView

Photon Voice 2 also comes with a WebRTC based DSP (Photon.Voice.Unity.WebRtcAudioDsp using Photon.Voice.WebRTCAudioProcessor).

Read more in the official documentation `here`.
You can download Photon Voice 2 `here`.

Chapter 2

Namespace Documentation

2.1 Photon Namespace Reference

2.2 Photon.Voice Namespace Reference

Classes

- class ArrayPool< T >

  Pool of Arrays with components of type T, with ObjectPool info being the array's size.
- class ArrayPoolSet< T >

  ArrayPool set of limited size.
- class ArraySegmentPool< T >

  Set of ArrayPool's of N^ 2-sized arrays used to produce an ArraySegment with the requested size.
- class AudioDesc
- class AudioInChangeNotifierNotSupported
- class **AudioInEnumeratorNotSupported**
- class AudioOutDelayControl
- class AudioOutDelayControl< T >
- class AudioOutDummy< T >
- class AudioReceptionSettings
- class AudioSenderSettings
- class AudioStreamReceiver

  Basic audio stream receiver Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)
- class AudioStreamSender

  Basic audio stream sender Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)
- class AudioSyncBuffer< T >
- class AudioUtil

  Collection of Audio Utility functions and classes.
- class BufferReaderPushAdapterAsyncPool< T >

  BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync.
- class BufferReaderPushAdapterAsyncPoolFloatToShort

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting float samples to short.

- class BufferReaderPushAdapterAsyncPoolShortToFloat

  BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting short samples to float.
- class BufferReaderPushAdapterBase< T >

  Adapter base reading data from IDataReader<T>.Read and pushing it to LocalVoice.
- class ByteStreamDecoder
- class ByteStreamEncoder
- class DecoderConfigFrame

  Stores the config frame and prevents other frames decoding until the decoder is ready.
- class DeviceEnumeratorBase
- class **DeviceEnumeratorNotSupported**
- class **DeviceEnumeratorSingleDevice**
- class DeviceFeatures
- struct DeviceInfo
- struct Flip
- struct FrameBuffer
- class FrameOut< T >
- class Framer< T >

  Utility class to re-frame packets.
- class FramerResampler< T >
- interface IAudioDesc

  Audio Source interface.
- interface IAudioInChangeNotifier
- interface IAudioOut< T >
- interface IAudioPusher< T >

  Audio Pusher interface.
- interface IAudioReader< T >

  Audio Reader interface.
- interface IDataReader< T >

  Interface for pulling data, in case this is more appropriate than pushing it.
- interface IDecoder

  Generic decoder interface.
- interface IDecoderDirect< B >

  Interface for an decoder which outputs data via explicit call.
- interface IDeviceEnumerator
- interface IEncoder

  Generic encoder interface.
- interface IEncoderDirect< B >

  Interface for an encoder which consumes input data via explicit call.
- interface IEncoderDirectImage

  Interface for an encoder which consumes images via explicit call.
- interface ILocalVoiceAudio

  Interface for an outgoing audio stream.
- interface ILogger
- struct ImageBufferInfo
- class ImageBufferNative
- class ImageBufferNativeAlloc
- class ImageBufferNativeGCHandleBytes
- class ImageBufferNativeGCHandleSinglePlane
- class ImageBufferNativePool< T >
- interface IProcessor< T >

Processor interface.

- interface IResettable
- interface IServiceable

  Interface for classes that want their Service() function to be called regularly in the context of a LocalVoice.
- interface IStreamReceiver

  Interface for component relying on a LoadBalancingTransport to receive a stream of data (audio, vidoe, ...)
- interface IStreamSender

  Interface for component relying on a LoadBalancingTransport to send a stream of data (audio, vidoe, ...)
- interface IStreamTransportManager

  Interface for components capable of providing a LoadBalancingTransport for an IStreamSender and/or IStreamReceiver Those components should forward the stream callbacks to their registered listeners:
- interface IStreamTransportManagerListener
- interface IVoiceTransport
- class LoadBalancingTransport

  Extends LoadBalancingClient with media streaming functionality.
- class LoadBalancingTransport2

  Variant of LoadBalancingTransport. Aims to be non-alloc at the cost of breaking compatibility with older clients.
- class LocalVoice

  Represents outgoing data stream.
- class LocalVoiceAudio< T >

  Outgoing audio stream.
- class LocalVoiceAudioDummy

  Dummy LocalVoiceAudio.
- class LocalVoiceAudioFloat

  Specialization of LocalVoiceAudio<T> for float audio.
- class LocalVoiceAudioShort

  Specialization of LocalVoiceAudio<T> for short audio.
- class LocalVoiceFramed< T >

  Typed re-framing LocalVoice.
- class MonoPInvokeCallbackAttribute
- interface ObjectFactory< TType, TInfo >

  Object factory with optional info useful in combination with ObjectPool<TType, TInfo>.
- class ObjectPool< TType, TInfo >

  Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).
- class OpusCodec
- class PermissionsRequester

  Launch successively permission requests (Android only for now)
- class PermissionWaiter

  Wait for a permission to be granted.
- class PhotonAppSettings

  Collection of connection-relevant settings, used internally by PhotonNetwork.ConnectUsingSettings.
- class **PhotonTransportProtocol**
- class Platform
- class RawCodec
- class **RemoteVoice**
- class RemoteVoiceInfo

  Information about a remote voice (incoming stream).
- struct RemoteVoiceOptions

  Event Actions and other options for a remote voice (incoming stream).
- struct SendFrameParams
- class **SpacingProfile**

- class StreamMatchmakingSettings
- class StreamTransportClient

  Basic implementation of a IRegisterableStreamTransportManager, initiating a LoadBalancingTransport and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses (that manage recorders, offer a deeper app life cycle management, ...)
- class StreamTransportClientSettings
- class StreamTransportManagerListener

  Base class for stream senders/receivers which rely on a IRegisterableStreamTransportManager to provide the underlying LoadBalancingTransport upon which the stream run Automatically tries to find _streamTransportManager IRegisterableStreamTransportManager, through StreamTransportObserver.FindStreamTransportManager.
- class StreamTransportObserver

  Wrapper handling LoadBalancingTransport callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the IRegisterableStreamTransportManager interface.
- struct StringSessionProperty
- class UnsupportedCodecException

  Exception thrown if an unsupported codec is encountered.
- class UnsupportedPlatformException

  Exception thrown if an unsupported platform is encountered.
- class UnsupportedSampleTypeException

  Exception thrown if an unsupported audio sample type is encountered.
- class **Util**
- class Utility
- class **VideoInEnumeratorNotSupported**
- class VoiceClient

  Voice client interact with other clients on network via IVoiceTransport.
- class VoiceConnectionMatchmaking

  Use matchmaking to connect and join a VoiceConnection.
- class VoiceConnectionPermissionWaiter

  Wait for Microphone access permissions to be available before letting VoiceConnection connect Note: recordWhen↩ Joined should be disabled on the recorder.
- struct VoiceCreateOptions

  Used to initialize optional properties of the LocalVoice instance at creation time.
- class VoiceEvent
- class VoiceFollowClient

  This class can be used to automatically sync client states between Leader and Voice clients.
- struct VoiceInfo

  Describes stream properties.
- class WebRTCAudioLib
- class WebRTCAudioProcessor

Enumerations

- enum **WebcamSelectionMode**
- enum **CameraFacing**
- enum AudioSampleType

  The type of samples used for audio processing.
- enum **LogLevel**
- enum **FrameFlags** : byte
- enum Codec

  Enum for Media Codecs supported by PhotonVoice.
- enum **ImageFormat**
- enum **Rotation**

2.2.1 Enumeration Type Documentation

2.2.1.1 Codec

`enum Codec`

Enum for Media Codecs supported by PhotonVoice.

Transmitted in VoiceInfo. Do not change the values of this Enum!

Enumerator

	AudioOpus	OPUS audio.

2.3 Photon.Voice.FMOD Namespace Reference

Classes

- class AudioInEnumerator
- class AudioInReader< T >
- class AudioOut< T >
- class AudioOutEvent< T >

2.4 Photon.Voice.Fusion Namespace Reference

Classes

- class FusionVoiceClient
- class VoiceNetworkObject

Typedefs

- using **PhotonAppSettings** = global::Fusion.Photon.Realtime.PhotonAppSettings
- using **LogLevel** = Photon.Voice.LogLevel

2.5 Photon.Voice.IOS Namespace Reference

Classes

- class AudioInChangeNotifier
- class AudioInPusher
- class AudioInReader
- struct AudioSessionParameters
- class AudioSessionParametersPresets

Enumerations

- enum AudioSessionCategory
- enum AudioSessionMode
- enum AudioSessionCategoryOption

2.5.1 Enumeration Type Documentation

2.5.1.1 AudioSessionCategory

```
enum AudioSessionCategory
```

Enumerator

Ambient	
SoloAmbient	
Playback	
Record	
PlayAndRecord	
AudioProcessing	
MultiRoute	Use this category to customize the usage of available audio accessories and built-in audio hardware. For exa

2.5.1.2 AudioSessionCategoryOption

`enum AudioSessionCategoryOption`

Enumerator

MixWithOthers	This allows an application to set whether or not other active audio apps will be interrupted or mixed with whe
DuckOthers	
AllowBluetooth	This allows an application to change the default behavio
DefaultToSpeaker	

2.5.1.3 AudioSessionMode

`enum AudioSessionMode`

Enumerator

Default	
VoiceChat	Only valid with AVAudioSessionCategoryPlayAndRecord. Appropriate for Voice over IP (VoIP) applicat
VideoRecording	
Measurement	
MoviePlayback	
VideoChat	Only valid with kAudioSessionCategory_PlayAndRecord. Reduces the number of allowable audio routes to be

2.6 Photon.Voice.MacOS Namespace Reference

Classes

- class AudioInChangeNotifier
- class AudioInEnumerator

  Enumerates microphones available on device.
- class AudioInPusher
- class AudioInReader

2.7 Photon.Voice.PUN Namespace Reference

Classes

- class PhotonVoiceView

  Component that should be attached to a networked PUN prefab that has PhotonView. It will bind remote Recorder with local Speaker of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.
- class PunVoiceClient

  This class can be used to automatically sync client states between PUN and Voice. It also finds the Speaker component for a character's voice. For this to work attach a PhotonVoiceView next to the PhotonView of your player's prefab.

2.8 Photon.Voice.PUN.UtilityScripts Namespace Reference

Classes

- class VoiceDebugScript

  Utility script to be attached next to PhotonVoiceView & PhotonView on the player prefab to be network instantiated. Call voiceDebugScript.CantHearYou() on the networked object of the remote (or local) player if you can't hear the corresponding player.

2.9 Photon.Voice.Unity Namespace Reference

Classes

- class AndroidAudioInAEC
- struct AndroidAudioInParameters
- class AudioChangesHandler

  This component is useful to handle audio device and config changes.
- class AudioClipWrapper
- class AudioInEnumerator
- class AudioOutCapture
- class Logger
- class MicWrapper
- class MicWrapperPusher
- class MicWrapperPusherOnAudioFilterRead
- class PhotonVoiceCreatedParams
- class Recorder

  Component representing outgoing audio stream in scene.
- class RecorderPreset
- class RemoteVoiceLink
- class Speaker
- class SpeakerAudioFilterRead
- class TimerWorker
- class UnityAudioOut
- class UnityLogger
- class UnityMicrophone

  A wrapper around UnityEngine.Microphone to be able to safely use Microphone and compile for WebGL.
- class UnityVoiceClient

Component that represents a Voice client and manages a simple Unity integration: a single Recorder and multiple remote speakers.

- class VideoInEnumerator
- class VoiceComponent
- class VoiceComponentImpl
- class VoiceConnection

  Component that represents a Voice client.
- class VoiceLogger
- class WebAudioAudioOut
- class WebAudioInEnumerator
- class WebAudioMicIn
- class WebDeviceEnumerator
- class WebRtcAudioDsp
- class WebVideoInEnumerator

2.10 Photon.Voice.Unity.FMOD Namespace Reference

Classes

- class FMODRecorderSetup
- class SpeakerFMOD

2.11 Photon.Voice.Unity.UtilityScripts Namespace Reference

Classes

- class ConnectAndJoin
- class MicAmplifier
- class MicAmplifierFloat
- class MicAmplifierShort
- class MicrophonePermission

  Helper to request Microphone permission on Android or iOS.
- class PhotonVoiceLagSimulationGui
- class PhotonVoiceStatsGui

  Basic GUI to show traffic and health statistics of the connection to Photon, toggled by shift+tab.
- class SaveIncomingStreamToFile
- class SaveOutgoingStreamToFile
- class TestTone
- class WaveWriter

2.12 Photon.Voice.UWP Namespace Reference

Classes

- class AudioInEnumerator
- class AudioInPusher
- class CaptureDevice
- class DeviceEnumerator
- class VideoInEnumerator

Functions

- delegate void **MediaCaptureInitConmpleted** (MediaCapture mediaCpture, bool ok)

2.13 Photon.Voice.Windows Namespace Reference

Classes

- class AudioInEnumerator

  Enumerates microphones available on device.
- class WindowsAudioInPusher

Chapter 3

Class Documentation

3.1 AndroidAudioInAEC Class Reference

Inherits IAudioPusher< T >, and IResettable.

Public Member Functions

- **AndroidAudioInAEC** (Voice.ILogger logger, bool enableAEC=false, bool enableAGC=false, bool enable↩ NS=false)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimal↩ FrameSize)
- void Reset ()
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.1.1 Member Function Documentation

3.1.1.1 Reset()

```
void Reset ()
```

Implements IResettable.

3.1.2 Property Documentation

3.1.2.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.1.2.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.1.2.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.2 AndroidAudioInParameters Struct Reference

Public Attributes

- bool **EnableAEC**
- bool **EnableAGC**
- bool **EnableNS**

Static Public Attributes

- static AndroidAudioInParameters **Default** = new AndroidAudioInParameters() { EnableAEC = true, Enable↩ AGC = true, EnableNS = true }

3.3 ArrayPool< T > Class Template Reference

Pool of Arrays with components of type T, with ObjectPool info being the array's size.

Inherits ObjectPool< TType, TInfo >.

Public Member Functions

- **ArrayPool** (int capacity, string name, int info)

Public Member Functions inherited from ObjectPool< TType, TInfo >

- ObjectPool (int capacity, string name, TInfo info)

  Create a new ObjectPool instance with the given info structure.
- TType New ()

  Acquire an existing object, or create a new one if none are available.
- TType New (TInfo info)

  Acquire an existing object (if info matches), or create a new one from the passed info.
- virtual bool Free (TType obj, TInfo objInfo)

  Returns object to pool.
- virtual bool Free (TType obj)

  Returns object to pool, or destroys it if the pool is full.
- void **Dispose** ()

  Free resources assoicated with this ObjectPool.

Protected Member Functions

- override T[ ] **createObject** (int info)
- override void **destroyObject** (T[ ] obj)
- override bool **infosMatch** (int i0, int i1)

Protected Member Functions inherited from ObjectPool< TType, TInfo >

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

Additional Inherited Members

Protected Attributes inherited from ObjectPool< TType, TInfo >

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

Properties inherited from ObjectPool< TType, TInfo >

- TInfo **Info** `[get]`

  The property (info) that objects in this Pool must match.

3.3.1 Detailed Description

Pool of Arrays with components of type T, with ObjectPool info being the array's size.

Template Parameters

	T	Array element type.

3.4 ArrayPoolSet< T > Class Template Reference

ArrayPool set of limited size.

Inherits ObjectFactory< TType, TInfo >.

Public Member Functions

- **ArrayPoolSet** (int capacity, string name, int defaultInfo, int setSize)
- T[ ] New ()
- bool **Free** (T[ ] obj)
- T[ ] **New** (int info)
- bool **Free** (T[ ] obj, int info)
- void **Dispose** ()

Public Member Functions inherited from ObjectFactory< TType, TInfo >

- TType New (TInfo info)
- bool Free (TType obj)
- bool Free (TType obj, TInfo info)

3.4.1 Detailed Description

ArrayPool set of limited size.

Template Parameters

	T	Array element type.

3.4.2 Member Function Documentation

3.4.2.1 New()

```
T[ ] New ()
```

Implements ObjectFactory< TType, TInfo >.

3.5 ArraySegmentPool< T > Class Template Reference

Set of ArrayPool's of N^2-sized arrays used to produce an ArraySegment with the requested size.

Inherits ObjectFactory< TType, TInfo >.

Public Member Functions

- **ArraySegmentPool** (int capacity, string name, int defaultInfo)
- ArraySegment< T > New ()
- ArraySegment< T > **New** (int info)
- bool **Free** (ArraySegment< T > obj, int info)
- bool **Free** (ArraySegment< T > obj)
- void **Dispose** ()

Public Member Functions inherited from ObjectFactory< TType, TInfo >

- TType New (TInfo info)
- bool Free (TType obj)
- bool Free (TType obj, TInfo info)

Static Public Member Functions

- static void **test** ()

3.5.1 Detailed Description

Set of ArrayPool's of N^2-sized arrays used to produce an ArraySegment with the requested size.

Template Parameters

	T	Array element type.

3.5.2 Member Function Documentation

3.5.2.1 New()

```
ArraySegment< T > New ()
```

Implements ObjectFactory< TType, TInfo >.

3.6 AudioChangesHandler Class Reference

This component is useful to handle audio device and config changes.

Inherits VoiceComponent.

Public Attributes

- bool **HandleDeviceChange** = true

  Try to react to device change notification when Recorder is started.
- bool **HandleDeviceChangeIOS**

  iOS: Try to react to device change notification when Recorder is started.
- bool **HandleDeviceChangeAndroid**

  Android: Try to react to device change notification when Recorder is started.

Protected Member Functions

- override void Awake ()

Additional Inherited Members

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.6.1 Detailed Description

This component is useful to handle audio device and config changes.

3.6.2 Member Function Documentation

3.6.2.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.7 AudioClipWrapper Class Reference

Inherits IAudioReader< T >.

Public Member Functions

- **AudioClipWrapper** (AudioClip audioClip)
- bool **Read** (float[ ] buffer)
- void **Dispose** ()

Public Member Functions inherited from IDataReader< T >

- bool Read (T[] buffer)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Properties

- bool **Loop** `[get, set]`
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- int Channels `[get]`

  Number of channels in the audio signal.
- string Error `[get]`

  If not null, audio object is in invalid state.

3.7.1 Property Documentation

3.7.1.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.7.1.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.7.1.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.8 AudioDesc Class Reference

Inherits IAudioDesc.

Public Member Functions

- **AudioDesc** (int samplingRate, int channels, string error)
- void **Dispose** ()

Properties

- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- int Channels `[get]`

  Number of channels in the audio signal.
- string Error `[get]`

  If not null, audio object is in invalid state.

3.8.1 Property Documentation

3.8.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.8.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.8.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.9 AudioInChangeNotifier Class Reference

Inherits IAudioInChangeNotifier.

Public Member Functions

- **AudioInChangeNotifier** (Action callback, ILogger logger)
- void **Dispose** ()

  Disposes enumerator. Call it to free native resources.

Properties

- bool IsSupported `[get]`
- string Error `[get]`

  If not null, the enumerator is in invalid state.

3.9.1 Property Documentation

3.9.1.1 Error

```
string Error [get]
```

If not null, the enumerator is in invalid state.

Implements IAudioInChangeNotifier.

3.9.1.2 IsSupported

```
bool IsSupported [get]
```

Implements IAudioInChangeNotifier.

3.10 AudioInChangeNotifier Class Reference

Inherits IAudioInChangeNotifier.

Public Member Functions

- **AudioInChangeNotifier** (Action callback, ILogger logger)
- void **Dispose** ()

  Disposes enumerator. Call it to free native resources.

Properties

- bool IsSupported `[get]`
- string Error `[get]`

  If not null, the enumerator is in invalid state.

3.10.1 Property Documentation

3.10.1.1 Error

```
string Error  [get]
```

If not null, the enumerator is in invalid state.

Implements IAudioInChangeNotifier.

3.10.1.2 IsSupported

```
bool IsSupported  [get]
```

Implements IAudioInChangeNotifier.

3.11 AudioInChangeNotifierNotSupported Class Reference

Inherits IAudioInChangeNotifier.

Public Member Functions

- **AudioInChangeNotifierNotSupported** (Action callback, ILogger logger)
- void **Dispose** ()

Properties

- bool IsSupported `[get]`
- string Error `[get]`

3.11.1 Property Documentation

3.11.1.1 Error

```
string Error  [get]
```

Implements IAudioInChangeNotifier.

3.11.1.2 IsSupported

```
bool IsSupported  [get]
```

Implements IAudioInChangeNotifier.

3.12 AudioInEnumerator Class Reference

Inherits DeviceEnumeratorBase.

Public Member Functions

- **AudioInEnumerator** (FMODLib.System coreSystem, ILogger logger)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.12.1 Member Function Documentation

3.12.1.1 Refresh()

```
override void Refresh ()
```

Implements IDeviceEnumerator.

3.13 AudioInEnumerator Class Reference

Enumerates microphones available on device.

Inherits DeviceEnumeratorBase.

Public Member Functions

- **AudioInEnumerator** (ILogger logger)
- override void Refresh ()

  Refreshes the microphones list.
- override void **Dispose** ()

  Disposes enumerator. Call it to free native resources.

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.13.1 Detailed Description

Enumerates microphones available on device.

3.13.2 Member Function Documentation

3.13.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements IDeviceEnumerator.

3.14 AudioInEnumerator Class Reference

Inherits DeviceEnumeratorBase.

Public Member Functions

- **AudioInEnumerator** (ILogger logger)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Properties

- override bool IsSupported `[get]`
- override string Error `[get]`

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

3.14.1 Member Function Documentation

3.14.1.1 Refresh()

```
override void Refresh ()
```

Implements IDeviceEnumerator.

3.14.2 Property Documentation

3.14.2.1 Error

```
override string Error  [get]
```

Implements IDeviceEnumerator.

3.14.2.2 IsSupported

```
override bool IsSupported  [get]
```

Implements IDeviceEnumerator.

3.15 AudioInEnumerator Class Reference

Inherits DeviceEnumerator.

Public Member Functions

- **AudioInEnumerator** (ILogger logger)

Public Member Functions inherited from DeviceEnumerator

- **DeviceEnumerator** (ILogger logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.16 AudioInEnumerator Class Reference

Enumerates microphones available on device.

Inherits DeviceEnumeratorBase.

Public Member Functions

- **AudioInEnumerator** (ILogger logger)
- override void Refresh ()

  Refreshes the microphones list.
- override void **Dispose** ()

  Disposes enumerator. Call it to free native resources.

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.16.1 Detailed Description

Enumerates microphones available on device.

3.16.2 Member Function Documentation

3.16.2.1 Refresh()

```
override void Refresh ()
```

Refreshes the microphones list.

Implements IDeviceEnumerator.

3.17 AudioInPusher Class Reference

Inherits IAudioPusher< T >, and IResettable.

Public Member Functions

- **AudioInPusher** (AudioSessionParameters sessParam, ILogger logger)
- void **SetCallback** (Action< float[]> callback, ObjectFactory< float[], int > bufferFactory, int optimalFrame↩ Size)
- void Reset ()
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.17.1 Member Function Documentation

3.17.1.1 Reset()

```
void Reset ()
```

Implements IResettable.

3.17.2 Property Documentation

3.17.2.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.17.2.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.17.2.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.18 AudioInPusher Class Reference

Inherits IAudioPusher< T >.

Public Member Functions

- **AudioInPusher** (int deviceID, ILogger logger)
- void **SetCallback** (Action< float[ ]> callback, ObjectFactory< float[ ], int > bufferFactory, int optimalFrame↩ Size)
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[ ]> callback, ObjectFactory< T[ ], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.18.1 Property Documentation

3.18.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.18.1.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.18.1.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.19 AudioInPusher Class Reference

Inherits IAudioPusher< T >.

Public Member Functions

- **AudioInPusher** (ILogger logger, int samplingRate, int channels, string deviceID)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimal↩ FrameSize)
- ArraySegment< byte > **DequeueOutput** (out FrameFlags flags)
- void **EndOfStream** ()
- I GetPlatformAPI< I > ()
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- int Channels [get]

  Number of channels in the audio signal.
- string Error [get]

  If not null, audio object is in invalid state.
- bool **ErrorAccess** [get]

3.19.1 Member Function Documentation

3.19.1.1 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Type Constraints

I* : *class

3.19.2 Property Documentation

3.19.2.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.19.2.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.19.2.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.20 AudioInReader Class Reference

Inherits IAudioReader< T >, and IResettable.

Public Member Functions

- **AudioInReader** (AudioSessionParameters sessParam, ILogger logger)
- void Reset ()
- void **Dispose** ()
- bool **Read** (float[ ] buf)

Public Member Functions inherited from IDataReader< T >

- bool Read (T[] buffer)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Properties

- int Channels [get]

  Number of channels in the audio signal.
- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- string Error [get]

  If not null, audio object is in invalid state.

3.20.1 Member Function Documentation

3.20.1.1 Reset()

```
void Reset ()
```

Implements IResettable.

3.20.2 Property Documentation

3.20.2.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.20.2.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.20.2.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.21 AudioInReader Class Reference

Inherits IAudioReader< T >.

Public Member Functions

- **AudioInReader** (int deviceID, ILogger logger)
- void **Dispose** ()
- bool **Read** (float[ ] buf)

Public Member Functions inherited from IDataReader< T >

- bool Read (T[ ] buffer)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Properties

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.21.1 Property Documentation

3.21.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.21.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.21.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.22 AudioInReader< T > Class Template Reference

Inherits IAudioReader< T >.

Public Member Functions

- **AudioInReader** (FMODLib.System coreSystem, int device, int suggestedFrequency, ILogger logger)
- void **Dispose** ()
- bool Read (T[] readBuf)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Public Attributes

- bool **isRecording**

Properties

- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- int Channels `[get]`

  Number of channels in the audio signal.
- string Error `[get]`

  If not null, audio object is in invalid state.

3.22.1 Member Function Documentation

3.22.1.1 Read()

```
bool Read (
            T[] buffer)
```

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Parameters

	buffer	Buffer to fill.

Returns

True if buffer was filled successfully, false otherwise.

Implements IDataReader< T >.

3.22.2 Property Documentation

3.22.2.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.22.2.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.22.2.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.23 AudioOut< T > Class Template Reference

Inherits AudioOutDelayControl< T >.

Inherited by AudioOutEvent< T >.

Public Member Functions

- **AudioOut** (FMODLib.System coreSystem, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int samplingRate, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[ ] frame, int offsetSamples)
- override void Stop ()

Public Member Functions inherited from AudioOutDelayControl< T >

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()

Protected Attributes

- int **channels**
- int **frequency**

Protected Attributes inherited from AudioOutDelayControl< T >

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

Properties

- FMODLib.Sound **Sound** `[get]`
- FMODLib.Channel **Channel** `[get]`
- override long **OutPos** `[get]`
- string **Error** `[get]`

Properties inherited from AudioOutDelayControl< T >

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

Additional Inherited Members

Static Public Attributes inherited from AudioOutDelayControl< T >

- const int **FRAME_POOL_CAPACITY** = 50

3.23.1 Member Function Documentation

3.23.1.1 Stop()

```
override void Stop ()  [virtual]
```

Reimplemented from AudioOutDelayControl< T >.

3.24 AudioOutCapture Class Reference

Inherits MonoBehaviour.

Events

- Action< float[ ], int > **OnAudioFrame**

3.25 AudioOutDelayControl Class Reference

Inherited by AudioOutDelayControl< T >.

Classes

- struct PlayDelayConfig

3.26 AudioOutDelayControl< T > Class Template Reference

Inherits AudioOutDelayControl, and IAudioOut< T >.

Inherited by AudioSyncBuffer< T >, AudioOut< T >, UnityAudioOut, and WebAudioAudioOut.

Public Member Functions

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()
- virtual void Stop ()

Static Public Attributes

- const int **FRAME_POOL_CAPACITY** = 50

Protected Attributes

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

Properties

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

3.26.1 Member Function Documentation

3.26.1.1 Flush()

```
void Flush ()
```

Implements IAudioOut< T >.

3.26.1.2 Push()

```
void Push (
            T[] frame)
```

Implements IAudioOut< T >.

3.26.1.3 Service()

```
void Service ()
```

Implements IAudioOut< T >.

3.26.1.4 Start()

```
void Start (
            int frequency,
            int channels,
            int frameSamples)
```

Implements IAudioOut< T >.

3.26.1.5 Stop()

```
virtual void Stop ()  [virtual]
```

Implements IAudioOut< T >.

3.26.2 Property Documentation

3.26.2.1 IsPlaying

```
bool IsPlaying  [get]
```

Implements IAudioOut< T >.

3.26.2.2 Lag

```
int Lag  [get]
```

Implements IAudioOut< T >.

3.27 AudioOutDummy< T > Class Template Reference

Inherits IAudioOut< T >.

Public Member Functions

- void Flush ()
- void Push (T[] frame)
- void Service ()
- void Start (int frequency, int channels, int frameSamplesPerChannel)
- void Stop ()

Properties

- bool IsPlaying [get]
- int Lag [get]

3.27.1 Member Function Documentation

3.27.1.1 Flush()

```
void Flush ()
```

Implements IAudioOut< T >.

3.27.1.2 Push()

```
void Push (
            T[] frame)
```

Implements IAudioOut< T >.

3.27.1.3 Service()

```
void Service ()
```

Implements IAudioOut< T >.

3.27.1.4 Start()

```
void Start (
            int frequency,
            int channels,
            int frameSamplesPerChannel)
```

Implements IAudioOut< T >.

3.27.1.5 Stop()

```
void Stop ()
```

Implements IAudioOut< T >.

3.27.2 Property Documentation

3.27.2.1 IsPlaying

```
bool IsPlaying  [get]
```

Implements IAudioOut< T >.

3.27.2.2 Lag

```
int Lag  [get]
```

Implements IAudioOut< T >.

3.28 AudioOutEvent< T > Class Template Reference

Inherits AudioOut< T >.

Public Member Functions

- **AudioOutEvent** (FMODLib.System coreSystem, FMODLib.Studio.EventInstance fmodEvent, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- override void **OutStart** ()
- override void Stop ()

Public Member Functions inherited from AudioOut< T >

- **AudioOut** (FMODLib.System coreSystem, PlayDelayConfig playDelayConfig, ILogger logger, string log↩ Prefix, bool debugInfo)
- override void **OutCreate** (int samplingRate, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[ ] frame, int offsetSamples)

Public Member Functions inherited from AudioOutDelayControl< T >

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()

Properties

- override long **OutPos** `[get]`

Properties inherited from AudioOut< T >

- FMODLib.Sound **Sound** `[get]`
- FMODLib.Channel **Channel** `[get]`
- override long **OutPos** `[get]`
- string **Error** `[get]`

Properties inherited from AudioOutDelayControl< T >

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

Additional Inherited Members

Static Public Attributes inherited from AudioOutDelayControl< T >

- const int **FRAME_POOL_CAPACITY** = 50

Protected Attributes inherited from AudioOut< T >

- int **channels**
- int **frequency**

Protected Attributes inherited from AudioOutDelayControl< T >

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

3.28.1 Member Function Documentation

3.28.1.1 Stop()

```
override void Stop ()  [virtual]
```

Reimplemented from AudioOut< T >.

3.29 AudioReceptionSettings Class Reference

Public Attributes

- int **AudioJitterBufferMs** = 200
- int **AudioJitterBufferMsHigh** = 400
- int **AudioJitterBufferMsMax** = 1000

3.30 AudioSenderSettings Class Reference

Public Attributes

- int **AudioBitrate** = 30000
- bool **AudioEcho** = false

Static Public Attributes

- const POpusCodec.Enums.SamplingRate **AudioEncoderSamplingRate** = POpusCodec.Enums.↩ SamplingRate.Sampling24000
- const int **AudioMicSamplingRate** = 24000

3.31 AudioSessionParameters Struct Reference

Public Member Functions

- int **CategoryOptionsToInt** ()
- override string **ToString** ()

Public Attributes

- AudioSessionCategory **Category**
- AudioSessionMode **Mode**
- AudioSessionCategoryOption[ ] **CategoryOptions**

3.32 AudioSessionParametersPresets Class Reference

Static Public Attributes

- static AudioSessionParameters Game
- static AudioSessionParameters VoIP

3.32.1 Member Data Documentation

3.32.1.1 Game

```
AudioSessionParameters Game  [static]
```

Initial value:

```
= new AudioSessionParameters()
        {
            Category = AudioSessionCategory.PlayAndRecord,
            Mode = AudioSessionMode.VoiceChat,
            CategoryOptions = new AudioSessionCategoryOption[] {
      AudioSessionCategoryOption.DefaultToSpeaker, AudioSessionCategoryOption.AllowBluetooth }
        }
```

3.32.1.2 VoIP

```
AudioSessionParameters VoIP  [static]
```

Initial value:

```
= new AudioSessionParameters()
        {
            Category = AudioSessionCategory.PlayAndRecord,
            Mode = AudioSessionMode.VoiceChat,


            CategoryOptions = new AudioSessionCategoryOption[] { AudioSessionCategoryOption.AllowBluetooth }
        }
```

3.33 AudioStreamReceiver Class Reference

Basic audio stream receiver Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

Inherits StreamTransportManagerListener, and IStreamReceiver.

Public Member Functions

- virtual void OnRemoteVoiceAdd (int channelId, int playerId, byte voiceId, VoiceInfo i, ref RemoteVoiceOptions options)

  Called when a remote stream (voice) has been detected.

Public Attributes

- AudioReceptionSettings **audioReceptionSettings** = new AudioReceptionSettings()

Public Attributes inherited from StreamTransportManagerListener

- GameObject **streamTransportGameObject** = null

Protected Member Functions

- override void **OnStreamTransportManagerFound** ()
- virtual void **Update** ()
- virtual void **StreamService** ()

Protected Member Functions inherited from StreamTransportManagerListener

- virtual void **Awake** ()
- virtual void **OnDestroy** ()
- virtual IEnumerator FindStreamTransportManagerCoroutine ()

  Try to find a IRegisterableStreamTransportManager, using StreamTransportObserver.FindStreamTransportManager, that will fill the _streamTransportManager field if found Retries a few times (around 10s) if not found, before giving up.
- void **OnStreamTransportManagerFound** ()

  Called when a _streamTransportManager is set, and so a trnasport can be used.

Protected Attributes

- List< IAudioOut< float > > **audioPlayers** = new List<IAudioOut<float>>()

Protected Attributes inherited from StreamTransportManagerListener

- IStreamTransportManager **_streamTransportManager** = null
- Photon.Voice.Unity.Logger **_logger** = new Photon.Voice.Unity.Logger(LogLevel.Info)

Properties

- IEnumerable< IAudioOut< float > > **AudioPlayers** `[get]`

Properties inherited from StreamTransportManagerListener

- GameObject StreamTransportGameObject `[get, set]`
- IStreamTransportManager StreamTransportManager `[get, set]`
- StreamTransport **Transport** `[get]`

Properties inherited from IStreamTransportManagerListener

- GameObject **gameObject** `[get]`

3.33.1 Detailed Description

Basic audio stream receiver Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

3.33.2 Member Function Documentation

3.33.2.1 OnRemoteVoiceAdd()

```
virtual void OnRemoteVoiceAdd (
            int channelId,
            int playerId,
            byte voiceId,
            VoiceInfo i,
            ref RemoteVoiceOptions options)  [virtual]
```

Called when a remote stream (voice) has been detected.

Implements IStreamReceiver.

3.34 AudioStreamSender Class Reference

Basic audio stream sender Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

Inherits StreamTransportManagerListener, and IStreamSender.

Public Types

- enum **Status**

Public Member Functions

- virtual bool **IsRecordingPossible** ()
- virtual void **StartSending** ()
- virtual void **StopSending** ()
- virtual void **ToggleSending** ()
- virtual void OnRoomJoined ()

  Called when the Photon Voice room is joined, and creating streams (voice) is possible.
- virtual void OnLeftRoom ()

Public Attributes

- AudioSenderSettings **audioSenderSettings** = new AudioSenderSettings()
- bool **sendOnJoinRoom** = true
- Status **status** = Status.NotSending

Public Attributes inherited from StreamTransportManagerListener

- GameObject **streamTransportGameObject** = null

Protected Member Functions

- override void **OnStreamTransportManagerFound** ()
- virtual void **Update** ()
- override void OnDestroy ()
- void **CreateLocalStreamAudio** ()
- void **RemoveLocalStreamAudio** ()
- void **CreateStream** ()
- virtual IEnumerator **CreateStreamCoroutine** ()

  Check if collecting is possible (by testing IsRecordingPossible()) before starting the recording If the recording is not possible for MaxWaitTimeforCollectingAvailability seconds, displays an error and gives up.
- virtual IEnumerator **WaitForRecordingBeingPossible** ()
- virtual void **WillTrytoRecord** ()
- virtual void **AskForPermissions** ()
- virtual void **OnPermissionGranted** (bool granted)

Protected Member Functions inherited from StreamTransportManagerListener

- virtual void **Awake** ()
- virtual IEnumerator FindStreamTransportManagerCoroutine ()

  Try to find a IRegisterableStreamTransportManager, using StreamTransportObserver.FindStreamTransportManager, that will fill the _streamTransportManager field if found Retries a few times (around 10s) if not found, before giving up.
- void **OnStreamTransportManagerFound** ()

  Called when a _streamTransportManager is set, and so a trnasport can be used.

Protected Attributes

- bool **_shouldSend** = false
- bool **_roomJoined** = false
- Photon.Voice.DeviceInfo **_micDevice** = Photon.Voice.DeviceInfo.Default
- IAudioDesc **_audioSource**
- bool **_permissionGranted** = false

Protected Attributes inherited from StreamTransportManagerListener

- IStreamTransportManager **_streamTransportManager** = null
- Photon.Voice.Unity.Logger **_logger** = new Photon.Voice.Unity.Logger(LogLevel.Info)

Properties

- virtual int **AudioChannels** `[get]`
- int **AudioTransportChannel** `[get]`
- virtual float **MaxWaitTimeforCollectingAvailability** `[get]`

Properties inherited from StreamTransportManagerListener

- GameObject StreamTransportGameObject `[get, set]`
- IStreamTransportManager StreamTransportManager `[get, set]`
- StreamTransport **Transport** `[get]`

Properties inherited from IStreamTransportManagerListener

- GameObject **gameObject** [get]

3.34.1 Detailed Description

Basic audio stream sender Here for demo/pedagogical purposes: prefer using FusionVoiceClient, UnityVoiceClient, ... aka the main VoiceConnection subclasses that directly manage the audio stream (they manage recorders, offer a deeper app life cycle management, ...)

3.34.2 Member Function Documentation

3.34.2.1 OnDestroy()

```
override void OnDestroy () [protected], [virtual]
```

Reimplemented from StreamTransportManagerListener.

3.34.2.2 OnLeftRoom()

```
virtual void OnLeftRoom () [virtual]
```

Implements IStreamSender.

3.34.2.3 OnRoomJoined()

```
virtual void OnRoomJoined () [virtual]
```

Called when the Photon Voice room is joined, and creating streams (voice) is possible.

Implements IStreamSender.

3.35 AudioSyncBuffer< T > Class Template Reference

Inherits AudioOutDelayControl< T >.

Public Member Functions

- **AudioSyncBuffer** (PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (T[] data, int offsetSamples)
- override void Stop ()
- void **Read** (T[] outBuf, int outChannels, int outSampleRate)

Public Member Functions inherited from AudioOutDelayControl< T >

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()

Properties

- override long **OutPos** `[get]`

Properties inherited from AudioOutDelayControl< T >

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

Additional Inherited Members

Static Public Attributes inherited from AudioOutDelayControl< T >

- const int **FRAME_POOL_CAPACITY** = 50

Protected Attributes inherited from AudioOutDelayControl< T >

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

3.35.1 Member Function Documentation

3.35.1.1 Stop()

```
override void Stop ()  [virtual]
```

Reimplemented from AudioOutDelayControl< T >.

3.36 AudioUtil Class Reference

Collection of Audio Utility functions and classes.

Classes

- class GeneratorPusher< T >

  IAudioPusher that provides a constant tone signal.
- class GeneratorReader< T >
- interface ILevelMeter

  Audio Level Metering interface.
- interface IVoiceDetector

  Voice Activity Detector interface.
- class LevelMeter< T >

  Audio Level Meter.
- class LevelMeterDummy

  Dummy Audio Level Meter that doesn't actually do anything.
- class LevelMeterFloat

  LevelMeter specialization for float audio.
- class LevelMeterShort

  LevelMeter specialization for short audio.
- class Resampler< T >

  Sample-rate conversion Audio Processor.
- class TempoUp< T >
- class ToneAudioPusher< T >

  IAudioPusher that provides a constant tone signal.
- class ToneAudioReader< T >

  IAudioReader that provides a constant tone signal.
- class VoiceDetector< T >

  Simple voice activity detector triggered by signal level.
- class VoiceDetectorCalibration< T >

  Calibration Utility for Voice Detector.
- class VoiceDetectorDummy

  Dummy VoiceDetector that doesn't actually do anything.
- class VoiceDetectorFloat

  VoiceDetector specialization for float audio.
- class VoiceDetectorShort

  VoiceDetector specialization for float audio.
- class VoiceLevelDetectCalibrate< T >

  Utility Audio Processor Voice Detection Calibration.
- class WaveformAudioPusher< T >

  IAudioPusher that provides the given waveform.
- class WaveformAudioReader< T >

  IAudioReader that provides the given waveform.

Static Public Member Functions

- static int **ToneToBuf**< **T** > (T[ ] buf, long timeSamples, int channels, double amp, double k, double phase↩ Mod=0)
- static int **ToneToBuf**< **T** > (T[ ] buf, int offset, int length, long timeSamples, int channels, double amp, double k, double phaseMod=0)
- static int **WaveformToBuf**< **T** > (T[ ] buf, T[ ] waveform, long timePos)
- static void Resample< T > (T[ ] src, T[ ] dst, int dstCount, int channels)

  Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer.

- static void **Resample**< **T** > (T[ ] src, int srcOffset, int srcCount, T[ ] dst, int dstOffset, int dstCount, int channels)
- static void **Resample**< **T** > (T[ ] src, int srcOffset, int srcCount, int srcChannels, T[ ] dst, int dstOffset, int dstCount, int dstChannels)
- static void ResampleAndConvert (short[ ] src, float[ ] dst, int dstCount, int channels)

  Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert short to float samples along the way.
- static void ResampleAndConvert (float[ ] src, short[ ] dst, int dstCount, int channels)

  Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert float to short samples along the way.
- static void Convert (float[ ] src, short[ ] dst, int dstCount)

  Convert audio buffer from float to short samples.
- static void Convert (short[ ] src, float[ ] dst, int dstCount)

  Convert audio buffer from short to float samples.
- static void ForceToStereo< T > (T[ ] src, T[ ] dst, int srcChannels)

  Convert audio buffer with arbitrary number of channels to stereo.

3.36.1 Detailed Description

Collection of Audio Utility functions and classes.

3.36.2 Member Function Documentation

3.36.2.1 Convert() [1/2]

```
void Convert (
            float[ ] src,
            short[ ] dst,
            int dstCount)  [static]
```

Convert audio buffer from float to short samples.

Parameters

	src	Source buffer.
	dst	Destination buffer.
dstCount		Size of destination buffer (in total samples), source buffer must be of same length or longer.

3.36.2.2 Convert() [2/2]

```
void Convert (
            short[ ] src,
            float[ ] dst,
            int dstCount)  [static]
```

Convert audio buffer from short to float samples.

Parameters

	src	Source buffer.
	dst	Destination buffer.
	dstCount	Size of destination buffer (in total samples), source buffer must be of same length or longer.

3.36.2.3 ForceToStereo< T >()

```
void ForceToStereo< T > (
            T[] src,
            T[] dst,
            int srcChannels)  [static]
```

Convert audio buffer with arbitrary number of channels to stereo.

For mono sources (srcChannels==1), the signal will be copied to both Left and Right stereo channels. For all others, the first two available channels will be used, any other channels will be discarded.

Parameters

	src	Source buffer.
	dst	Destination buffer.
	srcChannels	Number of (interleaved) channels in src.

3.36.2.4 Resample< T >()

```
void Resample< T > (
            T[] src,
            T[] dst,
            int dstCount,
            int channels)  [static]
```

Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

Parameters

	src	Source buffer.
	dst	Destination buffer.
	dstCount	Target size of destination buffer (in samples per channel).
	channels	Number of channels in the signal (1=mono, 2=stereo). Must be > 0.

3.36.2.5 ResampleAndConvert() [1/2]

```
void ResampleAndConvert (
            float[] src,
            short[] dst,
            int dstCount,
            int channels)  [static]
```

Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert float to short samples along the way.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

Parameters

	src	Source buffer.
	dst	Destination buffer.
	dstCount	Target size of destination buffer (in samples per channel).
	channels	Number of channels in the signal (1=mono, 2=stereo). Must be > 0.

3.36.2.6 ResampleAndConvert() [2/2]

```
void ResampleAndConvert (
            short[] src,
            float[] dst,
            int dstCount,
            int channels)  [static]
```

Resample audio data so that the complete src buffer fits into dstCount samples in the dst buffer, and convert short to float samples along the way.

This implements a primitive nearest-neighbor resampling algorithm for an arbitrary number of channels.

Parameters

	src	Source buffer.
	dst	Destination buffer.
	dstCount	Target size of destination buffer (in samples per channel).
	channels	Number of channels in the signal (1=mono, 2=stereo). Must be > 0.

3.37 BufferReaderPushAdapterAsyncPool< T > Class Template Reference

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync.

Inherits BufferReaderPushAdapterBase< T >.

Public Member Functions

- BufferReaderPushAdapterAsyncPool (IDataReader< T > reader)

  Create a new BufferReaderPushAdapter instance.
- override void Service (LocalVoice localVoice)

  Do the actual data read/push.

Public Member Functions inherited from BufferReaderPushAdapterBase< T >

- void Service (LocalVoice localVoice)

  Do the actual data read/push.
- BufferReaderPushAdapterBase (IDataReader< T > reader)

  Create a new BufferReaderPushAdapterBase instance.
- void **Dispose** ()

  Release resources associated with this instance.

Additional Inherited Members

Protected Attributes inherited from BufferReaderPushAdapterBase< T >

- IDataReader< T > **reader**

3.37.1 Detailed Description

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync.

Acquires a buffer from pool before each Read, releases buffer after last Read (brings Acquire/Release overhead). Expects localVoice to be a LocalVoiceFramed<T> of same T.

3.37.2 Member Function Documentation

3.37.2.1 BufferReaderPushAdapterAsyncPool()

```
BufferReaderPushAdapterAsyncPool (
            IDataReader< T > reader)
```

Create a new BufferReaderPushAdapter instance.

Parameters

	reader	DataReader to read from.

3.37.2.2 Service()

```
override void Service (
            LocalVoice localVoice)
```

Do the actual data read/push.

Parameters

	localVoice	LocalVoice instance to push data to. Must be a LocalVoiceFramed<T> of same T.

Implements IServiceable.

3.38 BufferReaderPushAdapterAsyncPoolFloatToShort Class Reference

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting float samples to short.

Inherits BufferReaderPushAdapterBase< T >.

Public Member Functions

- BufferReaderPushAdapterAsyncPoolFloatToShort (IDataReader< float > reader)

  Create a new BufferReaderPushAdapter instance.
- override void Service (LocalVoice localVoice)

  Do the actual data read/push.

Public Member Functions inherited from BufferReaderPushAdapterBase< T >

- void Service (LocalVoice localVoice)

  Do the actual data read/push.
- BufferReaderPushAdapterBase (IDataReader< T > reader)

  Create a new BufferReaderPushAdapterBase instance.
- void **Dispose** ()

  Release resources associated with this instance.

Additional Inherited Members

Protected Attributes inherited from BufferReaderPushAdapterBase< T >

- IDataReader< T > **reader**

3.38.1 Detailed Description

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting float samples to short.

This adapter works exactly like BufferReaderPushAdapterAsyncPool<T>, but it converts float samples to short. Acquires a buffer from pool before each Read, releases buffer after last Read.

Expects localVoice to be a LocalVoiceFramed<T> of same T.

3.38.2 Constructor & Destructor Documentation

3.38.2.1 BufferReaderPushAdapterAsyncPoolFloatToShort()

```
BufferReaderPushAdapterAsyncPoolFloatToShort (
            IDataReader< float > reader)
```

Create a new BufferReaderPushAdapter instance.

Parameters

	reader	DataReader to read from.

3.38.3 Member Function Documentation

3.38.3.1 Service()

```
override void Service (
            LocalVoice localVoice)
```

Do the actual data read/push.

Parameters

	localVoice	LocalVoice instance to push data to. Must be a LocalVoiceFramed<T> of same T.

Implements IServiceable.

3.39 BufferReaderPushAdapterAsyncPoolShortToFloat Class Reference

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting short samples to float.

Inherits BufferReaderPushAdapterBase< T >.

Public Member Functions

- BufferReaderPushAdapterAsyncPoolShortToFloat (IDataReader< short > reader)
  Create a new BufferReaderPushAdapter instance.
- override void Service (LocalVoice localVoice)
  Do the actual data read/push.

Public Member Functions inherited from BufferReaderPushAdapterBase< T >

- void Service (LocalVoice localVoice)
  Do the actual data read/push.
- BufferReaderPushAdapterBase (IDataReader< T > reader)
  Create a new BufferReaderPushAdapterBase instance.
- void **Dispose** ()
  Release resources associated with this instance.

Additional Inherited Members

Protected Attributes inherited from BufferReaderPushAdapterBase< T >

- IDataReader< T > **reader**

3.39.1 Detailed Description

BufferReaderPushAdapter<T> implementation using asynchronous LocalVoiceFramed<T>.PushDataAsync, converting short samples to float.

This adapter works exactly like BufferReaderPushAdapterAsyncPool<T>, but it converts short samples to float. Acquires a buffer from pool before each Read, releases buffer after last Read.

Expects localVoice to be a LocalVoiceFramed<T> of same T.

3.39.2 Constructor & Destructor Documentation

3.39.2.1 BufferReaderPushAdapterAsyncPoolShortToFloat()

```
BufferReaderPushAdapterAsyncPoolShortToFloat (
            IDataReader< short > reader)
```

Create a new BufferReaderPushAdapter instance.

Parameters

	reader	DataReader to read from.

3.39.3 Member Function Documentation

3.39.3.1 Service()

```
override void Service (
            LocalVoice localVoice)
```

Do the actual data read/push.

Parameters

	localVoice	LocalVoice instance to push data to. Must be a LocalVoiceFramed<T> of same T.

Implements IServiceable.

3.40 BufferReaderPushAdapterBase< T > Class Template Reference

Adapter base reading data from IDataReader<T>.Read and pushing it to LocalVoice.

Inherits IServiceable.

Inherited by BufferReaderPushAdapterAsyncPool< T >, BufferReaderPushAdapterAsyncPoolFloatToShort, and BufferReaderPushAdapterAsyncPoolShortToFloat.

Public Member Functions

- void Service (LocalVoice localVoice)
  Do the actual data read/push.
- BufferReaderPushAdapterBase (IDataReader< T > reader)
  Create a new BufferReaderPushAdapterBase instance.
- void **Dispose** ()
  Release resources associated with this instance.

Protected Attributes

- IDataReader< T > **reader**

3.40.1 Detailed Description

Adapter base reading data from IDataReader<T>.Read and pushing it to LocalVoice.

Use this with a LocalVoice of same T type.

3.40.2 Member Function Documentation

3.40.2.1 BufferReaderPushAdapterBase()

```
BufferReaderPushAdapterBase (
            IDataReader< T > reader)
```

Create a new BufferReaderPushAdapterBase instance.

Parameters

reader	DataReader to read from.

3.40.2.2 Service()

```
void Service (
            LocalVoice localVoice)  [abstract]
```

Do the actual data read/push.

Parameters

localVoice	LocalVoice instance to push data to.

Implements IServiceable.

3.41 ByteStreamDecoder Class Reference

Inherits IDecoder.

Public Member Functions

- delegate void **OutputDelegate** (ref FrameBuffer buf)
- **ByteStreamDecoder** (OutputDelegate output, Action onMissingFrame)
- void Input (ref FrameBuffer buf)

  Consumes the given encoded data.
- void Open (VoiceInfo info)

  Open (initialize) the decoder.
- void **Dispose** ()

Properties

- string Error `[get]`

  If not null, the object is in invalid state.

3.41.1 Member Function Documentation

3.41.1.1 Input()

```
void Input (
            ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements IDecoder.

3.41.1.2 Open()

```
void Open (
            VoiceInfo info)
```

Open (initialize) the decoder.

Parameters

	info	Properties of the data stream to decode.

Implements IDecoder.

3.41.2 Property Documentation

3.41.2.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implements IDecoder.

3.42 ByteStreamEncoder Class Reference

Inherits IEncoderDirect< B >.

Public Member Functions

- void **Input** (byte[ ] buf)
- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.
- void **Dispose** ()

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)

  Consumes the given raw data.

Properties

- string Error [get]

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output [set]

  Set callback encoder calls on each encoded data frame (if such output supported).

3.42.1 Member Function Documentation

3.42.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
            out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements IEncoder.

3.42.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements IEncoder.

3.42.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements IEncoder.

Type Constraints

I* : *class

3.42.2 Property Documentation

3.42.2.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implements IEncoder.

3.42.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output  [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements IEncoder.

3.43 CaptureDevice Class Reference

Public Types

- enum **Media**

Public Member Functions

- **CaptureDevice** (ILogger logger, Media media, string deviceID)
- void **Initialize** ()
- void **InitializeAsync** ()
- async Task **CleanUpAsync** ()

  Asynchronous method cleaning up resources and stopping recording if necessary.
- async Task< IMediaEncodingProperties > SelectPreferredCameraStreamSettingAsync (MediaStreamType mediaStreamType, Func< IMediaEncodingProperties, bool > filterSettings)

  Allow selection of camera settings.
- async Task StartRecordingAsync (MediaEncodingProfile encodingProfile, Action< byte[], FrameFlags > encoderCallback)

  Starts media recording asynchronously.
- async Task **StopRecordingAsync** ()

  Stops recording asynchronously.

Static Public Member Functions

- static async Task< bool > **CheckForRecordingDeviceAsync** ()

Properties

- MediaCapture **CaptureSource** `[get]`

  Creates url object from MediaCapture.

Events

- EventHandler< MediaCaptureFailedEventArgs > **CaptureFailed**

3.43.1 Member Function Documentation

3.43.1.1 SelectPreferredCameraStreamSettingAsync()

```
async Task< IMediaEncodingProperties > SelectPreferredCameraStreamSettingAsync (
            MediaStreamType mediaStreamType,
            Func< IMediaEncodingProperties, bool > filterSettings)
```

Allow selection of camera settings.

Parameters

mediaStreamType		Type of a the media stream.
	filterSettings	A predicate function, which will be called to filter the correct settings.

3.43.1.2 StartRecordingAsync()

```
async Task StartRecordingAsync (
            MediaEncodingProfile encodingProfile,
            Action< byte[], FrameFlags > encoderCallback)
```

Starts media recording asynchronously.

Parameters

encodingProfile	Encoding profile used for the recording session

3.44 ConnectAndJoin Class Reference

Inherits MonoBehaviour, IConnectionCallbacks, and IMatchmakingCallbacks.

Public Member Functions

- void **ConnectNow** ()
- void **OnCreatedRoom** ()
- void **OnCreateRoomFailed** (short returnCode, string message)
- void **OnFriendListUpdate** (List< FriendInfo > friendList)
- void **OnJoinedRoom** ()
- void **OnJoinRandomFailed** (short returnCode, string message)
- void **OnJoinRoomFailed** (short returnCode, string message)
- void **OnLeftRoom** ()
- void **OnConnected** ()
- void **OnConnectedToMaster** ()
- void **OnDisconnected** (DisconnectCause cause)
- void **OnRegionListReceived** (RegionHandler regionHandler)
- void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- void **OnCustomAuthenticationFailed** (string debugMessage)

Public Attributes

- bool **RandomRoom** = true
- string **RoomName**

Properties

- bool **IsConnected** `[get]`

3.45 VoiceClient.CreateOptions Struct Reference

Public Attributes

- byte **VoiceIDMin**
- byte **VoiceIDMax**

Static Public Attributes

- static CreateOptions Default

3.45.1 Member Data Documentation

3.45.1.1 Default

```
CreateOptions Default  [static]
```

Initial value:

```
= new CreateOptions()
            {
                VoiceIDMin = 1,
                VoiceIDMax = 15
            }
```

3.46 OpusCodec.Decoder< T > Class Template Reference

Inherits IDecoder.

Public Member Functions

- **Decoder** (Action< FrameOut< T > > output, ILogger logger)
- void Open (VoiceInfo i)

  Open (initialize) the decoder.
- void **Dispose** ()
- void Input (ref FrameBuffer buf)

  Consumes the given encoded data.

Protected Attributes

- OpusDecoder< T > **decoder**

Properties

- string Error `[get]`

  If not null, the object is in invalid state.

3.46.1 Member Function Documentation

3.46.1.1 Input()

```
void Input (
            ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements IDecoder.

3.46.1.2 Open()

```
void Open (
            VoiceInfo info)
```

Open (initialize) the decoder.

Parameters

	info	Properties of the data stream to decode.

Implements IDecoder.

3.46.2 Property Documentation

3.46.2.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implements IDecoder.

3.47 RawCodec.Decoder< T > Class Template Reference

Inherits IDecoder.

Public Member Functions

- **Decoder** (Action< FrameOut< T > > output)
- void Open (VoiceInfo info)

  Open (initialize) the decoder.
- void Input (ref FrameBuffer byteBuf)

  Consumes the given encoded data.
- void **Dispose** ()

Properties

- string Error [get]

  If not null, the object is in invalid state.

3.47.1 Member Function Documentation

3.47.1.1 Input()

```
void Input (
            ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implements IDecoder.

3.47.1.2 Open()

```
void Open (
            VoiceInfo info)
```

Open (initialize) the decoder.

Parameters

	info	Properties of the data stream to decode.

Implements IDecoder.

3.47.2 Property Documentation

3.47.2.1 Error

```
string Error [get]
```

If not null, the object is in invalid state.

Implements IDecoder.

3.48 DecoderConfigFrame Class Reference

Stores the config frame and prevents other frames decoding until the decoder is ready.

Inherits IDisposable.

Public Member Functions

- **DecoderConfigFrame** (ILogger logger, IDecoder decoder)
- bool TryConfigure (ref FrameBuffer buf, bool decoderReady)

  Call it in Input().
- void **Dispose** ()

3.48.1 Detailed Description

Stores the config frame and prevents other frames decoding until the decoder is ready.

3.48.2 Member Function Documentation

3.48.2.1 TryConfigure()

```
bool TryConfigure (
            ref FrameBuffer buf,
            bool decoderReady)
```

Call it in Input().

Parameters

	buf	Data frame.
decoderReady		True if the decoder is ready.

Returns

True if decoder is allowed to decode the current frame.

3.49 DeviceEnumerator Class Reference

Inherits DeviceEnumeratorBase.

Inherited by AudioInEnumerator, and VideoInEnumerator.

Public Member Functions

- **DeviceEnumerator** (ILogger logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.49.1 Member Function Documentation

3.49.1.1 Refresh()

```
override void Refresh ()
```

Implements IDeviceEnumerator.

3.50 DeviceEnumeratorBase Class Reference

Inherits IDeviceEnumerator.

Inherited by , , AudioInEnumerator, AudioInEnumerator, DeviceEnumerator, AudioInEnumerator, and VideoInEnumerator.

Public Member Functions

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Protected Attributes

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.50.1 Member Function Documentation

3.50.1.1 Refresh()

```
void Refresh ()  [abstract]
```

Implements IDeviceEnumerator.

3.50.2 Property Documentation

3.50.2.1 Error

```
virtual string Error  [get], [protected set]
```

Implements IDeviceEnumerator.

3.50.2.2 IsSupported

```
virtual bool IsSupported  [get]
```

Implements IDeviceEnumerator.

3.50.2.3 OnReady

```
Action OnReady  [protected get], [set]
```

Implements IDeviceEnumerator.

3.51 DeviceFeatures Class Reference

Public Member Functions

- **DeviceFeatures** (CameraFacing facing)

Properties

- CameraFacing **CameraFacing** `[get]`

3.52 DeviceInfo Struct Reference

Public Member Functions

- **DeviceInfo** (int id, string name, DeviceFeatures features=null)
- **DeviceInfo** (string id, string name, DeviceFeatures features=null)
- **DeviceInfo** (string name, DeviceFeatures features=null)
- override bool **Equals** (object obj)
- override int **GetHashCode** ()
- override string **ToString** ()

Static Public Member Functions

- static bool **operator==** (DeviceInfo d1, DeviceInfo d2)
- static bool **operator!=** (DeviceInfo d1, DeviceInfo d2)

Public Attributes

- DeviceFeatures **features**

Static Public Attributes

- static readonly DeviceInfo **Default** = new DeviceInfo(true, -128, "", "[Default]")

Properties

- bool **IsDefault** `[get]`
- int **IDInt** `[get]`
- string **IDString** `[get]`
- string **Name** `[get]`
- DeviceFeatures **Features** `[get]`

3.53 RecorderPreset.DSP Struct Reference

Public Attributes

- bool **AEC**
- bool **VAD**

3.54 OpusCodec.Encoder< T > Class Template Reference

Inherits IEncoderDirect< B >.

Inherited by OpusCodec.EncoderFloat, and OpusCodec.EncoderShort.

Public Member Functions

- void **Input** (T[] buf)
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.
- void **Dispose** ()

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)

  Consumes the given raw data.

Protected Member Functions

- **Encoder** (VoiceInfo i, ILogger logger)
- void **encodeTyped** (T[] buf)

Protected Attributes

- OpusEncoder **encoder**
- bool **disposed**

Properties

- string Error `[get]`

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output `[get, set]`

  Set callback encoder calls on each encoded data frame (if such output supported).

3.54.1 Member Function Documentation

3.54.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
            out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements IEncoder.

3.54.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements IEncoder.

3.54.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements IEncoder.

Type Constraints

I* : *class

3.54.2 Property Documentation

3.54.2.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implements IEncoder.

3.54.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output  [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements IEncoder.

3.55 RawCodec.Encoder< T > Class Template Reference

Inherits IEncoderDirect< B >.

Public Member Functions

- ArraySegment< byte > DequeueOutput (out FrameFlags flags)
  Returns next encoded data frame (if such output supported).
- void EndOfStream ()
  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- I GetPlatformAPI< I > ()
  Returns an platform-specific interface.
- void **Dispose** ()
- void **Input** (T[] buf)

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)
  Consumes the given raw data.

Properties

- string Error [get]
  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output [get, set]
  Set callback encoder calls on each encoded data frame (if such output supported).

3.55.1 Member Function Documentation

3.55.1.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
            out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implements IEncoder.

3.55.1.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implements IEncoder.

3.55.1.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implements IEncoder.

Type Constraints

I* : *class

3.55.2 Property Documentation

3.55.2.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implements IEncoder.

3.55.2.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output  [get], [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implements IEncoder.

3.56 OpusCodec.EncoderFloat Class Reference

Inherits OpusCodec.Encoder< T >.

Protected Member Functions

- override void **encodeTyped** (float[ ] buf)

Protected Member Functions inherited from OpusCodec.Encoder< T >

- **Encoder** (VoiceInfo i, ILogger logger)
- void **encodeTyped** (T[ ] buf)

Additional Inherited Members

Public Member Functions inherited from OpusCodec.Encoder< T >

- void **Input** (T[ ] buf)
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.
- void **Dispose** ()

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)

  Consumes the given raw data.

Protected Attributes inherited from OpusCodec.Encoder< T >

- OpusEncoder **encoder**
- bool **disposed**

Properties inherited from OpusCodec.Encoder< T >

- string Error `[get]`

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output `[get, set]`

  Set callback encoder calls on each encoded data frame (if such output supported).

3.57 OpusCodec.EncoderShort Class Reference

Inherits OpusCodec.Encoder< T >.

Protected Member Functions

- override void **encodeTyped** (short[ ] buf)

Protected Member Functions inherited from OpusCodec.Encoder< T >

- **Encoder** (VoiceInfo i, ILogger logger)
- void **encodeTyped** (T[ ] buf)

Additional Inherited Members

Public Member Functions inherited from OpusCodec.Encoder< T >

- void **Input** (T[ ] buf)
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.
- void **Dispose** ()

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)

  Consumes the given raw data.

Protected Attributes inherited from OpusCodec.Encoder< T >

- OpusEncoder **encoder**
- bool **disposed**

Properties inherited from OpusCodec.Encoder< T >

- string Error `[get]`

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output `[get, set]`

  Set callback encoder calls on each encoded data frame (if such output supported).

3.58 OpusCodec.Factory Class Reference

Static Public Member Functions

- static IEncoder **CreateEncoder**< **B** > (VoiceInfo i, ILogger logger)

3.59 Flip Struct Reference

Public Member Functions

- override bool **Equals** (object obj)
- override int **GetHashCode** ()

Static Public Member Functions

- static bool **operator==** (Flip f1, Flip f2)
- static bool **operator!=** (Flip f1, Flip f2)
- static Flip **operator*** (Flip f1, Flip f2)

Static Public Attributes

- static Flip **None**
- static Flip **Vertical** = new Flip() { IsVertical = true }
- static Flip **Horizontal** = new Flip() { IsHorizontal = true }
- static Flip **Both** = Vertical * Horizontal

Properties

- bool **IsVertical** `[get]`
- bool **IsHorizontal** `[get]`

3.60 FMODRecorderSetup Class Reference

Inherits VoiceComponent.

Protected Member Functions

- override void Awake ()

Additional Inherited Members

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.60.1 Member Function Documentation

3.60.1.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.61 FrameBuffer Struct Reference

Public Member Functions

- **FrameBuffer** (byte[ ] array, int offset, int count, FrameFlags flags, byte frameNum, IDisposable disposer)
- **FrameBuffer** (byte[ ] array, FrameFlags flags, byte frameNum)
- **FrameBuffer** (FrameBuffer from, int offset, int count, FrameFlags flags, byte frameNum)
- void **Retain** ()
- void **Release** ()
- override string **ToString** ()

Public Attributes

- readonly byte[ ] **array**
- readonly int **offset**
- readonly int **count**
- readonly IDisposable **disposer**
- bool **disposed**
- int **refCnt**
- GCHandle **gcHandle**
- IntPtr **ptr**
- bool **pinned**

Properties

- IntPtr **Ptr** `[get]`
- byte[ ] **Array** `[get]`
- int **Length** `[get]`
- int **Offset** `[get]`
- FrameFlags **Flags** `[get]`
- byte **FrameNum** `[get]`
- bool **IsFEC** `[get]`
- bool **IsConfig** `[get]`
- bool **IsKeyframe** `[get]`

3.62 FrameOut< T > Class Template Reference

Public Member Functions

- **FrameOut** (T[ ] buf, bool endOfStream)
- FrameOut< T > **Set** (T[ ] buf, bool endOfStream)

Properties

- T[ ] **Buf** `[get]`
- bool **EndOfStream** `[get]`

3.63 Framer< T > Class Template Reference

Utility class to re-frame packets.

Inherited by FramerResampler< T >.

Public Member Functions

- **Framer** (int frameSize)

  Create new Framer instance.
- virtual IEnumerable< T[ ]> Frame (T[ ] buf)

  Append arbitrary-sized buffer and return available full frames.

Protected Attributes

- T[ ] **frame**
- int **sizeofT**
- int **framePos** = 0

3.63.1 Detailed Description

Utility class to re-frame packets.

3.63.2 Member Function Documentation

3.63.2.1 Frame()

```
virtual IEnumerable< T[ ]> Frame (
            T[ ] buf)  [virtual]
```

Append arbitrary-sized buffer and return available full frames.

Parameters

	buf	Array of samples to add.

Returns

Enumerator of full frames (might be none).

Reimplemented in FramerResampler< T >.

3.64 FramerResampler< T > Class Template Reference

Inherits Framer< T >.

Public Member Functions

- **FramerResampler** (int frameSize, int channels, int resampleNum, int resampleDen, bool interpolate)
- override IEnumerable< T[ ]> Frame (T[ ] bufT)

  Append arbitrary-sized buffer and return available full frames.

Public Member Functions inherited from Framer< T >

- **Framer** (int frameSize)

  Create new Framer instance.

Protected Attributes

- bool **TisFloat**
- bool **interpolate**
- int **channels**
- int **resampleNum**
- int **resampleDen**
- float **resampleRatioInv**
- int **delta**

Protected Attributes inherited from Framer< T >

- T[ ] **frame**
- int **sizeofT**
- int **framePos** = 0

3.64.1 Member Function Documentation

3.64.1.1 Frame()

```
override IEnumerable< T[ ]> Frame (
            T[ ] buf)  [virtual]
```

Append arbitrary-sized buffer and return available full frames.

Parameters

	buf	Array of samples to add.

Returns

Enumerator of full frames (might be none).

Reimplemented from Framer< T >.

3.65 FusionVoiceClient Class Reference

Inherits VoiceFollowClient, and INetworkRunnerCallbacks.

Public Attributes

- bool **UseFusionAppSettings** = true

  Whether or not to use the Voice AppId and all the other AppSettings from Fusion's RealtimeAppSettings Scriptable↩ Object singleton in the Voice client/app.
- bool **UseFusionAuthValues** = true

  Whether or not to use the same AuthenticationValues used in Fusion client/app in Voice client/app as well. This means that the same UserID will be used in both clients. If custom authentication is used and setup in Fusion AppId from dashboard, the same configuration should be done for the Voice AppId.

Public Attributes inherited from VoiceFollowClient

- bool **AutoConnectAndJoin** = true

  Auto connect voice client and join a voice room when Leader client is joined to a Leader room.

Public Attributes inherited from VoiceConnection

- AppSettings **Settings**

  Settings to be used by this Voice Client.

Protected Member Functions

- override void Start ()
- override void **Awake** ()
- override void OnDestroy ()
- override Speaker InstantiateSpeakerForRemoteVoice (int playerId, byte voiceId, object userData)
- override string **GetVoiceRoomName** ()
- override bool **ConnectVoice** ()

Protected Member Functions inherited from VoiceFollowClient

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- override void OnDestroy ()
- override void OnOperationResponseReceived (OperationResponse operationResponse)
- void **LeaderStateChanged** (ClientState toState)
- override void OnVoiceStateChanged (ClientState fromState, ClientState toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

Protected Member Functions inherited from VoiceConnection

- override void **Awake** ()
- virtual void **Update** ()
- void **CalcStatistics** ()

Properties

- override bool **LeaderInRoom** `[get]`
- override bool **LeaderOfflineMode** `[get]`

Properties inherited from VoiceFollowClient

- bool **LeaderInRoom** `[get]`
- bool **LeaderOfflineMode** `[get]`

Properties inherited from VoiceConnection

- virtual bool **AlwaysUsePrimaryRecorder** `[get]`
- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- new LoadBalancingTransport **Client** `[get]`
- VoiceClient **VoiceClient** `[get]`

  Returns underlying Photon Voice client.
- ClientState **ClientState** `[get]`

  Returns Photon Voice client state.
- float **FramesReceivedPerSecond** `[get]`

  Number of frames received per second.
- float **FramesLostPerSecond** `[get]`

  Number of frames lost per second.
- float **FramesLostPercent** `[get]`

  Percentage of lost frames.
- GameObject **SpeakerPrefab** `[get, set]`

  Prefab that contains Speaker component to be instantiated when receiving a new remote audio source info.
- Recorder **PrimaryRecorder** `[get, set]`

  Primary Recorder to be used by VoiceConnection implementations directly or via integration objects.
- bool **UsePrimaryRecorder** `[get]`

  Use VoiceConnection.PrimaryRecorder directly.
- string **BestRegionSummaryInPreferences** `[get, set]`

  Used to store and access the "Best Region Summary" in the Player Preferences.

Additional Inherited Members

Public Member Functions inherited from VoiceFollowClient

- bool ConnectAndJoinRoom ()

  Connect voice client to Photon servers and join a Voice room.
- void **Disconnect** ()

  Disconnect voice client from all Photon servers.

Public Member Functions inherited from VoiceConnection

- virtual bool ConnectUsingSettings (AppSettings overwriteSettings=null)

  Connect to Photon server using Settings
- bool AddSpeaker (Speaker speaker, object userData)

  Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.
- Speaker InstantiateSpeakerPrefab (GameObject parent, bool destroyOnRemove)

  Instantiates SpeakerPrefab, optionally attaches it to the provided parent.
- bool **AddRecorder** (Recorder rec)
- void **RemoveRecorder** (Recorder rec)

Static Public Attributes inherited from VoiceConnection

- const int **ChannelAudio** = 1

  Recommended Photon Transport channel for audio. Chosen not to interfere with video and default channel.
- const int **ChannelVideo** = 2

  Recommended Photon Transport channel for video. Chosen not to interfere with audio and default channel.

Events inherited from VoiceConnection

- Action< Speaker > **SpeakerLinked**

  Fires when a speaker has been linked to a remote audio stream.
- Action< RemoteVoiceLink > **RemoteVoiceAdded**

  Fires when a remote voice stream is added.

3.65.1 Member Function Documentation

3.65.1.1 InstantiateSpeakerForRemoteVoice()

```
override Speaker InstantiateSpeakerForRemoteVoice (
            int playerId,
            byte voiceId,
            object userData)  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.65.1.2 OnDestroy()

```
override void OnDestroy ()  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.65.1.3 Start()

```
override void Start ()  [protected], [virtual]
```

Reimplemented from VoiceFollowClient.

3.66 AudioUtil.GeneratorPusher< T > Class Template Reference

IAudioPusher that provides a constant tone signal.

Inherits IAudioPusher< T >.

Inherited by AudioUtil.ToneAudioPusher< T >, and AudioUtil.WaveformAudioPusher< T >.

Public Member Functions

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)
  Set the callback function used for pushing data.
- void **Dispose** ()

Protected Member Functions

- int **Gen** (T[] buf, long timeSamples)

Protected Attributes

- long **timeSamples**

Properties

- int Channels `[get]`
  Number of channels in the audio signal.
- int SamplingRate `[get]`
  Sampling rate of the audio signal (in Hz).
- string Error `[get]`
  If not null, audio object is in invalid state.

3.66.1 Detailed Description

IAudioPusher that provides a constant tone signal.

3.66.2 Member Function Documentation

3.66.2.1 SetCallback()

```
void SetCallback (
            Action< T[]> callback,
            ObjectFactory< T[], int > bufferFactory,
            int optimalFrameSize)
```

Set the callback function used for pushing data.

Parameters

	callback	Callback function to use
	bufferFactory	Buffer factory used to create the buffer that is pushed to the callback

Implements IAudioPusher< T >.

3.66.3 Property Documentation

3.66.3.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.66.3.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.66.3.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.67 AudioUtil.GeneratorReader< T > Class Template Reference

Inherits IAudioReader< T >.

Inherited by AudioUtil.ToneAudioReader< T >, and AudioUtil.WaveformAudioReader< T >.

Public Member Functions

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool Read (T[ ] buf)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Protected Member Functions

- int **Gen** (T[] buf, long timeSamples)

Properties

- int Channels [get]
  Number of channels in the audio signal.
- int SamplingRate [get]
  Sampling rate of the audio signal (in Hz).
- string Error [get]
  If not null, audio object is in invalid state.

3.67.1 Member Function Documentation

3.67.1.1 Read()

```
bool Read (
            T[] buffer)
```

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Parameters

	buffer	Buffer to fill.

Returns

True if buffer was filled successfully, false otherwise.

Implements IDataReader< T >.

3.67.2 Property Documentation

3.67.2.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.67.2.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.67.2.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.68 IAudioDesc Interface Reference

Audio Source interface.

Inherits IDisposable.

Inherited by AudioDesc, IAudioPusher< T >, and IAudioReader< T >.

Properties

- int SamplingRate [get]
  Sampling rate of the audio signal (in Hz).
- int Channels [get]
  Number of channels in the audio signal.
- string Error [get]
  If not null, audio object is in invalid state.

3.68.1 Detailed Description

Audio Source interface.

3.68.2 Property Documentation

3.68.2.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implemented in AudioDesc, AudioUtil.GeneratorPusher< T >, AudioUtil.GeneratorReader< T >, AudioInReader< T >, AudioInPusher, AudioInReader, AudioInPusher, AudioInReader, AndroidAudioInAEC, AudioClipWrapper, MicWrapper, MicWrapperPusher, WebAudioMicIn, AudioInPusher, and WindowsAudioInPusher.

3.68.2.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implemented in AudioDesc, AudioUtil.GeneratorPusher< T >, AudioUtil.GeneratorReader< T >, AudioInReader< T >, AudioInPusher, AudioInReader, AudioInPusher, AudioInReader, AndroidAudioInAEC, AudioClipWrapper, MicWrapper, MicWrapperPusher, WebAudioMicIn, AudioInPusher, and WindowsAudioInPusher.

3.68.2.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implemented in AudioDesc, AudioUtil.GeneratorPusher< T >, AudioUtil.GeneratorReader< T >, AudioInReader< T >, AudioInPusher, AudioInReader, AudioInPusher, AudioInReader, AndroidAudioInAEC, AudioClipWrapper, MicWrapper, MicWrapperPusher, WebAudioMicIn, AudioInPusher, and WindowsAudioInPusher.

3.69 IAudioInChangeNotifier Interface Reference

Inherits IDisposable.

Inherited by AudioInChangeNotifierNotSupported, AudioInChangeNotifier, and AudioInChangeNotifier.

Properties

- bool **IsSupported** [get]
- string Error [get]

3.69.1 Property Documentation

3.69.1.1 Error

```
string Error  [get]
```

Implemented in AudioInChangeNotifier, and AudioInChangeNotifier.

3.70 IAudioOut< T > Interface Template Reference

Inherited by AudioOutDelayControl< T >, and AudioOutDummy< T >.

Public Member Functions

- void **Start** (int frequency, int channels, int frameSamplesPerChannel)
- void **Flush** ()
- void **Stop** ()
- void **Push** (T[] frame)
- void **Service** ()

Properties

- bool **IsPlaying** [get]
- int **Lag** [get]

3.71 IAudioPusher< T > Interface Template Reference

Audio Pusher interface.

Inherits IAudioDesc.

Inherited by AudioUtil.GeneratorPusher< T >, AudioInPusher, AudioInPusher, AudioInPusher, AndroidAudioInAEC, MicWrapperPusher, WebAudioMicIn, and WindowsAudioInPusher.

Public Member Functions

- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Additional Inherited Members

Properties inherited from IAudioDesc

- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- int Channels [get]

  Number of channels in the audio signal.
- string Error [get]

  If not null, audio object is in invalid state.

3.71.1 Detailed Description

Audio Pusher interface.

Opposed to an IAudioReader (which will deliver audio data when it is "pulled"), an IAudioPusher will push its audio data whenever it is ready,

3.71.2 Member Function Documentation

3.71.2.1 SetCallback()

```
void SetCallback (
            Action< T[]> callback,
            ObjectFactory< T[], int > bufferFactory,
            int optimalFrameSize)
```

Set the callback function used for pushing data.

Parameters

	callback	Callback function to use.
	bufferFactory	Buffer factory used to create the buffer that is pushed to the callback

Implemented in AudioUtil.GeneratorPusher< T >.

3.72 IAudioReader< T > Interface Template Reference

Audio Reader interface.

Inherits IDataReader< T >, and IAudioDesc.

Inherited by AudioUtil.GeneratorReader< T >, AudioInReader< T >, AudioInReader, AudioInReader, AudioClipWrapper, and MicWrapper.

Additional Inherited Members

Public Member Functions inherited from IDataReader< T >

- bool Read (T[] buffer)
  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Properties inherited from IAudioDesc

- int SamplingRate `[get]`
  Sampling rate of the audio signal (in Hz).
- int Channels `[get]`
  Number of channels in the audio signal.
- string Error `[get]`
  If not null, audio object is in invalid state.

3.72.1 Detailed Description

Audio Reader interface.

Opposed to an IAudioPusher (which will push its audio data whenever it is ready), an IAudioReader will deliver audio data when it is "pulled" (it's Read function is called).

3.73 IDataReader< T > Interface Template Reference

Interface for pulling data, in case this is more appropriate than pushing it.

Inherits IDisposable.

Inherited by IAudioReader< T >.

Public Member Functions

- bool Read (T[] buffer)
  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

3.73.1 Detailed Description

Interface for pulling data, in case this is more appropriate than pushing it.

3.73.2 Member Function Documentation

3.73.2.1 Read()

```
bool Read (
            T[] buffer)
```

Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Parameters

	buffer	Buffer to fill.

Returns

True if buffer was filled successfully, false otherwise.

Implemented in AudioUtil.GeneratorReader< T >, and AudioInReader< T >.

3.74 IDecoder Interface Reference

Generic decoder interface.

Inherits IDisposable.

Inherited by ByteStreamDecoder, IDecoderDirect< B >, OpusCodec.Decoder< T >, and RawCodec.Decoder< T >.

Public Member Functions

- void Open (VoiceInfo info)
  Open (initialize) the decoder.
- void Input (ref FrameBuffer buf)
  Consumes the given encoded data.

Properties

- string Error `[get]`
  If not null, the object is in invalid state.

3.74.1 Detailed Description

Generic decoder interface.

3.74.2 Member Function Documentation

3.74.2.1 Input()

```
void Input (
            ref FrameBuffer buf)
```

Consumes the given encoded data.

The callee can call buf.Retain() to prevent the caller from disposing the buffer. In this case, the callee should call buf.Release() when buffer is no longer needed.

Implemented in ByteStreamDecoder, OpusCodec.Decoder< T >, and RawCodec.Decoder< T >.

3.74.2.2 Open()

```
void Open (
            VoiceInfo info)
```

Open (initialize) the decoder.

Parameters

	info	Properties of the data stream to decode.

Implemented in ByteStreamDecoder, OpusCodec.Decoder< T >, and RawCodec.Decoder< T >.

3.74.3 Property Documentation

3.74.3.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implemented in ByteStreamDecoder, OpusCodec.Decoder< T >, and RawCodec.Decoder< T >.

3.75 IDecoderDirect< B > Interface Template Reference

Interface for an decoder which outputs data via explicit call.

Inherits IDecoder.

Properties

- Action< B > **Output** `[set]`
  Callback to call when a new decoded data buffer is available.

Properties inherited from IDecoder

- string Error `[get]`
  If not null, the object is in invalid state.

Additional Inherited Members

Public Member Functions inherited from IDecoder

- void Open (VoiceInfo info)
  Open (initialize) the decoder.
- void Input (ref FrameBuffer buf)
  Consumes the given encoded data.

3.75.1 Detailed Description

Interface for an decoder which outputs data via explicit call.

3.76 IDeviceEnumerator Interface Reference

Inherits IDisposable, and IEnumerable.

Inherited by DeviceEnumeratorBase.

Public Member Functions

- void Refresh ()

Properties

- bool **IsSupported** [get]
- Action **OnReady** [set]
- string **Error** [get]

3.76.1 Member Function Documentation

3.76.1.1 Refresh()

```
void Refresh ()
```

Implemented in AudioInEnumerator, and AudioInEnumerator.

3.77 IEncoder Interface Reference

Generic encoder interface.

Inherits IDisposable.

Inherited by IEncoderDirect< B >.

Public Member Functions

- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.

Properties

- string Error [get]

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output [set]

  Set callback encoder calls on each encoded data frame (if such output supported).

3.77.1 Detailed Description

Generic encoder interface.

Depending on implementation, encoder should either call Output on eaach data frame or return next data frame in DequeueOutput() call.

3.77.2 Member Function Documentation

3.77.2.1 DequeueOutput()

```
ArraySegment< byte > DequeueOutput (
            out FrameFlags flags)
```

Returns next encoded data frame (if such output supported).

Implemented in ByteStreamEncoder, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

3.77.2.2 EndOfStream()

```
void EndOfStream ()
```

Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).

Implemented in ByteStreamEncoder, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

3.77.2.3 GetPlatformAPI< I >()

```
I GetPlatformAPI< I > ()
```

Returns an platform-specific interface.

Implemented in ByteStreamEncoder, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

Type Constraints

I* : *class

3.77.3 Property Documentation

3.77.3.1 Error

```
string Error  [get]
```

If not null, the object is in invalid state.

Implemented in ByteStreamEncoder, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

3.77.3.2 Output

```
Action<ArraySegment<byte>, FrameFlags> Output  [set]
```

Set callback encoder calls on each encoded data frame (if such output supported).

Implemented in ByteStreamEncoder, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

3.78 IEncoderDirect< B > Interface Template Reference

Interface for an encoder which consumes input data via explicit call.

Inherits IEncoder.

Inherited by ByteStreamEncoder, IEncoderDirectImage, OpusCodec.Encoder< T >, and RawCodec.Encoder< T >.

Public Member Functions

- void Input (B buf)

  Consumes the given raw data.

Public Member Functions inherited from IEncoder

- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.

Additional Inherited Members

Properties inherited from IEncoder

- string Error `[get]`

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output `[set]`

  Set callback encoder calls on each encoded data frame (if such output supported).

3.78.1 Detailed Description

Interface for an encoder which consumes input data via explicit call.

3.78.2 Member Function Documentation

3.78.2.1 Input()

```
void Input (
            B buf)
```

Consumes the given raw data.

Parameters

	buf	Array containing raw data (e.g. audio samples).

3.79 IEncoderDirectImage Interface Reference

Interface for an encoder which consumes images via explicit call.

Inherits IEncoderDirect< B >.

Properties

- ImageFormat **ImageFormat** `[get]`

  Recommended encoder input image format. Encoder may support other formats.

Properties inherited from IEncoder

- string Error `[get]`

  If not null, the object is in invalid state.
- Action< ArraySegment< byte >, FrameFlags > Output `[set]`

  Set callback encoder calls on each encoded data frame (if such output supported).

Additional Inherited Members

Public Member Functions inherited from IEncoderDirect< B >

- void Input (B buf)

  Consumes the given raw data.

Public Member Functions inherited from IEncoder

- ArraySegment< byte > DequeueOutput (out FrameFlags flags)

  Returns next encoded data frame (if such output supported).
- void EndOfStream ()

  Forces an encoder to flush and produce frame with EndOfStream flag (in output queue).
- I GetPlatformAPI< I > ()

  Returns an platform-specific interface.

3.79.1 Detailed Description

Interface for an encoder which consumes images via explicit call.

3.80 AudioUtil.ILevelMeter Interface Reference

Audio Level Metering interface.

Inherited by AudioUtil.LevelMeter< T >, and AudioUtil.LevelMeterDummy.

Public Member Functions

- void ResetAccumAvgPeakAmp ()
  Reset AccumAvgPeakAmp.

Properties

- float CurrentAvgAmp [get]
  Average amplitude value over last half second.
- float CurrentPeakAmp [get]
  Maximum amplitude value over last half second sec.
- float AccumAvgPeakAmp [get]
  Average of CurrentPeakAmps since last reset.

3.80.1 Detailed Description

Audio Level Metering interface.

3.80.2 Member Function Documentation

3.80.2.1 ResetAccumAvgPeakAmp()

```
void ResetAccumAvgPeakAmp ()
```

Reset AccumAvgPeakAmp.

Implemented in AudioUtil.LevelMeter< T >, and AudioUtil.LevelMeterDummy.

3.80.3 Property Documentation

3.80.3.1 AccumAvgPeakAmp

```
float AccumAvgPeakAmp [get]
```

Average of CurrentPeakAmps since last reset.

Implemented in AudioUtil.LevelMeter< T >, and AudioUtil.LevelMeterDummy.

3.80.3.2 CurrentAvgAmp

```
float CurrentAvgAmp [get]
```

Average amplitude value over last half second.

Implemented in AudioUtil.LevelMeter< T >, and AudioUtil.LevelMeterDummy.

3.80.3.3 CurrentPeakAmp

```
float CurrentPeakAmp [get]
```

Maximum amplitude value over last half second sec.

Implemented in AudioUtil.LevelMeter< T >, and AudioUtil.LevelMeterDummy.

3.81 ILocalVoiceAudio Interface Reference

Interface for an outgoing audio stream.

Inherited by LocalVoiceAudio< T >, and LocalVoiceAudioDummy.

Public Member Functions

- void VoiceDetectorCalibrate (int durationMs, Action< float > onCalibrated=null)
  Trigger voice detector calibration process.

Properties

- AudioUtil.IVoiceDetector VoiceDetector `[get]`
  The VoiceDetector in use.
- AudioUtil.ILevelMeter LevelMeter `[get]`
  The LevelMeter utility in use.
- bool VoiceDetectorCalibrating `[get]`
  If true, voice detector calibration is in progress.

3.81.1 Detailed Description

Interface for an outgoing audio stream.

A LocalVoice always brings a LevelMeter and a VoiceDetector, which you can access using this interface.

3.81.2 Member Function Documentation

3.81.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
            int durationMs,
            Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. Voice detector sets threshold based on measured backgroud noise level.

Parameters

	durationMs	Duration of calibration (in milliseconds).
	onCalibrated	Called when calibration is complete. Parameter is new threshold value.

Implemented in LocalVoiceAudio< T >, and LocalVoiceAudioDummy.

3.81.3 Property Documentation

3.81.3.1 LevelMeter

```
AudioUtil.ILevelMeter LevelMeter  [get]
```

The LevelMeter utility in use.

Implemented in LocalVoiceAudio< T >, and LocalVoiceAudioDummy.

3.81.3.2 VoiceDetector

```
AudioUtil.IVoiceDetector VoiceDetector  [get]
```

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implemented in LocalVoiceAudio< T >, and LocalVoiceAudioDummy.

3.81.3.3 VoiceDetectorCalibrating

```
bool VoiceDetectorCalibrating  [get]
```

If true, voice detector calibration is in progress.

Implemented in LocalVoiceAudio< T >, and LocalVoiceAudioDummy.

3.82 ILogger Interface Reference

Inherited by Logger.

Public Member Functions

- void **Log** (LogLevel level, string msg)
- void **Log** (LogLevel level, string fmt, params object[ ] args)

Properties

- LogLevel **Level** `[get]`

3.83 ImageBufferInfo Struct Reference

Classes

- struct StrideSet

Public Member Functions

- **ImageBufferInfo** (int width, int height, StrideSet stride, ImageFormat format)

Properties

- int **Width** `[get]`
- int **Height** `[get]`
- StrideSet **Stride** `[get]`
- ImageFormat **Format** `[get]`
- Rotation **Rotation** `[get, set]`
- Flip **Flip** `[get, set]`

3.84 ImageBufferNative Class Reference

Inherited by ImageBufferNativeAlloc, ImageBufferNativeGCHandleBytes, and ImageBufferNativeGCHandleSinglePlane.

Classes

- struct PlaneSet

Public Member Functions

- **ImageBufferNative** (ImageBufferInfo info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()
- virtual void **Dispose** ()

Public Attributes

- ImageBufferInfo **Info**
- PlaneSet **Planes**

Protected Member Functions

- virtual void **Reset** ()
- virtual void **Free** ()

3.85 ImageBufferNativeAlloc Class Reference

Inherits ImageBufferNative, and IDisposable.

Public Member Functions

- **ImageBufferNativeAlloc** (ImageBufferNativePool< ImageBufferNativeAlloc > pool, ImageBufferInfo info)
- override void Dispose ()

Public Member Functions inherited from ImageBufferNative

- **ImageBufferNative** (ImageBufferInfo info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()

Protected Member Functions

- override void Free ()

Protected Member Functions inherited from ImageBufferNative

- virtual void **Reset** ()

Additional Inherited Members

Public Attributes inherited from ImageBufferNative

- ImageBufferInfo **Info**
- PlaneSet **Planes**

3.85.1 Member Function Documentation

3.85.1.1 Dispose()

```
override void Dispose ()  [virtual]
```

Reimplemented from ImageBufferNative.

3.85.1.2 Free()

```
override void Free ()  [protected], [virtual]
```

Reimplemented from ImageBufferNative.

3.86 ImageBufferNativeGCHandleBytes Class Reference

Inherits ImageBufferNative, and IDisposable.

Public Member Functions

- **ImageBufferNativeGCHandleBytes** (ImageBufferNativePool< ImageBufferNativeGCHandleBytes > pool, ImageBufferInfo info)
- override void Dispose ()

Public Member Functions inherited from ImageBufferNative

- **ImageBufferNative** (ImageBufferInfo info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()

Protected Member Functions

- override void Free ()

Protected Member Functions inherited from ImageBufferNative

- virtual void **Reset** ()

Properties

- byte[][] **PlaneBytes** `[get]`

Additional Inherited Members

Public Attributes inherited from ImageBufferNative

- ImageBufferInfo **Info**
- PlaneSet **Planes**

3.86.1 Member Function Documentation

3.86.1.1 Dispose()

```
override void Dispose ()  [virtual]
```

Reimplemented from ImageBufferNative.

3.86.1.2 Free()

```
override void Free ()  [protected], [virtual]
```

Reimplemented from ImageBufferNative.

3.87 ImageBufferNativeGCHandleSinglePlane Class Reference

Inherits ImageBufferNative, and IDisposable.

Public Member Functions

- **ImageBufferNativeGCHandleSinglePlane** (ImageBufferNativePool< ImageBufferNativeGCHandleSinglePlane > pool, ImageBufferInfo info)
- void **PinPlane** (byte[] plane)
- override void Dispose ()

Public Member Functions inherited from ImageBufferNative

- **ImageBufferNative** (ImageBufferInfo info)
- **ImageBufferNative** (IntPtr buf, int width, int height, int stride, ImageFormat imageFormat)
- void **Retain** ()
- void **Retain** (int times)
- void **Release** ()

Protected Member Functions

- override void Free ()

Protected Member Functions inherited from ImageBufferNative

- virtual void **Reset** ()

Additional Inherited Members

Public Attributes inherited from ImageBufferNative

- ImageBufferInfo **Info**
- PlaneSet **Planes**

3.87.1 Member Function Documentation

3.87.1.1 Dispose()

```
override void Dispose ()  [virtual]
```

Reimplemented from ImageBufferNative.

3.87.1.2 Free()

```
override void Free ()  [protected], [virtual]
```

Reimplemented from ImageBufferNative.

3.88 ImageBufferNativePool< T > Class Template Reference

Inherits ObjectPool< TType, TInfo >.

Public Member Functions

- delegate T **Factory** (ImageBufferNativePool< T > pool, ImageBufferInfo info)
- **ImageBufferNativePool** (int capacity, Factory factory, string name, ImageBufferInfo info)

Public Member Functions inherited from ObjectPool< TType, TInfo >

- ObjectPool (int capacity, string name, TInfo info)

  Create a new ObjectPool instance with the given info structure.
- TType New ()

  Acquire an existing object, or create a new one if none are available.
- TType New (TInfo info)

  Acquire an existing object (if info matches), or create a new one from the passed info.
- virtual bool Free (TType obj, TInfo objInfo)

  Returns object to pool.
- virtual bool Free (TType obj)

  Returns object to pool, or destroys it if the pool is full.
- void **Dispose** ()

  Free resources assoicated with this ObjectPool.

Protected Member Functions

- override T **createObject** (ImageBufferInfo info)
- override void **destroyObject** (T obj)
- override bool **infosMatch** (ImageBufferInfo i0, ImageBufferInfo i1)

Protected Member Functions inherited from ObjectPool< TType, TInfo >

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

Additional Inherited Members

Protected Attributes inherited from ObjectPool< TType, TInfo >

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

Properties inherited from ObjectPool< TType, TInfo >

- TInfo **Info** `[get]`

  The property (info) that objects in this Pool must match.

3.89 IProcessor< T > Interface Template Reference

Processor interface.

Inherits IDisposable.

Inherited by AudioUtil.LevelMeter< T >, AudioUtil.Resampler< T >, AudioUtil.VoiceDetector< T >, AudioUtil.VoiceDetectorCalibrati
AudioUtil.VoiceLevelDetectCalibrate< T >, MicAmplifierFloat, MicAmplifierShort, and WebRTCAudioProcessor.

Public Member Functions

- T[ ] Process (T[ ] buf)

  Process a frame of data.

3.89.1 Detailed Description

Processor interface.

3.89.2 Member Function Documentation

3.89.2.1 Process()

```
T[] Process (
            T[] buf)
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implemented in AudioUtil.LevelMeter< T >, AudioUtil.Resampler< T >, AudioUtil.VoiceDetector< T >, AudioUtil.VoiceDetectorCalibration< T >, and AudioUtil.VoiceLevelDetectCalibrate< T >.

3.90 IResettable Interface Reference

Inherited by AudioInPusher, AudioInReader, and AndroidAudioInAEC.

Public Member Functions

- void **Reset** ()

3.91 IServiceable Interface Reference

Interface for classes that want their Service() function to be called regularly in the context of a LocalVoice.

Inherited by BufferReaderPushAdapterBase< T >.

Public Member Functions

- void Service (LocalVoice localVoice)
  Service function that should be called regularly.

3.91.1 Detailed Description

Interface for classes that want their Service() function to be called regularly in the context of a LocalVoice.

3.91.2 Member Function Documentation

3.91.2.1 Service()

```
void Service (
            LocalVoice localVoice)
```

Service function that should be called regularly.

Implemented in BufferReaderPushAdapterAsyncPool< T >, BufferReaderPushAdapterAsyncPoolFloatToShort, BufferReaderPushAdapterAsyncPoolShortToFloat, and BufferReaderPushAdapterBase< T >.

3.92 IStreamReceiver Interface Reference

Interface for component relying on a LoadBalancingTransport to receive a stream of data (audio, vidoe, ...)

Inherited by AudioStreamReceiver.

Public Member Functions

- void OnRemoteVoiceAdd (int channelId, int playerId, byte voiceId, VoiceInfo i, ref RemoteVoiceOptions options)

  Called when a remote stream (voice) has been detected.

3.92.1 Detailed Description

Interface for component relying on a LoadBalancingTransport to receive a stream of data (audio, vidoe, ...)

3.92.2 Member Function Documentation

3.92.2.1 OnRemoteVoiceAdd()

```
void OnRemoteVoiceAdd (
            int channelId,
            int playerId,
            byte voiceId,
            VoiceInfo i,
            ref RemoteVoiceOptions options)
```

Called when a remote stream (voice) has been detected.

Implemented in AudioStreamReceiver.

3.93 IStreamSender Interface Reference

Interface for component relying on a LoadBalancingTransport to send a stream of data (audio, vidoe, ...)

Inherited by AudioStreamSender.

Public Member Functions

- void OnRoomJoined ()

  Called when the Photon Voice room is joined, and creating streams (voice) is possible.
- void **OnLeftRoom** ()

3.93.1 Detailed Description

Interface for component relying on a LoadBalancingTransport to send a stream of data (audio, vidoe, ...)

3.93.2 Member Function Documentation

3.93.2.1 OnRoomJoined()

```
void OnRoomJoined ()
```

Called when the Photon Voice room is joined, and creating streams (voice) is possible.

Implemented in AudioStreamSender.

3.94 IStreamTransportManager Interface Reference

Interface for components capable of providing a LoadBalancingTransport for an IStreamSender and/or IStreamReceiver Those components should forward the stream callbacks to their registered listeners:

Inherited by StreamTransportClient, and StreamTransportObserver.

Public Member Functions

- LogLevel **StreamLogLevel** ()
- void **RegisterStreamSender** (IStreamSender streamSender)
- void **RegisterStreamReceiver** (IStreamReceiver streamReceiver)

Properties

- StreamTransport **Transport** `[get]`
- long **RoundTripTime** `[get]`
- long **BytesOut** `[get]`
- long **PacketsOut** `[get]`
- long **BytesIn** `[get]`
- long **PacketsIn** `[get]`

3.94.1 Detailed Description

Interface for components capable of providing a LoadBalancingTransport for an IStreamSender and/or IStreamReceiver Those components should forward the stream callbacks to their registered listeners:

- OnRoomJoined / OnLeftRoom to IStreamSender
- OnRemoteVoiceAdd to IStreamReceiver

3.95 IStreamTransportManagerListener Interface Reference

Inherited by StreamTransportManagerListener.

Properties

- GameObject **gameObject** `[get]`
- GameObject **StreamTransportGameObject** `[get, set]`
- IStreamTransportManager **StreamTransportManager** `[get, set]`

3.96 AudioUtil.IVoiceDetector Interface Reference

Voice Activity Detector interface.

Inherited by AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

Properties

- bool On `[get, set]`

  If true, voice detection enabled.
- float Threshold `[get, set]`

  Voice detected as soon as signal level exceeds threshold.
- bool Detected `[get]`

  If true, voice detected.
- DateTime DetectedTime `[get]`

  Last time when switched to detected state.
- int ActivityDelayMs `[get, set]`

  Keep detected state during this time after signal level dropped below threshold.

Events

- Action **OnDetected**

  Called when switched to detected state.

3.96.1 Detailed Description

Voice Activity Detector interface.

3.96.2 Property Documentation

3.96.2.1 ActivityDelayMs

```
int ActivityDelayMs  [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implemented in AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

3.96.2.2 Detected

```
bool Detected  [get]
```

If true, voice detected.

Implemented in AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

3.96.2.3 DetectedTime

```
DateTime DetectedTime  [get]
```

Last time when switched to detected state.

Implemented in AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

3.96.2.4 On

```
bool On  [get], [set]
```

If true, voice detection enabled.

Implemented in AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

3.96.2.5 Threshold

```
float Threshold  [get], [set]
```

Voice detected as soon as signal level exceeds threshold.

Implemented in AudioUtil.VoiceDetector< T >, and AudioUtil.VoiceDetectorDummy.

3.97 IVoiceTransport Interface Reference

Inherited by LoadBalancingTransport.

Public Member Functions

- bool **IsChannelJoined** (int channelId)
- void **SendVoiceInfo** (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void **SendVoiceRemove** (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void **SendFrame** (ArraySegment< byte > data, FrameFlags flags, ushort evNumber, byte frNumber, byte voiceId, int channelId, SendFrameParams par)
- string **ChannelIdStr** (int channelId)
- string **PlayerIdStr** (int playerId)
- int **GetPayloadFragmentSize** (SendFrameParams par)

3.98 AudioUtil.LevelMeter< T > Class Template Reference

Audio Level Meter.

Inherits IProcessor< T >, and AudioUtil.ILevelMeter.

Inherited by AudioUtil.LevelMeterFloat, and AudioUtil.LevelMeterShort.

Public Member Functions

- void ResetAccumAvgPeakAmp ()
  Reset AccumAvgPeakAmp.
- T[ ] Process (T[ ] buf)
  Process a frame of data.
- void **Dispose** ()

Protected Attributes

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[ ] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

Properties

- float CurrentAvgAmp `[get]`
  Average amplitude value over last half second.
- float CurrentPeakAmp `[get, protected set]`
  Maximum amplitude value over last half second sec.
- float AccumAvgPeakAmp `[get]`
  Average of CurrentPeakAmps since last reset.

3.98.1 Detailed Description

Audio Level Meter.

3.98.2 Member Function Documentation

3.98.2.1 Process()

```
T[] Process (
            T[] buf)  [abstract]
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements IProcessor< T >.

3.98.2.2 ResetAccumAvgPeakAmp()

```
void ResetAccumAvgPeakAmp ()
```

Reset AccumAvgPeakAmp.

Implements AudioUtil.ILevelMeter.

3.98.3 Property Documentation

3.98.3.1 AccumAvgPeakAmp

```
float AccumAvgPeakAmp  [get]
```

Average of CurrentPeakAmps since last reset.

Implements AudioUtil.ILevelMeter.

3.98.3.2 CurrentAvgAmp

```
float CurrentAvgAmp  [get]
```

Average amplitude value over last half second.

Implements AudioUtil.ILevelMeter.

3.98.3.3 CurrentPeakAmp

```
float CurrentPeakAmp  [get], [protected set]
```

Maximum amplitude value over last half second sec.

Implements AudioUtil.ILevelMeter.

3.99 AudioUtil.LevelMeterDummy Class Reference

Dummy Audio Level Meter that doesn't actually do anything.

Inherits AudioUtil.ILevelMeter.

Public Member Functions

- void ResetAccumAvgPeakAmp ()
  Reset AccumAvgPeakAmp.

Properties

- float CurrentAvgAmp `[get]`
  Average amplitude value over last half second.
- float CurrentPeakAmp `[get]`
  Maximum amplitude value over last half second sec.
- float AccumAvgPeakAmp `[get]`
  Average of CurrentPeakAmps since last reset.

3.99.1 Detailed Description

Dummy Audio Level Meter that doesn't actually do anything.

3.99.2 Member Function Documentation

3.99.2.1 ResetAccumAvgPeakAmp()

```
void ResetAccumAvgPeakAmp ()
```

Reset AccumAvgPeakAmp.

Implements AudioUtil.ILevelMeter.

3.99.3 Property Documentation

3.99.3.1 AccumAvgPeakAmp

```
float AccumAvgPeakAmp  [get]
```

Average of CurrentPeakAmps since last reset.

Implements AudioUtil.ILevelMeter.

3.99.3.2 CurrentAvgAmp

```
float CurrentAvgAmp  [get]
```

Average amplitude value over last half second.

Implements AudioUtil.ILevelMeter.

3.99.3.3 CurrentPeakAmp

```
float CurrentPeakAmp [get]
```

Maximum amplitude value over last half second sec.

Implements AudioUtil.ILevelMeter.

3.100 AudioUtil.LevelMeterFloat Class Reference

LevelMeter specialization for float audio.

Inherits AudioUtil.LevelMeter< T >.

Public Member Functions

- LevelMeterFloat (int samplingRate, int numChannels)
  Create new LevelMeterFloat instance.
- override float[ ] **Process** (float[ ] buf)

Public Member Functions inherited from AudioUtil.LevelMeter< T >

- void ResetAccumAvgPeakAmp ()
  Reset AccumAvgPeakAmp.
- T[ ] Process (T[ ] buf)
  Process a frame of data.
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from AudioUtil.LevelMeter< T >

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[ ] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

Properties inherited from AudioUtil.LevelMeter< T >

- float CurrentAvgAmp `[get]`
  Average amplitude value over last half second.
- float CurrentPeakAmp `[get, protected set]`
  Maximum amplitude value over last half second sec.
- float AccumAvgPeakAmp `[get]`
  Average of CurrentPeakAmps since last reset.

3.100.1 Detailed Description

LevelMeter specialization for float audio.

3.100.2 Constructor & Destructor Documentation

3.100.2.1 LevelMeterFloat()

```
LevelMeterFloat (
            int samplingRate,
            int numChannels)
```

Create new LevelMeterFloat instance.

Parameters

	samplingRate	Sampling rate of the audio signal (in Hz).
	numChannels	Number of channels in the audio signal.

3.101 AudioUtil.LevelMeterShort Class Reference

LevelMeter specialization for short audio.

Inherits AudioUtil.LevelMeter< T >.

Public Member Functions

- LevelMeterShort (int samplingRate, int numChannels)
  Create new LevelMeterShort instance.
- override short[ ] **Process** (short[ ] buf)

Public Member Functions inherited from AudioUtil.LevelMeter< T >

- void ResetAccumAvgPeakAmp ()
  Reset AccumAvgPeakAmp.
- T[ ] Process (T[ ] buf)
  Process a frame of data.
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from AudioUtil.LevelMeter< T >

- float **ampSum**
- float **ampPeak**
- int **bufferSize**
- float[ ] **prevValues**
- int **prevValuesHead**
- float **accumAvgPeakAmpSum**
- int **accumAvgPeakAmpCount**
- float **currentPeakAmp**
- float **norm**

Properties inherited from AudioUtil.LevelMeter< T >

- float CurrentAvgAmp [get]

  Average amplitude value over last half second.
- float CurrentPeakAmp [get, protected set]

  Maximum amplitude value over last half second sec.
- float AccumAvgPeakAmp [get]

  Average of CurrentPeakAmps since last reset.

3.101.1 Detailed Description

LevelMeter specialization for short audio.

3.101.2 Constructor & Destructor Documentation

3.101.2.1 LevelMeterShort()

```
LevelMeterShort (
            int samplingRate,
            int numChannels)
```

Create new LevelMeterShort instance.

Parameters

	samplingRate	Sampling rate of the audio signal (in Hz).
	numChannels	Number of channels in the audio signal.

3.102 LoadBalancingTransport Class Reference

Extends LoadBalancingClient with media streaming functionality.

Inherits LoadBalancingClient, IVoiceTransport, and IDisposable.

Inherited by LoadBalancingTransport2.

Public Member Functions

- virtual int GetPayloadFragmentSize (SendFrameParams par)
- bool IsChannelJoined (int channelId)
- LoadBalancingTransport (ILogger logger=null, ConnectionProtocol connectionProtocol=Connection↩ Protocol.Udp, bool cppCompatibilityMode=false)

  Initializes a new LoadBalancingTransport.
- new void **Service** ()

  This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2 to 20 times a second).
- virtual bool **ChangeAudioGroups** (byte[ ] groupsToRemove, byte[ ] groupsToAdd)
- void SendVoiceInfo (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void SendVoiceRemove (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void SendFrame (ArraySegment< byte > data, FrameFlags flags, ushort evNumber, byte frNumber, byte voiceId, int channelId, SendFrameParams par)
- string ChannelIdStr (int channelId)
- string PlayerIdStr (int playerId)
- void **Dispose** ()

  Releases all resources used by the LoadBalancingTransport instance.

Protected Member Functions

- virtual object **buildFrameMessage** (byte voiceId, ushort evNumber, byte frNumber, ArraySegment< byte > data, FrameFlags flags)
- virtual void **onEventActionVoiceClient** (EventData ev)

Protected Attributes

- VoiceClient **voiceClient**
- readonly bool **cppCompatibilityMode**
- readonly ILogger **logger**

Properties

- VoiceClient **VoiceClient** `[get]`

  The VoiceClient implementation associated with this LoadBalancingTransport.
- virtual byte **FrameCode** `[get]`

3.102.1 Detailed Description

Extends LoadBalancingClient with media streaming functionality.

Use your normal LoadBalancing workflow to join a Voice room. All standard LoadBalancing features are available. Use VoiceClient to work with media streams.

3.102.2 Constructor & Destructor Documentation

3.102.2.1 LoadBalancingTransport()

```
LoadBalancingTransport (
            ILogger logger = null,
            ConnectionProtocol connectionProtocol = ConnectionProtocol::Udp,
            bool cppCompatibilityMode = false)
```

Initializes a new LoadBalancingTransport.

Parameters

	logger	ILogger instance. If null, this instance LoadBalancingClient.DebugReturn implementation is used.Conne
	connectionProtocol	Connection protocol (UDP or TCP). Conne
	cppCompatibilityMode	Use a protocol compatible with V

3.102.3 Member Function Documentation

3.102.3.1 ChannelIdStr()

```
string ChannelIdStr (
            int channelId)
```

Implements IVoiceTransport.

3.102.3.2 GetPayloadFragmentSize()

```
virtual int GetPayloadFragmentSize (
            SendFrameParams par)  [virtual]
```

Implements IVoiceTransport.

3.102.3.3 IsChannelJoined()

```
bool IsChannelJoined (
            int channelId)
```

Implements IVoiceTransport.

3.102.3.4 PlayerIdStr()

```
string PlayerIdStr (
            int playerId)
```

Implements IVoiceTransport.

3.102.3.5 SendFrame()

```
void SendFrame (
            ArraySegment< byte > data,
            FrameFlags flags,
            ushort evNumber,
            byte frNumber,
            byte voiceId,
            int channelId,
            SendFrameParams par)
```

Implements IVoiceTransport.

3.102.3.6 SendVoiceInfo()

```
void SendVoiceInfo (
            LocalVoice voice,
            int channelId,
            bool targetMe,
            int[] targetPlayers)
```

Implements IVoiceTransport.

3.102.3.7 SendVoiceRemove()

```
void SendVoiceRemove (
            LocalVoice voice,
            int channelId,
            bool targetMe,
            int[] targetPlayers)
```

Implements IVoiceTransport.

3.103 LoadBalancingTransport2 Class Reference

Variant of LoadBalancingTransport. Aims to be non-alloc at the cost of breaking compatibility with older clients.

Inherits LoadBalancingTransport.

Public Member Functions

- **LoadBalancingTransport2** (ILogger logger=null, ConnectionProtocol connectionProtocol=Connection↩ Protocol.Udp, bool cppCompatibilityMode=false)
- override int GetPayloadFragmentSize (SendFrameParams par)

Public Member Functions inherited from LoadBalancingTransport

- bool IsChannelJoined (int channelId)
- LoadBalancingTransport (ILogger logger=null, ConnectionProtocol connectionProtocol=Connection↩ Protocol.Udp, bool cppCompatibilityMode=false)

  Initializes a new LoadBalancingTransport.
- new void **Service** ()

  This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2 to 20 times a second).
- virtual bool **ChangeAudioGroups** (byte[ ] groupsToRemove, byte[ ] groupsToAdd)
- void SendVoiceInfo (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void SendVoiceRemove (LocalVoice voice, int channelId, bool targetMe, int[ ] targetPlayers)
- void SendFrame (ArraySegment< byte > data, FrameFlags flags, ushort evNumber, byte frNumber, byte voiceId, int channelId, SendFrameParams par)
- string ChannelIdStr (int channelId)
- string PlayerIdStr (int playerId)
- void **Dispose** ()

  Releases all resources used by the LoadBalancingTransport instance.

Protected Member Functions

- override object buildFrameMessage (byte voiceId, ushort evNumber, byte frNumber, ArraySegment< byte > data, FrameFlags flags)
- override void onEventActionVoiceClient (EventData ev)

Properties

- override byte **FrameCode** `[get]`

Properties inherited from LoadBalancingTransport

- VoiceClient **VoiceClient** `[get]`

  The VoiceClient implementation associated with this LoadBalancingTransport.
- virtual byte **FrameCode** `[get]`

Additional Inherited Members

Protected Attributes inherited from LoadBalancingTransport

- VoiceClient **voiceClient**
- readonly bool **cppCompatibilityMode**
- readonly ILogger **logger**

3.103.1 Detailed Description

Variant of LoadBalancingTransport. Aims to be non-alloc at the cost of breaking compatibility with older clients.

3.103.2 Member Function Documentation

3.103.2.1 buildFrameMessage()

```
override object buildFrameMessage (
            byte voiceId,
            ushort evNumber,
            byte frNumber,
            ArraySegment< byte > data,
            FrameFlags flags)  [protected], [virtual]
```

Reimplemented from LoadBalancingTransport.

3.103.2.2 GetPayloadFragmentSize()

```
override int GetPayloadFragmentSize (
            SendFrameParams par)  [virtual]
```

Reimplemented from LoadBalancingTransport.

3.103.2.3 onEventActionVoiceClient()

```
override void onEventActionVoiceClient (
            EventData ev)  [protected], [virtual]
```

Reimplemented from LoadBalancingTransport.

3.104 LocalVoice Class Reference

Represents outgoing data stream.

Inherits IDisposable.

Inherited by LocalVoiceAudioDummy, and LocalVoiceFramed< T >.

Public Member Functions

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.
- virtual void Dispose ()

Static Public Attributes

- const int **DATA_POOL_CAPACITY** = 50

Protected Member Functions

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Protected Attributes

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

Properties

- VoiceInfo **Info** `[get]`

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** `[get, set]`

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if stream broadcasts.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentFragmented** `[get]`

  Sent fragmented frames counter.
- int **FramesSentFragments** `[get]`

  Sent frames fragments counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- bool **Reliable** `[get, set]`

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** `[get, set]`

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** `[get, set]`

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.

- int **FEC** `[get, set]`

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** `[get, set]`

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** `[get, set]`
- byte InterestGroup `[get, set]`

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[] TargetPlayers `[get, set]`

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** `[get]`
- int **SendSpacingProfileMax** `[get]`

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** `[get]`
- int **EventBufferSize** `[get]`
- string **shortName** `[get]`
- string **Name** `[get]`
- string **LogPrefix** `[get]`
- bool **isJoined** `[get]`

3.104.1 Detailed Description

Represents outgoing data stream.

3.104.2 Member Function Documentation

3.104.2.1 Dispose()

```
virtual void Dispose ()  [virtual]
```

Reimplemented in LocalVoiceFramed< T >.

3.104.3 Property Documentation

3.104.3.1 DebugEchoMode

```
bool DebugEchoMode  [get], [set]
```

If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.

This functionality availability depends on transport.

3.104.3.2 InterestGroup

```
byte InterestGroup  [get], [set]
```

If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.

A remote voice is created even if the remote player is not subscribed to the InterestGroup. Use VoiceCreateOptions.TargetPlayers and TargetPlayers to completely hide the existence of LocalVoice from the remote player./>

3.104.3.3 TargetPlayers

```
int [] TargetPlayers  [get], [set]
```

If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.

TargetPlayers update triggers the sending of voice info to added players and voice remove to removed. Depending on the transport, TargetPlayers may disregard InterestGroup: the remote player whos number is in TargetPlayers, receives the stream even if not subscribed to the interest group. If the local player number is in TargetPlayers, it works like DebugEchoMode. Using both DebugEchoMode and TargetPlayers at the same time to route the stream back to the sender leads to an inconsistent state after one of the options is switched off: the voice is removed but the frames are still delivered.

3.105 LocalVoiceAudio< T > Class Template Reference

Outgoing audio stream.

Inherits LocalVoiceFramed< T >, and ILocalVoiceAudio.

Inherited by LocalVoiceAudioFloat, and LocalVoiceAudioShort.

Public Member Functions

- void VoiceDetectorCalibrate (int durationMs, Action< float > onCalibrated=null)

  Trigger voice detector calibration process.

Public Member Functions inherited from LocalVoiceFramed< T >

- void AddPostProcessor (params IProcessor< T >[] processors)

  Adds processors after any built-in processors and everything added with AddPreProcessor.
- void AddPreProcessor (params IProcessor< T >[] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void RemoveProcessor (params IProcessor< T >[] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void **ClearProcessors** ()

  Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.
- void **PushDataAsync** (T[] buf)

  Asynchronously push data into this stream.
- void **PushData** (T[] buf)

  Synchronously push data into this stream.
- override void Dispose ()

  Releases resources used by the LocalVoiceFramed<T> instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.

Protected Member Functions

- void **initBuiltinProcessors** ()

Protected Member Functions inherited from LocalVoiceFramed< T >

- T[ ] **processFrame** (T[ ] buf, int p0, int p1)

Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Protected Attributes

- AudioUtil.VoiceDetector< T > **voiceDetector**
- AudioUtil.VoiceDetectorCalibration< T > **voiceDetectorCalibration**
- AudioUtil.LevelMeter< T > **levelMeter**
- int **channels**

Protected Attributes inherited from LocalVoice

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

Properties

- virtual AudioUtil.IVoiceDetector VoiceDetector `[get]`

  The VoiceDetector in use.
- virtual AudioUtil.ILevelMeter LevelMeter `[get]`

  The LevelMeter utility in use.
- bool VoiceDetectorCalibrating `[get]`

  True if the VoiceDetector is currently calibrating.

Properties inherited from LocalVoiceFramed< T >

- int **OptimalSourceFrameSize** `[get]`
- ObjectFactory< T[], int > **BufferFactory** `[get]`

  <see cref="PushData(T[])" and />.
- bool **PushDataAsyncReady** `[get]`

  Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.

Properties inherited from LocalVoice

- VoiceInfo **Info** `[get]`

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** `[get, set]`

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if stream broadcasts.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentFragmented** `[get]`

  Sent fragmented frames counter.
- int **FramesSentFragments** `[get]`

  Sent frames fragments counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- bool **Reliable** `[get, set]`

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** `[get, set]`

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** `[get, set]`

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.
- int **FEC** `[get, set]`

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** `[get, set]`

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** `[get, set]`
- byte InterestGroup `[get, set]`

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[] TargetPlayers `[get, set]`

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** `[get]`
- int **SendSpacingProfileMax** `[get]`

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** `[get]`
- int **EventBufferSize** `[get]`
- string **shortName** `[get]`
- string **Name** `[get]`
- string **LogPrefix** `[get]`
- bool **isJoined** `[get]`

Additional Inherited Members

Static Public Attributes inherited from LocalVoice

- const int **DATA_POOL_CAPACITY** = 50

3.105.1 Detailed Description

Outgoing audio stream.

3.105.2 Member Function Documentation

3.105.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
            int durationMs,
            Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. Voice detector sets threshold basing on measured backgroud noise level.

Parameters

	durationMs	Duration of calibration in milliseconds.
	onCalibrated	Called when calibration is complete. Parameter is new threshold value.

Implements ILocalVoiceAudio.

3.105.3 Property Documentation

3.105.3.1 LevelMeter

```
virtual AudioUtil.ILevelMeter LevelMeter  [get]
```

The LevelMeter utility in use.

Implements ILocalVoiceAudio.

3.105.3.2 VoiceDetector

```
virtual AudioUtil.IVoiceDetector VoiceDetector  [get]
```

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implements ILocalVoiceAudio.

3.105.3.3 VoiceDetectorCalibrating

```
bool VoiceDetectorCalibrating [get]
```

True if the VoiceDetector is currently calibrating.

Implements ILocalVoiceAudio.

3.106 LocalVoiceAudioDummy Class Reference

Dummy LocalVoiceAudio.

Inherits LocalVoice, and ILocalVoiceAudio.

Public Member Functions

- void VoiceDetectorCalibrate (int durationMs, Action< float > onCalibrated=null)

  Trigger voice detector calibration process.

Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.
- virtual void Dispose ()

Static Public Attributes

- static LocalVoiceAudioDummy **Dummy** = new LocalVoiceAudioDummy()

  A Dummy LocalVoiceAudio instance.

Static Public Attributes inherited from LocalVoice

- const int **DATA_POOL_CAPACITY** = 50

Properties

- AudioUtil.IVoiceDetector VoiceDetector [get]

  The VoiceDetector in use.
- AudioUtil.ILevelMeter LevelMeter [get]

  The LevelMeter utility in use.
- bool VoiceDetectorCalibrating [get]

  If true, voice detector calibration is in progress.

Properties inherited from LocalVoice

- VoiceInfo **Info** `[get]`

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** `[get, set]`

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if stream broadcasts.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentFragmented** `[get]`

  Sent fragmented frames counter.
- int **FramesSentFragments** `[get]`

  Sent frames fragments counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- bool **Reliable** `[get, set]`

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** `[get, set]`

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** `[get, set]`

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.
- int **FEC** `[get, set]`

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** `[get, set]`

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** `[get, set]`
- byte InterestGroup `[get, set]`

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[ ] TargetPlayers `[get, set]`

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** `[get]`
- int **SendSpacingProfileMax** `[get]`

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** `[get]`
- int **EventBufferSize** `[get]`
- string **shortName** `[get]`
- string **Name** `[get]`
- string **LogPrefix** `[get]`
- bool **isJoined** `[get]`

Additional Inherited Members

Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Protected Attributes inherited from LocalVoice

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

3.106.1 Detailed Description

Dummy LocalVoiceAudio.

For testing, this LocalVoiceAudio implementation features a AudioUtil.VoiceDetectorDummy and a AudioUtil.LevelMeterDummy

3.106.2 Member Function Documentation

3.106.2.1 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
            int durationMs,
            Action< float > onCalibrated = null)
```

Trigger voice detector calibration process.

While calibrating, keep silence. Voice detector sets threshold based on measured backgroud noise level.

Parameters

	durationMs	Duration of calibration (in milliseconds).
	onCalibrated	Called when calibration is complete. Parameter is new threshold value.

Implements ILocalVoiceAudio.

3.106.3 Property Documentation

3.106.3.1 LevelMeter

```
AudioUtil.ILevelMeter LevelMeter  [get]
```

The LevelMeter utility in use.

Implements ILocalVoiceAudio.

3.106.3.2 VoiceDetector

```
AudioUtil.IVoiceDetector VoiceDetector  [get]
```

The VoiceDetector in use.

Use it to enable or disable voice detector and set its parameters.

Implements ILocalVoiceAudio.

3.106.3.3 VoiceDetectorCalibrating

```
bool VoiceDetectorCalibrating  [get]
```

If true, voice detector calibration is in progress.

Implements ILocalVoiceAudio.

3.107 LocalVoiceAudioFloat Class Reference

Specialization of LocalVoiceAudio<T> for float audio.

Inherits LocalVoiceAudio< T >.

Additional Inherited Members

Public Member Functions inherited from LocalVoiceAudio< T >

- void VoiceDetectorCalibrate (int durationMs, Action< float > onCalibrated=null)

  Trigger voice detector calibration process.

Public Member Functions inherited from LocalVoiceFramed< T >

- void AddPostProcessor (params IProcessor< T >[ ] processors)

  Adds processors after any built-in processors and everything added with AddPreProcessor.
- void AddPreProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void RemoveProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void **ClearProcessors** ()

  Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.
- void **PushDataAsync** (T[ ] buf)

  Asynchronously push data into this stream.
- void **PushData** (T[ ] buf)

  Synchronously push data into this stream.
- override void Dispose ()

  Releases resources used by the LocalVoiceFramed<T> instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.

Static Public Attributes inherited from LocalVoice

- const int **DATA_POOL_CAPACITY** = 50

Protected Member Functions inherited from LocalVoiceAudio< T >

- void **initBuiltinProcessors** ()

Protected Member Functions inherited from LocalVoiceFramed< T >

- T[ ] **processFrame** (T[ ] buf, int p0, int p1)

Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Protected Attributes inherited from LocalVoiceAudio< T >

- AudioUtil.VoiceDetector< T > **voiceDetector**
- AudioUtil.VoiceDetectorCalibration< T > **voiceDetectorCalibration**
- AudioUtil.LevelMeter< T > **levelMeter**
- int **channels**

Protected Attributes inherited from LocalVoice

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

Properties inherited from LocalVoiceAudio< T >

- virtual AudioUtil.IVoiceDetector VoiceDetector `[get]`

  The VoiceDetector in use.
- virtual AudioUtil.ILevelMeter LevelMeter `[get]`

  The LevelMeter utility in use.
- bool VoiceDetectorCalibrating `[get]`

  True if the VoiceDetector is currently calibrating.

Properties inherited from LocalVoiceFramed< T >

- int **OptimalSourceFrameSize** [get]
- ObjectFactory< T[], int > **BufferFactory** [get]

  <see cref="PushData(T[])" and />.
- bool **PushDataAsyncReady** [get]

  Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.

Properties inherited from LocalVoice

- VoiceInfo **Info** [get]

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** [get, set]

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** [get]

  Returns true if stream broadcasts.
- int **FramesSent** [get]

  Sent frames counter.
- int **FramesSentFragmented** [get]

  Sent fragmented frames counter.
- int **FramesSentFragments** [get]

  Sent frames fragments counter.
- int **FramesSentBytes** [get]

  Sent frames bytes counter.
- bool **Reliable** [get, set]

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** [get, set]

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** [get, set]

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.
- int **FEC** [get, set]

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** [get, set]

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** [get, set]
- byte InterestGroup [get, set]

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode [get, set]

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[] TargetPlayers [get, set]

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** [get]
- int **SendSpacingProfileMax** [get]

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** [get]
- int **EventBufferSize** [get]
- string **shortName** [get]
- string **Name** [get]
- string **LogPrefix** [get]
- bool **isJoined** [get]

3.107.1 Detailed Description

Specialization of LocalVoiceAudio<T> for float audio.

3.108 LocalVoiceAudioShort Class Reference

Specialization of LocalVoiceAudio<T> for short audio.

Inherits LocalVoiceAudio< T >.

Additional Inherited Members

Public Member Functions inherited from LocalVoiceAudio< T >

- void VoiceDetectorCalibrate (int durationMs, Action< float > onCalibrated=null)

  Trigger voice detector calibration process.

Public Member Functions inherited from LocalVoiceFramed< T >

- void AddPostProcessor (params IProcessor< T >[ ] processors)

  Adds processors after any built-in processors and everything added with AddPreProcessor.
- void AddPreProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void RemoveProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void **ClearProcessors** ()

  Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.
- void **PushDataAsync** (T[ ] buf)

  Asynchronously push data into this stream.
- void **PushData** (T[ ] buf)

  Synchronously push data into this stream.
- override void Dispose ()

  Releases resources used by the LocalVoiceFramed<T> instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.

Static Public Attributes inherited from LocalVoice

- const int **DATA_POOL_CAPACITY** = 50

Protected Member Functions inherited from LocalVoiceAudio< T >

- void **initBuiltinProcessors** ()

Protected Member Functions inherited from LocalVoiceFramed< T >

- T[ ] **processFrame** (T[ ] buf, int p0, int p1)

Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Protected Attributes inherited from LocalVoiceAudio< T >

- AudioUtil.VoiceDetector< T > **voiceDetector**
- AudioUtil.VoiceDetectorCalibration< T > **voiceDetectorCalibration**
- AudioUtil.LevelMeter< T > **levelMeter**
- int **channels**

Protected Attributes inherited from LocalVoice

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

Properties inherited from LocalVoiceAudio< T >

- virtual AudioUtil.IVoiceDetector VoiceDetector `[get]`

  The VoiceDetector in use.
- virtual AudioUtil.ILevelMeter LevelMeter `[get]`

  The LevelMeter utility in use.
- bool VoiceDetectorCalibrating `[get]`

  True if the VoiceDetector is currently calibrating.

Properties inherited from LocalVoiceFramed< T >

- int **OptimalSourceFrameSize** `[get]`
- ObjectFactory< T[ ], int > **BufferFactory** `[get]`

  <see cref="PushData(T[ ])" and />.
- bool **PushDataAsyncReady** `[get]`

  Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.

Properties inherited from LocalVoice

- VoiceInfo **Info** `[get]`

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** `[get, set]`

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if stream broadcasts.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentFragmented** `[get]`

  Sent fragmented frames counter.
- int **FramesSentFragments** `[get]`

  Sent frames fragments counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- bool **Reliable** `[get, set]`

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** `[get, set]`

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** `[get, set]`

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.
- int **FEC** `[get, set]`

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** `[get, set]`

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** `[get, set]`
- byte InterestGroup `[get, set]`

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[ ] TargetPlayers `[get, set]`

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** `[get]`
- int **SendSpacingProfileMax** `[get]`

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** `[get]`
- int **EventBufferSize** `[get]`
- string **shortName** `[get]`
- string **Name** `[get]`
- string **LogPrefix** `[get]`
- bool **isJoined** `[get]`

3.108.1 Detailed Description

Specialization of LocalVoiceAudio<T> for short audio.

3.109 LocalVoiceFramed< T > Class Template Reference

Typed re-framing LocalVoice.

Inherits LocalVoice.

Inherited by LocalVoiceAudio< T >.

Public Member Functions

- void AddPostProcessor (params IProcessor< T >[ ] processors)

  Adds processors after any built-in processors and everything added with AddPreProcessor.
- void AddPreProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void RemoveProcessor (params IProcessor< T >[ ] processors)

  Adds processors before built-in processors and everything added with AddPostProcessor.
- void **ClearProcessors** ()

  Clears all processors in pipeline including built-in resampling. User should add at least resampler processor after call.
- void **PushDataAsync** (T[ ] buf)

  Asynchronously push data into this stream.
- void **PushData** (T[ ] buf)

  Synchronously push data into this stream.
- override void Dispose ()

  Releases resources used by the LocalVoiceFramed< T > instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Public Member Functions inherited from LocalVoice

- void **SendSpacingProfileStart** ()
- void **RemoveSelf** ()

  Remove this voice from it's VoiceClient (using VoiceClient.RemoveLocalVoice.

Protected Member Functions

- T[ ] **processFrame** (T[ ] buf, int p0, int p1)

Protected Member Functions inherited from LocalVoice

- bool **targetExits** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceInfoAndConfigFrame** (bool targetMe, int[ ] targetPlayers)
- void **sendVoiceRemove** (bool targetMe, int[ ] targetPlayers)

Properties

- int **OptimalSourceFrameSize** `[get]`
- ObjectFactory< T[ ], int > **BufferFactory** `[get]`

  <see cref="PushData(T[])" and />.
- bool **PushDataAsyncReady** `[get]`

  Wether this LocalVoiceFramed has capacity for more data buffers to be pushed asynchronously.

Properties inherited from LocalVoice

- VoiceInfo **Info** `[get]`

  Returns Info structure assigned on local voice cration.
- bool **TransmitEnabled** `[get, set]`

  If true, stream data broadcasted.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if stream broadcasts.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentFragmented** `[get]`

  Sent fragmented frames counter.
- int **FramesSentFragments** `[get]`

  Sent frames fragments counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- bool **Reliable** `[get, set]`

  Send data reliable. See also VoiceCreateOptions.Reliable.
- bool **Encrypt** `[get, set]`

  Send data encrypted. See also VoiceCreateOptions.Encrypt.
- bool **Fragment** `[get, set]`

  Split frames into fragments according to the size provided by the Transport. See also VoiceCreateOptions.Fragment.
- int **FEC** `[get, set]`

  Forward Error Correction control. See also VoiceCreateOptions.FEC.
- IServiceable **LocalUserServiceable** `[get, set]`

  Optional user object attached to LocalVoice. its Service() will be called at each VoiceClient.Service() call.
- byte **Group** `[get, set]`
- byte InterestGroup `[get, set]`

  If InterestGroup != 0, streaming only to the players subscribed to this group (if supported by the transport). See also VoiceCreateOptions.InterestGroup.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams. See also VoiceCreateOptions.DebugEchoMode.
- int[ ] TargetPlayers `[get, set]`

  If TargetPlayers is not null, sending voice info and streaming only to clients having player numbers specified in the array (if supported by transport). See also VoiceCreateOptions.TargetPlayers.
- string **SendSpacingProfileDump** `[get]`
- int **SendSpacingProfileMax** `[get]`

  Logs input frames time spacing profiling results. Do not call frequently.
- byte **ID** `[get]`
- int **EventBufferSize** `[get]`
- string **shortName** `[get]`
- string **Name** `[get]`
- string **LogPrefix** `[get]`
- bool **isJoined** `[get]`

Additional Inherited Members

Static Public Attributes inherited from LocalVoice

- const int **DATA_POOL_CAPACITY** = 50

Protected Attributes inherited from LocalVoice

- int[ ] **targetPlayers_**
- VoiceInfo **info**
- IEncoder **encoder**
- VoiceClient **voiceClient**
- bool **threadingEnabled**
- ArraySegment< byte > **configFrame**
- volatile bool **disposed**
- object **disposeLock** = new object()

3.109.1 Detailed Description

Typed re-framing LocalVoice.

Consumes data in array buffers of arbitrary length. Repacks them in frames of VoiceInfo.FrameSize length for further processing and encoding.

3.109.2 Member Function Documentation

3.109.2.1 AddPostProcessor()

```
void AddPostProcessor (
            params IProcessor< T >[] processors)
```

Adds processors after any built-in processors and everything added with AddPreProcessor.

Parameters

	processors

3.109.2.2 AddPreProcessor()

```
void AddPreProcessor (
            params IProcessor< T >[] processors)
```

Adds processors before built-in processors and everything added with AddPostProcessor.

Parameters

	processors

3.109.2.3 Dispose()

```
override void Dispose ()  [virtual]
```

Releases resources used by the LocalVoiceFramed<T> instance. Buffers used for asynchronous push will be disposed in encoder thread's 'finally'.

Reimplemented from LocalVoice.

3.109.2.4 RemoveProcessor()

```
void RemoveProcessor (
            params IProcessor< T >[] processors)
```

Adds processors before built-in processors and everything added with AddPostProcessor.

Parameters

	processors

3.110 Logger Class Reference

Inherits ILogger.

Public Member Functions

- **Logger** (LogLevel level=LogLevel.Debug)
- void Log (LogLevel level, string msg)
- void Log (LogLevel level, string fmt, params object[] args)

Properties

- LogLevel Level `[get, set]`

3.110.1 Member Function Documentation

3.110.1.1 Log() `[1/2]`

```
void Log (
            LogLevel level,
            string fmt,
            params object[] args)
```

Implements ILogger.

3.110.1.2 Log() `[2/2]`

```
void Log (
            LogLevel level,
            string msg)
```

Implements ILogger.

3.110.2 Property Documentation

3.110.2.1 Level

```
LogLevel Level  [get], [set]
```

Implements ILogger.

3.111 MicAmplifier Class Reference

Inherits VoiceComponent.

Properties

- float **AmplificationFactor** `[get, set]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

Additional Inherited Members

Protected Member Functions inherited from VoiceComponent

- virtual void **Awake** ()

3.112 MicAmplifierFloat Class Reference

Inherits IProcessor< T >.

Public Member Functions

- **MicAmplifierFloat** (float amplificationFactor)
- float[ ] **Process** (float[ ] buf)
- void **Dispose** ()

Public Member Functions inherited from IProcessor< T >

- T[ ] Process (T[ ] buf)

  Process a frame of data.

Properties

- float **AmplificationFactor** `[get, set]`
- bool **Disabled** `[get, set]`

3.113 MicAmplifierShort Class Reference

Inherits IProcessor< T >.

Public Member Functions

- **MicAmplifierShort** (float amplificationFactor)
- short[ ] **Process** (short[ ] buf)
- void **Dispose** ()

Public Member Functions inherited from IProcessor< T >

- T[ ] Process (T[ ] buf)

  Process a frame of data.

Properties

- float **AmplificationFactor** `[get, set]`
- bool **Disabled** `[get, set]`

3.114 MicrophonePermission Class Reference

Helper to request Microphone permission on Android or iOS.

Inherits VoiceComponent.

Public Member Functions

- void **InitVoice** ()

Protected Member Functions

- override void Awake ()

Properties

- bool **HasPermission** `[get]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

Events

- static Action< bool > **MicrophonePermissionCallback**

3.114.1 Detailed Description

Helper to request Microphone permission on Android or iOS.

3.114.2 Member Function Documentation

3.114.2.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.115 MicWrapper Class Reference

Inherits IAudioReader< T >.

Public Member Functions

- **MicWrapper** (string device, int suggestedFrequency, ILogger logger)
- void **Dispose** ()
- bool **Read** (float[ ] buffer)

Public Member Functions inherited from IDataReader< T >

- bool Read (T[ ] buffer)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Properties

- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- int Channels `[get]`

  Number of channels in the audio signal.
- string Error `[get]`

  If not null, audio object is in invalid state.

3.115.1 Property Documentation

3.115.1.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.115.1.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.115.1.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.116 MicWrapperPusher Class Reference

Inherits IAudioPusher< T >.

Public Member Functions

- **MicWrapperPusher** (GameObject parent, string device, int suggestedFrequency, ILogger logger)
- void **SetCallback** (Action< float[ ]> callback, ObjectFactory< float[ ], int > bufferFactory, int optimalFrame↩ Size)
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[ ]> callback, ObjectFactory< T[ ], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- int Channels [get]

  Number of channels in the audio signal.
- string Error [get]

  If not null, audio object is in invalid state.

3.116.1 Property Documentation

3.116.1.1 Channels

```
int Channels  [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.116.1.2 Error

```
string Error  [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.116.1.3 SamplingRate

```
int SamplingRate  [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.117 MicWrapperPusherOnAudioFilterRead Class Reference

Inherits MonoBehaviour.

Events

- Action< float[ ], int > **OnAudioFrame**

3.118 MonoPInvokeCallbackAttribute Class Reference

Inherits Attribute.

Public Member Functions

- **MonoPInvokeCallbackAttribute** (Type t)

3.119 ObjectFactory< TType, TInfo > Interface Template Reference

Object factory with optional info useful in combination with ObjectPool<TType, TInfo>.

Inherits IDisposable.

Inherited by ArrayPoolSet< T >, ArraySegmentPool< T >, and ObjectPool< TType, TInfo >.

Public Member Functions

- TType New ()
- TType New (TInfo info)
- bool Free (TType obj)
- bool Free (TType obj, TInfo info)

3.119.1 Detailed Description

Object factory with optional info useful in combination with ObjectPool<TType, TInfo>.

Template Parameters

	TType	Object type.
	TInfo	Type of property used to check 2 objects identity (like integral length of array).

3.119.2 Member Function Documentation

3.119.2.1 Free() [1/2]

```
bool Free (
            TType obj)
```

Implemented in ObjectPool< TType, TInfo >.

3.119.2.2 Free() [2/2]

```
bool Free (
            TType obj,
            TInfo info)
```

Implemented in ObjectPool< TType, TInfo >.

3.119.2.3 New() [1/2]

```
TType New ()
```

Implemented in ObjectPool< TType, TInfo >.

3.119.2.4 New() [2/2]

```
TType New (
            TInfo info)
```

Implemented in ObjectPool< TType, TInfo >.

3.120 ObjectPool< TType, TInfo > Class Template Reference

Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).

Inherits ObjectFactory< TType, TInfo >.

Inherited by ArrayPool< T >, and ImageBufferNativePool< T >.

Public Member Functions

- ObjectPool (int capacity, string name, TInfo info)

  Create a new ObjectPool instance with the given info structure.
- TType New ()

  Acquire an existing object, or create a new one if none are available.
- TType New (TInfo info)

  Acquire an existing object (if info matches), or create a new one from the passed info.
- virtual bool Free (TType obj, TInfo objInfo)

  Returns object to pool.
- virtual bool Free (TType obj)

  Returns object to pool, or destroys it if the pool is full.
- void **Dispose** ()

  Free resources assoicated with this ObjectPool.

Protected Member Functions

- TType **createObject** (TInfo info)
- void **destroyObject** (TType obj)
- bool **infosMatch** (TInfo i0, TInfo i1)

Protected Attributes

- int **capacity**
- TInfo **info**
- int **pos**
- string **name**

Properties

- TInfo **Info** [get]

  The property (info) that objects in this Pool must match.

3.120.1 Detailed Description

Generic Pool to re-use objects of a certain type (TType) that optionally match a certain property or set of properties (TInfo).

Template Parameters

TType		Object type.
TInfo	Type of parameter used to check 2 objects identity (like integral length of array).	

3.120.2 Member Function Documentation

3.120.2.1 Free() [1/2]

```
virtual bool Free (
            TType obj)  [virtual]
```

Returns object to pool, or destroys it if the pool is full.

Parameters

obj	The object to return to the pool.	

Implements ObjectFactory< TType, TInfo >.

3.120.2.2 Free() [2/2]

```
virtual bool Free (
            TType obj,
            TInfo objInfo)  [virtual]
```

Returns object to pool.

Parameters

obj	The object to return to the pool.	
objInfo	The info structure about obj.	

obj is returned to the pool only if objInfo matches this pool's info. Else, it is destroyed.

Implements ObjectFactory< TType, TInfo >.

3.120.2.3 New() [1/2]

```
TType New ()
```

Acquire an existing object, or create a new one if none are available.

If it fails to get one from the pool, this will create from the info given in this pool's constructor.

Implements ObjectFactory< TType, TInfo >.

3.120.2.4 New() [2/2]

```
TType New (
            TInfo info)
```

Acquire an existing object (if info matches), or create a new one from the passed info.

Parameters

	info	Info structure to match, or create a new object with.

Implements ObjectFactory< TType, TInfo >.

3.120.2.5 ObjectPool()

```
ObjectPool (
            int capacity,
            string name,
            TInfo info)
```

Create a new ObjectPool instance with the given info structure.

Parameters

	capacity	Capacity (size) of the object pool.
	name	Name of the object pool.
	info	Info about this Pool's objects.

3.121 OpusCodec Class Reference

Classes

- class Decoder< T >
- class Encoder< T >
- class EncoderFloat
- class EncoderShort
- class Factory
- class Util

Public Types

- enum **FrameDuration**

Properties

- static string **Version** `[get]`

3.122 PermissionsRequester.PermissionRequest Class Reference

Launch a permission request.

Public Member Functions

- **PermissionRequest** (string permissionName, float delayBeforeRequest=DefaultDelayBeforeRequest)
- async void **CheckPermission** ()
- void **UpdatePermission** ()

Public Attributes

- string **permissionName** = ""
- bool **hasBeenChecked** = false

  Tells if the permission has received an answer. Set by callbacks on Android, and by the HasPermission setter on other platforms.
- bool **isRequesting** = false
- float **delayBeforeRequest** = DefaultDelayBeforeRequest
- float **delayBeforeCallback** = 0
- UnityEvent< bool > **permissionCallback** = new UnityEvent<bool>()
- UnityEvent **permissionGrantedCallback** = new UnityEvent()
- UnityEvent **permissionCheckStartedCallback** = new UnityEvent()
- bool **isUnityApplicationAuthorization** = false

Static Public Attributes

- const float **DefaultDelayBeforeRequest** = 0.05f

Protected Member Functions

- void **AndroidCheckPermission** ()
- IEnumerator UnityCheckAuthorization ()

  UserAuthorization based authorization, for iOS and webGL.

Properties

- bool **HasPermission** `[get]`

  Modify hasPermission, and when setting values, ensure to set hasBeenChecked/isRequesting, to consider the request as handled.

3.122.1 Detailed Description

Launch a permission request.

3.122.2 Member Function Documentation

3.122.2.1 UnityCheckAuthorization()

```
IEnumerator UnityCheckAuthorization ()  [protected]
```

UserAuthorization based authorization, for iOS and webGL.

Returns

3.123 PermissionsRequester Class Reference

Launch successively permission requests (Android only for now)

Inherits MonoBehaviour.

Classes

- class PermissionRequest

  Launch a permission request.

Public Member Functions

- string **PermissionNameForUserAuthoriation** (UserAuthorization userAuthoriation)
- async Task **LaunchPermissionRequests** ()
- bool **TryFindConfiguredPermissionRequest** (string permission, out PermissionRequest requester)
- void **AddMicrophoneRequest** (UnityAction< bool > permissionCallback=null, UnityAction permission↩ CheckStartedCallback=null)
- void **AddWebcamRequest** (UnityAction< bool > permissionCallback=null, UnityAction permissionCheck↩ StartedCallback=null)
- void **AddAuthorizationRequest** (UserAuthorization authorization, UnityAction< bool > permission↩ Callback=null, UnityAction permissionCheckStartedCallback=null)

  Queue a permission request for authorization UserAuthorization (for iOS or webGL) Will be converted to a string (WebcamAuthorizationPermissionName or MicrophoneAuthorizationPermissionName), and passed to Add↩ PermissionRequest.
- async void **AddPermissionRequest** (string permission, UnityAction< bool > permissionCallback=null, UnityAction permissionCheckStartedCallback=null)

  Queue a permission request for permission string (can be either an Android request, or, for iOS and webGL WebcamAuthorizationPermissionName or MicrophoneAuthorizationPermissionName values)

Static Public Member Functions

- static bool **TryConvertToUserAuthorization** (string permissionName, out UserAuthorization user↩ Authoriation)

  Convert a string permission to a UserAuthorization (using MicrophoneAuthorizationPermissionName and Webcam↩ AuthorizationPermissionName values)
- static bool **HasMicrophonePermission** ()
- static bool **HasWebcamPermission** ()
- static async Task **Delay** (int milliseconds)

Public Attributes

- List< PermissionRequest > **permissionRequests** = new List<PermissionRequest>()
- UnityEvent **allPermissionCallback** = new UnityEvent()
- List< GameObject > **objectsToActivateAfterPermissionChecks** = new List<GameObject>()
- bool **requestOnStart** = true
- bool **isLaunchingRequests** = false

Static Public Attributes

- const int **EXECUTION_ORDER** = -20_000
- static PermissionsRequester **_sharedInstance**
- const string **MicrophoneAuthorizationPermissionName** = MicrophoneAndroidPermissionName
- const string **WebcamAuthorizationPermissionName** = WebcamAndroidPermissionName
- const string **MicrophoneAndroidPermissionName** = "android.permission.RECORD_AUDIO"
- const string **WebcamAndroidPermissionName** = "android.permission.CAMERA"

Properties

- static PermissionsRequester **SharedInstance** `[get]`

3.123.1 Detailed Description

Launch successively permission requests (Android only for now)

3.124 PermissionWaiter Class Reference

Wait for a permission to be granted.

Inherits MonoBehaviour.

Inherited by VoiceConnectionPermissionWaiter.

Public Member Functions

- virtual void OnPermissionGranted ()

  Should be called manually (for instance in a PermissionsRequester callback) if checkPermisisonGrantAutomatically is not checked Otherwise, automatically called when relevant in Update.

Public Attributes

- bool **checkPermisisonGrantAutomatically** = true
- UnityEvent< bool > **onPermissionChanged** = new UnityEvent<bool>()
- UnityEvent **onPermissionGranted** = new UnityEvent()

Static Public Attributes

- const int **EXECUTION_ORDER** = PermissionsRequester.EXECUTION_ORDER + 1

Protected Member Functions

- virtual void **Awake** ()
- virtual void OnPermissionRequired ()

  Called when the permission is required.
- virtual void **OnPermissionChanged** (bool isGranted)
- virtual void **Update** ()

Protected Attributes

- bool **waitingForPermission** = false

Properties

- bool **IsWaitingForPermission** `[get]`
- virtual string **PermissionName** `[get]`
- bool **IsPermissionGranted** `[get]`

3.124.1 Detailed Description

Wait for a permission to be granted.

Should be used alongside a PermissionsRequester, to align permissions on the desired timing (to avoid conflicting requests) Can create one if missing

Subclass should have a [DefaultExecutionOrder(PermissionWaiter.EXECUTION_ORDER)] (PermissionWaiter.↩ EXECUTION_ORDER or any higher value - to wait for the Permissionsrequester Awake, so that the singleton is detected) tag to be sure to trigger early, if needed

3.124.2 Member Function Documentation

3.124.2.1 OnPermissionGranted()

```
virtual void OnPermissionGranted ()  [virtual]
```

Should be called manually (for instance in a PermissionsRequester callback) if checkPermisisonGrantAutomatically is not checked Otherwise, automatically called when relevant in Update.

Reimplemented in VoiceConnectionPermissionWaiter.

3.124.2.2 OnPermissionRequired()

```
virtual void OnPermissionRequired ()  [protected], [virtual]
```

Called when the permission is required.

Reimplemented in VoiceConnectionPermissionWaiter.

3.125 PhotonAppSettings Class Reference

Collection of connection-relevant settings, used internally by PhotonNetwork.ConnectUsingSettings.

Inherits ScriptableObject.

Public Member Functions

- void **UseCloud** (string cloudAppid, string code="")

  Sets appid and region code in the AppSettings. Used in Editor.
- override string **ToString** ()

  String summary of the AppSettings.

Static Public Member Functions

- static void **LoadOrCreateSettings** ()

Public Attributes

- AppSettings **AppSettings**

Properties

- static PhotonAppSettings **Instance** `[get]`

3.125.1 Detailed Description

Collection of connection-relevant settings, used internally by PhotonNetwork.ConnectUsingSettings.

Includes the AppSettings class from the Realtime APIs plus some other, PUN-relevant, settings.

3.126 PhotonVoiceCreatedParams Class Reference

Properties

- Voice.LocalVoice **Voice** `[get, set]`
- Voice.IAudioDesc **AudioDesc** `[get, set]`

3.127 PhotonVoiceLagSimulationGui Class Reference

Inherits MonoBehaviour.

Public Member Functions

- void **Start** ()

3.128 PhotonVoiceStatsGui Class Reference

Basic GUI to show traffic and health statistics of the connection to Photon, toggled by shift+tab.

Inherits MonoBehaviour.

3.128.1 Detailed Description

Basic GUI to show traffic and health statistics of the connection to Photon, toggled by shift+tab.

The shown health values can help identify problems with connection losses or performance. Example: If the time delta between two consecutive SendOutgoingCommands calls is a second or more, chances rise for a disconnect being caused by this (because acknowledgments to the server need to be sent in due time).

3.129 PhotonVoiceView Class Reference

Component that should be attached to a networked PUN prefab that has PhotonView. It will bind remote Recorder with local Speaker of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.

Inherits VoiceComponent.

Protected Member Functions

- override void Awake ()

Properties

- Recorder **RecorderInUse** `[get]`

  The Recorder component currently used by this PhotonVoiceView.
- Speaker **SpeakerInUse** `[get]`

  The Speaker component currently used by this PhotonVoiceView.
- bool **IsSpeaking** `[get]`

  If true, this PhotonVoiceView has a Speaker that is currently playing received audio frames from remote audio source.
- bool **IsRecording** `[get]`

  If true, this PhotonVoiceView has a Recorder that is currently transmitting audio stream from local audio source.

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.129.1 Detailed Description

Component that should be attached to a networked PUN prefab that has PhotonView. It will bind remote Recorder with local Speaker of the same networked prefab. This component makes automatic voice stream routing easy for players' characters/avatars.

3.129.2 Member Function Documentation

3.129.2.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.130 ImageBufferNative.PlaneSet Struct Reference

Public Member Functions

- **PlaneSet** (int length, IntPtr p0=default(IntPtr), IntPtr p1=default(IntPtr), IntPtr p2=default(IntPtr), IntPtr p3=default(IntPtr))

Properties

- IntPtr **this[int key]** `[get, set]`
- int **Length** `[get]`

3.131 Platform Class Reference

Static Public Member Functions

- static IDeviceEnumerator **CreateAudioInEnumerator** (ILogger logger)
- static IAudioInChangeNotifier **CreateAudioInChangeNotifier** (Action callback, ILogger logger)
- static IEncoder **CreateDefaultAudioEncoder**< **T** > (ILogger logger, VoiceInfo info)
- static IAudioDesc **CreateDefaultAudioSource** (ILogger logger, DeviceInfo dev, int samplingRate, int channels, object otherParams=null)
- static IAudioOut< float > **CreateUnityAudioOut** (UnityEngine.AudioSource audioSource, AudioOutDelay↩Control.PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- static IDeviceEnumerator **CreateVideoInEnumerator** (ILogger logger)
- static IEncoderDirectImage **CreateDefaultVideoEncoder** (ILogger logger, VoiceInfo info)
- static IDecoderDirect< ImageBufferNative > **CreateDefaultVideoDecoder** (ILogger logger, VoiceInfo info)
- static IVideoRecorder **CreateDefaultVideoRecorder** (ILogger logger, VoiceInfo info, DeviceInfo camDevice, Action< IVideoRecorder > onReady)
- static IVideoPlayer **CreateDefaultVideoPlayer** (ILogger logger, VoiceInfo info, Action< IVideoPlayer > on↩Ready)
- static IPreviewManager **CreateDefaultPreviewManager** (ILogger logger)
- static IVideoRecorder **CreateVideoRecorderUnityTexture** (ILogger logger, VoiceInfo info, DeviceInfo cam↩Device, Action< IVideoRecorder > onReady)
- static IVideoPlayer **CreateVideoPlayerUnityTexture** (ILogger logger, VoiceInfo info, Action< IVideoPlayer > onReady)
- static IPreviewManager **CreatePreviewManagerUnityTexture** (ILogger logger)

3.132 AudioOutDelayControl.PlayDelayConfig Struct Reference

Public Attributes

- int **Low**
- int **High**
- int **Max**
- int **SpeedUpPerc**

Static Public Attributes

- static PlayDelayConfig Default

Properties

- int **Delay** `[get, set]`

3.132.1 Member Data Documentation

3.132.1.1 Default

```
PlayDelayConfig Default  [static]
```

Initial value:

```
= new PlayDelayConfig()
            {
                Low = 200,
                High = 200,
                Max = 1000,
                SpeedUpPerc = 5,


            }
```

3.133 PunVoiceClient Class Reference

This class can be used to automatically sync client states between PUN and Voice. It also finds the Speaker component for a character's voice. For this to work attach a PhotonVoiceView next to the PhotonView of your player's prefab.

Inherits VoiceFollowClient.

Static Public Attributes

- const string **VoiceRoomNameSuffix** = "_voice_"

  Suffix for voice room names appended to Leader room names.

Static Public Attributes inherited from VoiceConnection

- const int **ChannelAudio** = 1

  Recommended Photon Transport channel for audio. Chosen not to interfere with video and default channel.
- const int **ChannelVideo** = 2

  Recommended Photon Transport channel for video. Chosen not to interfere with audio and default channel.

Protected Member Functions

- override void Start ()
- override void OnDestroy ()
- override Speaker InstantiateSpeakerForRemoteVoice (int playerId, byte voiceId, object userData)
- override string **GetVoiceRoomName** ()
- override bool **ConnectVoice** ()

Protected Member Functions inherited from VoiceFollowClient

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- override void OnDestroy ()
- override void OnOperationResponseReceived (OperationResponse operationResponse)
- void **LeaderStateChanged** (ClientState toState)
- override void OnVoiceStateChanged (ClientState fromState, ClientState toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

Protected Member Functions inherited from VoiceConnection

- override void **Awake** ()
- virtual void **Update** ()
- void **CalcStatistics** ()

Properties

- override bool **LeaderInRoom** `[get]`
- override bool **LeaderOfflineMode** `[get]`
- static PunVoiceClient **Instance** `[get]`

  Singleton instance for PunVoiceClient.
- bool **UsePunAppSettings** `[get, set]`

  Whether or not to use the Voice AppId and all the other AppSettings from PUN's PhotonServerSettings Scriptable↩ Object singleton in the Voice client/app.
- bool **UsePunAuthValues** `[get, set]`

  Whether or not to use the same PhotonNetwork.AuthValues in PunVoiceClient.Instance.Client.AuthValues. This means that the same UserID will be used in both clients. If custom authentication is used and setup in PUN app, the same configuration should be done for the Voice app.

Properties inherited from VoiceFollowClient

- bool **LeaderInRoom** `[get]`
- bool **LeaderOfflineMode** `[get]`

Properties inherited from VoiceConnection

- virtual bool **AlwaysUsePrimaryRecorder** `[get]`
- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- new LoadBalancingTransport **Client** `[get]`
- VoiceClient **VoiceClient** `[get]`

  Returns underlying Photon Voice client.
- ClientState **ClientState** `[get]`

  Returns Photon Voice client state.
- float **FramesReceivedPerSecond** `[get]`

  Number of frames received per second.
- float **FramesLostPerSecond** `[get]`

  Number of frames lost per second.
- float **FramesLostPercent** `[get]`

  Percentage of lost frames.
- GameObject **SpeakerPrefab** `[get, set]`

  Prefab that contains Speaker component to be instantiated when receiving a new remote audio source info.
- Recorder **PrimaryRecorder** `[get, set]`

  Primary Recorder to be used by VoiceConnection implementations directly or via integration objects.
- bool **UsePrimaryRecorder** `[get]`

  Use VoiceConnection.PrimaryRecorder directly.
- string **BestRegionSummaryInPreferences** `[get, set]`

  Used to store and access the "Best Region Summary" in the Player Preferences.

Additional Inherited Members

Public Member Functions inherited from VoiceFollowClient

- bool ConnectAndJoinRoom ()

  Connect voice client to Photon servers and join a Voice room.
- void **Disconnect** ()

  Disconnect voice client from all Photon servers.

Public Member Functions inherited from VoiceConnection

- virtual bool ConnectUsingSettings (AppSettings overwriteSettings=null)

  Connect to Photon server using Settings
- bool AddSpeaker (Speaker speaker, object userData)

  Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.
- Speaker InstantiateSpeakerPrefab (GameObject parent, bool destroyOnRemove)

  Instantiates SpeakerPrefab, optionally attaches it to the provided parent.
- bool **AddRecorder** (Recorder rec)
- void **RemoveRecorder** (Recorder rec)

Public Attributes inherited from VoiceFollowClient

- bool **AutoConnectAndJoin** = true

  Auto connect voice client and join a voice room when Leader client is joined to a Leader room.

Public Attributes inherited from VoiceConnection

- AppSettings **Settings**

  Settings to be used by this Voice Client.

Events inherited from VoiceConnection

- Action< Speaker > **SpeakerLinked**

  Fires when a speaker has been linked to a remote audio stream.
- Action< RemoteVoiceLink > **RemoteVoiceAdded**

  Fires when a remote voice stream is added.

3.133.1 Detailed Description

This class can be used to automatically sync client states between PUN and Voice. It also finds the Speaker component for a character's voice. For this to work attach a PhotonVoiceView next to the PhotonView of your player's prefab.

3.133.2 Member Function Documentation

3.133.2.1 InstantiateSpeakerForRemoteVoice()

```
override Speaker InstantiateSpeakerForRemoteVoice (
            int playerId,
            byte voiceId,
            object userData)  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.133.2.2 OnDestroy()

```
override void OnDestroy ()  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.133.2.3 Start()

```
override void Start ()  [protected], [virtual]
```

Reimplemented from VoiceFollowClient.

3.134 RawCodec Class Reference

Classes

- class Decoder< T >
- class Encoder< T >
- class ShortToFloat

3.135 Recorder Class Reference

Component representing outgoing audio stream in scene.

Inherits VoiceComponent.

Public Types

- enum **InputSourceType**
- enum **MicType**

Public Member Functions

- bool **RestartRecording** ()

  Restarts recording if Recorder.RecordingEnabled is true.
- void VoiceDetectorCalibrate (int durationMs, Action< float > detectionEndedCallback=null)

  Trigger voice detector calibration process. While calibrating, keep silence. Voice detector sets threshold basing on measured background noise level.
- bool SetIosAudioSessionParameters (IOS.AudioSessionParameters asp)

  Sets the AudioSessionParameters for iOS audio initialization when Photon MicrophoneType is used.
- bool SetIosAudioSessionParameters (IOS.AudioSessionCategory category, IOS.AudioSessionMode mode, IOS.AudioSessionCategoryOption[ ] options)

  Sets the AudioSessionParameters for iOS audio initialization when Photon MicrophoneType is used.
- bool SetAndroidNativeMicrophoneSettings (bool aec=false, bool agc=false, bool ns=false)

  Sets the native Android audio input settings when the Photon microphone type is used.
- bool ResetLocalAudio ()

  Resets audio session and parameters locally to fix broken recording due to system configuration modifications or audio interruptions or audio routing changes.

Static Public Attributes

- const int **MIN_OPUS_BITRATE** = 6000
- const int **MAX_OPUS_BITRATE** = 510000

Protected Member Functions

- virtual void **SendPhotonVoiceCreatedMessage** ()
- void **Update** ()

Protected Member Functions inherited from VoiceComponent

- virtual void **Awake** ()

Properties

- bool **TransmitEnabled** `[get, set]`

  If true, audio transmission is enabled.
- bool Encrypt `[get, set]`

  If true, voice stream is sent encrypted.
- bool DebugEchoMode `[get, set]`

  If true, outgoing stream routed back to client via server same way as for remote client's streams.
- bool ReliableMode `[get, set]`

  If true, stream data sent in reliable mode.
- bool **VoiceDetection** `[get, set]`

  If true, voice detection enabled.
- float **VoiceDetectionThreshold** `[get, set]`

  Voice detection threshold (0..1, where 1 is full amplitude).
- int **VoiceDetectionDelayMs** `[get, set]`

  Keep detected state during this time after signal level dropped below threshold. Default is 500ms.
- object **UserData** `[get, set]`

  Custom user object to be sent in the voice stream info event.
- Func< IAudioDesc > **InputFactory** `[get, set]`

  Set the method returning new Voice.IAudioDesc instance to be assigned to a new voice created with Source set to Factory.
- AudioUtil.IVoiceDetector **VoiceDetector** `[get]`

  Returns voice activity detector for recorder's audio stream.
- byte InterestGroup `[get, set]`

  Target interest group that will receive transmitted audio.
- int[ ] TargetPlayers `[get, set]`

  Target players which will receive transmitted audio.
- bool **IsCurrentlyTransmitting** `[get]`

  Returns true if audio stream broadcasts.
- AudioUtil.ILevelMeter **LevelMeter** `[get]`

  Level meter utility.
- bool **VoiceDetectorCalibrating** `[get]`

  If true, voice detector calibration is in progress.
- ILocalVoiceAudio **voiceAudio** `[get]`
- InputSourceType **SourceType** `[get, set]`

  Audio data source.
- MicType **MicrophoneType** `[get, set]`

  Which microphone API to use when the Source is set to Microphone.
- AudioClip **AudioClip** `[get, set]`

  Source audio clip.
- bool **LoopAudioClip** `[get, set]`

  Loop playback for audio clip sources.
- SamplingRate **SamplingRate** `[get, set]`

  Outgoing audio stream sampling rate.
- OpusCodec.FrameDuration **FrameDuration** `[get, set]`

  Outgoing audio stream encoder delay.
- int **Bitrate** `[get, set]`

  Outgoing audio stream bitrate.
- bool **RecordingEnabled** `[get, set]`

  Gets or sets whether this Recorder is recording audio to be transmitted.
- bool **StopRecordingWhenPaused** `[get, set]`

If true, stop recording when paused resume/restart when un-paused.

- bool UseOnAudioFilterRead `[get, set]`

  If true, recording will make use of Unity's OnAudioFitlerRead callback from a muted local AudioSource.
- bool **UseMicrophoneTypeFallback** `[get, set]`

  If true, if recording fails to start with Unity microphone type, Photon microphone type is used -if available- as a fallback and vice versa.
- bool **RecordWhenJoined** `[get, set]`

  If true, recording starts when joining the room and stops when leaving the room.
- DeviceInfo **MicrophoneDevice** `[get, set]`
- bool **AndroidMicrophoneAGC** `[get]`
- bool **AndroidMicrophoneAEC** `[get]`
- bool **AndroidMicrophoneNS** `[get]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.135.1 Detailed Description

Component representing outgoing audio stream in scene.

3.135.2 Member Function Documentation

3.135.2.1 ResetLocalAudio()

```
bool ResetLocalAudio ()
```

Resets audio session and parameters locally to fix broken recording due to system configuration modifications or audio interruptions or audio routing changes.

Returns

If reset is done.

3.135.2.2 SetAndroidNativeMicrophoneSettings()

```
bool SetAndroidNativeMicrophoneSettings (
            bool aec = false,
            bool agc = false,
            bool ns = false)
```

Sets the native Android audio input settings when the Photon microphone type is used.

Parameters

	aec	Acoustic Echo Cancellation
	agc	Automatic Gain Control
	ns	Noise Suppression

Returns

If a change has been made.

3.135.2.3 SetIosAudioSessionParameters() [1/2]

```
bool SetIosAudioSessionParameters (
            IOS.AudioSessionCategory category,
            IOS.AudioSessionMode mode,
            IOS.AudioSessionCategoryOption[] options)
```

Sets the AudioSessionParameters for iOS audio initialization when Photon MicrophoneType is used.

Parameters

	category	Audio session category to be used.
	mode	Audio session mode to be used.
	options	Audio session category options to be used

Returns

If a change has been made.

3.135.2.4 SetIosAudioSessionParameters() [2/2]

```
bool SetIosAudioSessionParameters (
            IOS.AudioSessionParameters asp)
```

Sets the AudioSessionParameters for iOS audio initialization when Photon MicrophoneType is used.

Parameters

	asp	You can use custom value or one from presets, IOS.AudioSessionParametersPresets

Returns

If a change has been made.

3.135.2.5 VoiceDetectorCalibrate()

```
void VoiceDetectorCalibrate (
            int durationMs,
            Action< float > detectionEndedCallback = null)
```

Trigger voice detector calibration process. While calibrating, keep silence. Voice detector sets threshold basing on measured background noise level.

Parameters

	durationMs	Duration of calibration in milliseconds.
	detectionEndedCallback	Callback when VAD calibration ends.

3.135.3 Property Documentation

3.135.3.1 DebugEchoMode

```
bool DebugEchoMode  [get], [set]
```

If true, outgoing stream routed back to client via server same way as for remote client's streams.

Initialized with serialized field and can be updated by Editor at runtime.

3.135.3.2 Encrypt

```
bool Encrypt  [get], [set]
```

If true, voice stream is sent encrypted.

Initialized with serialized field and can be updated by Editor at runtime.

3.135.3.3 InterestGroup

```
byte InterestGroup  [get], [set]
```

Target interest group that will receive transmitted audio.

If InterestGroup != 0, recorder's audio data is sent only to clients listening to this group. Initialized with serialized field and can be updated by Editor at runtime.

3.135.3.4 ReliableMode

```
bool ReliableMode  [get], [set]
```

If true, stream data sent in reliable mode.

Initialized with serialized field and can be updated by Editor at runtime.

3.135.3.5 TargetPlayers

```
int [] TargetPlayers  [get], [set]
```

Target players which will receive transmitted audio.

Initialized with serialized field and can be updated by Editor at runtime

3.135.3.6 UseOnAudioFilterRead

```
bool UseOnAudioFilterRead  [get], [set]
```

If true, recording will make use of Unity's OnAudioFitlerRead callback from a muted local AudioSource.

If enabled, 3D sounds and voice positioning can be lost.

3.136 RecorderPreset Class Reference

Inherits VoiceComponent.

Classes

- struct DSP

Public Attributes

- RuntimePlatform **Platform**
- MicType **MicrophoneType**
- bool **DSPEnabled**
- DSP **DSPSettings**

Protected Member Functions

- override void Awake ()

Additional Inherited Members

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.136.1 Member Function Documentation

3.136.1.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.137 RemoteVoiceInfo Class Reference

Information about a remote voice (incoming stream).

Properties

- VoiceInfo **Info** `[get]`

  Remote voice info.
- int **ChannelId** `[get]`

  ID of channel used for transmission.
- int **PlayerId** `[get]`

  Player ID of voice owner.
- byte **VoiceId** `[get]`

  Voice ID (unique in the room).

3.137.1 Detailed Description

Information about a remote voice (incoming stream).

3.138 RemoteVoiceLink Class Reference

Public Member Functions

- **RemoteVoiceLink** (VoiceInfo info, int playerId, byte voiceId, int channelId, ref RemoteVoiceOptions options)
- override string **ToString** ()

Public Attributes

- readonly VoiceInfo **VoiceInfo**
- readonly int **PlayerId**
- readonly byte **VoiceId**
- readonly int **ChannelId**

Events

- Action< FrameOut< float > > **FloatFrameDecoded**
- Action **RemoteVoiceRemoved**

3.139 RemoteVoiceOptions Struct Reference

Event Actions and other options for a remote voice (incoming stream).

Public Member Functions

- **RemoteVoiceOptions** (ILogger logger, string logPrefix, VoiceInfo voiceInfo)
- void **SetOutput** (Action< FrameOut< float > > output)

  Create default audio decoder and register a method to be called when a data frame is decoded.
- void **SetOutput** (Action< FrameOut< short > > output)

  Create default audio decoder and register a method to be called when a data frame is decoded.

Properties

- Action **OnRemoteVoiceRemoveAction** `[get, set]`

  Register a method to be called when the remote voice is removed.
- IDecoder **Decoder** `[get, set]`

  Remote voice data decoder. Use to set decoder options or override it with user decoder.

3.139.1 Detailed Description

Event Actions and other options for a remote voice (incoming stream).

3.140 AudioUtil.Resampler< T > Class Template Reference

Sample-rate conversion Audio Processor.

Inherits IProcessor< T >.

Public Member Functions

- Resampler (int dstSize, int channels)

  Create a new Resampler instance.
- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Protected Attributes

- T[ ] **frameResampled**

3.140.1 Detailed Description

Sample-rate conversion Audio Processor.

This processor converts the sample-rate of the source stream. Internally, it uses AudioUtil.Resample<T>(T[ ], T[ ], int, int).

3.140.2 Member Function Documentation

3.140.2.1 Process()

```
T[] Process (
            T[] buf)
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements IProcessor< T >.

3.140.2.2 Resampler()

```
Resampler (
            int dstSize,
            int channels)
```

Create a new Resampler instance.

Parameters

	dstSize	Frame size of a destination frame. Determins output rate.
	channels	Number of audio channels expected in both in- and output.

3.141 SaveIncomingStreamToFile Class Reference

Inherits VoiceComponent.

Protected Member Functions

- override void Awake ()

Additional Inherited Members

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.141.1 Member Function Documentation

3.141.1.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.142 SaveOutgoingStreamToFile Class Reference

Inherits VoiceComponent.

Additional Inherited Members

Protected Member Functions inherited from VoiceComponent

- virtual void **Awake** ()

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.143 SendFrameParams Struct Reference

Public Member Functions

- **SendFrameParams** (bool targetMe, int[ ] targetPlayers, byte interestGroup, bool reliable, bool encrypt)

Properties

- bool **TargetMe** `[get]`
- int[ ] **TargetPlayers** `[get]`
- byte **InterestGroup** `[get]`
- bool **Reliable** `[get]`
- bool **Encrypt** `[get]`

3.144 RawCodec.ShortToFloat Class Reference

Public Member Functions

- **ShortToFloat** (Action< FrameOut< float > > output)
- void **Output** (FrameOut< short > shortBuf)

3.145 Speaker Class Reference

Inherits VoiceComponent.

Inherited by SpeakerFMOD, and SpeakerAudioFilterRead.

Public Member Functions

- void RestartPlayback ()

  Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.

Protected Member Functions

- override void Awake ()
- virtual IAudioOut< float > **CreateAudioOut** ()
- virtual void **OnDestroy** ()
- void **Update** ()

Protected Attributes

- IAudioOut< float > **audioOutput**
- AudioOutDelayControl.PlayDelayConfig **playDelayConfig** = AudioOutDelayControl.PlayDelayConfig.Default
- bool **restartOnDeviceChange** = true

Properties

- bool **IsPlaying** `[get]`

  Is the speaker playing right now.
- int **Lag** `[get]`

  The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.
- Action< Speaker > **OnRemoteVoiceRemoveAction** `[get, set]`

  Register a method to be called when remote voice removed.
- RemoteVoiceLink **RemoteVoice** `[get]`
- bool **IsLinked** `[get]`

  Whether or not this Speaker has been linked to a remote voice stream.
- AudioOutDelayControl.PlayDelayConfig PlayDelayConfig `[get, set]`

  Gets or sets jitter buffer config.
- int PlayDelay `[get, set]`

  Gets or sets jitter buffer size in ms.
- bool **RestartOnDeviceChange** `[get, set]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.145.1 Member Function Documentation

3.145.1.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.145.1.2 RestartPlayback()

```
void RestartPlayback ()
```

Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.

Returns

True if playback is successfully restarted.

3.145.2 Property Documentation

3.145.2.1 PlayDelay

```
int PlayDelay  [get], [set]
```

Gets or sets jitter buffer size in ms.

The method updates PlayDelayConfig with reasonable values based on the single value provided. Use PlayDelayConfig for more precise control.

3.145.2.2 PlayDelayConfig

```
AudioOutDelayControl.PlayDelayConfig PlayDelayConfig  [get], [set]
```

Gets or sets jitter buffer config.

Make sure that the new value is fully initialized or built from AudioOutDelayControl.PlayDelayConfig.Default.

3.146 SpeakerAudioFilterRead Class Reference

Inherits Speaker.

Protected Member Functions

- override IAudioOut< float > CreateAudioOut ()

Protected Member Functions inherited from Speaker

- override void Awake ()
- virtual void **OnDestroy** ()
- void **Update** ()

Additional Inherited Members

Public Member Functions inherited from Speaker

- void RestartPlayback ()

  Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.

Protected Attributes inherited from Speaker

- IAudioOut< float > **audioOutput**
- AudioOutDelayControl.PlayDelayConfig **playDelayConfig** = AudioOutDelayControl.PlayDelayConfig.Default
- bool **restartOnDeviceChange** = true

Properties inherited from Speaker

- bool **IsPlaying** `[get]`

  Is the speaker playing right now.
- int **Lag** `[get]`

  The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.
- Action< Speaker > **OnRemoteVoiceRemoveAction** `[get, set]`

  Register a method to be called when remote voice removed.
- RemoteVoiceLink **RemoteVoice** `[get]`
- bool **IsLinked** `[get]`

  Whether or not this Speaker has been linked to a remote voice stream.
- AudioOutDelayControl.PlayDelayConfig PlayDelayConfig `[get, set]`

  Gets or sets jitter buffer config.
- int PlayDelay `[get, set]`

  Gets or sets jitter buffer size in ms.
- bool **RestartOnDeviceChange** `[get, set]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.146.1 Member Function Documentation

3.146.1.1 CreateAudioOut()

```
override IAudioOut< float > CreateAudioOut ()  [protected], [virtual]
```

Reimplemented from Speaker.

3.147 SpeakerFMOD Class Reference

Inherits Speaker.

Protected Member Functions

- override IAudioOut< float > CreateAudioOut ()

Protected Member Functions inherited from Speaker

- override void Awake ()
- virtual void **OnDestroy** ()
- void **Update** ()

Additional Inherited Members

Public Member Functions inherited from Speaker

- void RestartPlayback ()

  Restarts the audio playback of the linked incoming remote audio stream via AudioSource component.

Protected Attributes inherited from Speaker

- IAudioOut< float > **audioOutput**
- AudioOutDelayControl.PlayDelayConfig **playDelayConfig** = AudioOutDelayControl.PlayDelayConfig.Default
- bool **restartOnDeviceChange** = true

Properties inherited from Speaker

- bool **IsPlaying** `[get]`

  Is the speaker playing right now.
- int **Lag** `[get]`

  The current difference between positions in the buffer of (jittery) stream writer and (clock-driven) audio output reader in ms.
- Action< Speaker > **OnRemoteVoiceRemoveAction** `[get, set]`

  Register a method to be called when remote voice removed.
- RemoteVoiceLink **RemoteVoice** `[get]`
- bool **IsLinked** `[get]`

  Whether or not this Speaker has been linked to a remote voice stream.
- AudioOutDelayControl.PlayDelayConfig PlayDelayConfig `[get, set]`

  Gets or sets jitter buffer config.
- int PlayDelay `[get, set]`

  Gets or sets jitter buffer size in ms.
- bool **RestartOnDeviceChange** `[get, set]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.147.1 Member Function Documentation

3.147.1.1 CreateAudioOut()

```
override IAudioOut< float > CreateAudioOut ()  [protected], [virtual]
```

Reimplemented from Speaker.

3.148 StreamMatchmakingSettings Class Reference

Public Member Functions

- virtual PhotonJoinRandomRoomArgs **RandomRoomParams** ()
- virtual PhotonEnterRoomArgs **EnterRoomParams** ()

Public Attributes

- int **maxPlayers** = 20
- List< StringSessionProperty > **matchmakingSessionProperties** = new List<StringSessionProperty>()

3.149 StreamTransportClient Class Reference

Basic implementation of a IRegisterableStreamTransportManager, initiating a LoadBalancingTransport and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using FusionVoice↩ Client, UnityVoiceClient, ... aka the main VoiceConnection subclasses (that manage recorders, offer a deeper app life cycle management, ...)

Inherits MonoBehaviour, IStreamTransportManager, IConnectionCallbacks, and IMatchmakingCallbacks.

Public Member Functions

- virtual void **Disconnect** ()
- virtual void **StartConnectionWithDefaultSettings** ()
- virtual void **StartConnection** (StreamTransportClientSettings startVideoClientSettings)
- virtual void **StartConnection** (AppSettings appSettings)
- virtual AppSettings FindAppSettings ()

  Find the AppSettings used to connect. Behaviour:
- virtual AppSettings FindAppSettings (StreamTransportClientSettings startVideoClientSettings)

  Find the AppSettings used to connect. Behaviour:
- LogLevel StreamLogLevel ()
- virtual void RegisterStreamSender (IStreamSender streamSender)
- virtual void RegisterStreamReceiver (IStreamReceiver streamReceiver)
- virtual void **JoinRoom** ()

  Join the room once connected to the server. The default implementation relies on random matchmaking, using EnterRoomParams parameters.
- virtual void **OnConnectedToMaster** ()
- virtual void **OnDisconnected** (DisconnectCause cause)
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)

Public Attributes

- bool **autoStartWithAppSettings** = true
- bool **UseSharedAppSettings** = false
- StreamTransportClientSettings **transportClientSettings** = new StreamTransportClientSettings()
- StreamMatchmakingSettings **matchmakingSettings** = new StreamMatchmakingSettings()
- LogLevel **logLevel** = LogLevel.Info
- UnityEvent **onStartConnection** = new UnityEvent()
- bool **isRoomJoined** = false
- string **joinedRoomName** = ""
- List< StringSessionProperty > **roomCustomPropertiesPreview** = new List<StringSessionProperty>()

Protected Member Functions

- virtual void **Awake** ()
- virtual void **Start** ()
- virtual void **Update** ()
- virtual void **OnApplicationQuit** ()
- virtual void **OnDestroy** ()
- virtual void **Service** ()
- virtual void **NotifiyJoinedRoom** ()
- virtual void **NotifyLeftRoom** ()
- virtual void **NotifyRemoteVoiceAdd** (int channelId, int playerId, byte voiceId, VoiceInfo i, ref RemoteVoiceOptions options)
- virtual void **CheckJoinStatus** ()
- virtual void **OnRemoteVoiceAdd** (int channelId, int playerId, byte voiceId, VoiceInfo i, ref RemoteVoiceOptions options)

Protected Attributes

- Photon.Voice.Unity.Logger **_logger**
- StreamTransport **_client** = new StreamTransport()
- List< IStreamSender > **_registeredStreamSenders** = new List<IStreamSender>()
- List< IStreamReceiver > **_registeredStreamReceivers** = new List<IStreamReceiver>()

Properties

- ConnectionProtocol **Protocol** `[get]`
- ConnectionProtocol **UsedProtocol** `[get]`
- virtual AuthenticationValues **AuthValues** `[get]`
- StreamTransport Transport `[get]`

Properties inherited from IStreamTransportManager

- long **RoundTripTime** `[get]`
- long **BytesOut** `[get]`
- long **PacketsOut** `[get]`
- long **BytesIn** `[get]`
- long **PacketsIn** `[get]`

3.149.1 Detailed Description

Basic implementation of a IRegisterableStreamTransportManager, initiating a LoadBalancingTransport and forwarding its callbacks to the sender/receiver listeners Here for demo/pedagogical purposes: prefer using FusionVoice↩ Client, UnityVoiceClient, ... aka the main VoiceConnection subclasses (that manage recorders, offer a deeper app life cycle management, ...)

3.149.2 Member Function Documentation

3.149.2.1 FindAppSettings() `[1/2]`

```
virtual AppSettings FindAppSettings ()  [virtual]
```

Find the AppSettings used to connect. Behaviour:

- if UseSharedAppSettings == true => use the global VoiceAppSettings (PhotonAppSettings). If its AppId is empty, log an error and return null.
- if UseSharedAppSettings == false => use transportClientSettings. If its appId is empty, log an error and return null.

Returns

3.149.2.2 FindAppSettings() [2/2]

```
virtual AppSettings FindAppSettings (
            StreamTransportClientSettings startVideoClientSettings)  [virtual]
```

Find the AppSettings used to connect. Behaviour:

- if UseSharedAppSettings == true => use the global VoiceAppSettings (PhotonAppSettings). If its AppId is empty, log an error and return null.
- if UseSharedAppSettings == false => use the local startVideoClientSettings. If its appId is empty, log an error and return null.

3.149.2.3 RegisterStreamReceiver()

```
virtual void RegisterStreamReceiver (
            IStreamReceiver streamReceiver)  [virtual]
```

Implements IStreamTransportManager.

3.149.2.4 RegisterStreamSender()

```
virtual void RegisterStreamSender (
            IStreamSender streamSender)  [virtual]
```

Implements IStreamTransportManager.

3.149.2.5 StreamLogLevel()

```
LogLevel StreamLogLevel ()
```

Implements IStreamTransportManager.

3.149.3 Property Documentation

3.149.3.1 Transport

```
StreamTransport Transport  [get]
```

Implements IStreamTransportManager.

3.150 StreamTransportClientSettings Class Reference

Public Attributes

- string **appId**
- string **appVersion** = "1"
- string **region** = ""
- ConnectionProtocol **protocol** = ConnectionProtocol.Udp

3.151 StreamTransportManagerListener Class Reference

Base class for stream senders/receivers which rely on a IRegisterableStreamTransportManager to provide the underlying LoadBalancingTransport upon which the stream run Automatically tries to find _streamTransportManager IRegisterableStreamTransportManager, through StreamTransportObserver.FindStreamTransportManager.

Inherits MonoBehaviour, and IStreamTransportManagerListener.

Inherited by AudioStreamReceiver, and AudioStreamSender.

Public Attributes

- GameObject **streamTransportGameObject** = null

Protected Member Functions

- virtual void **Awake** ()
- virtual void **OnDestroy** ()
- virtual IEnumerator FindStreamTransportManagerCoroutine ()

  Try to find a IRegisterableStreamTransportManager, using StreamTransportObserver.FindStreamTransportManager, that will fill the _streamTransportManager field if found Retries a few times (around 10s) if not found, before giving up.
- void **OnStreamTransportManagerFound** ()

  Called when a _streamTransportManager is set, and so a trnasport can be used.

Protected Attributes

- IStreamTransportManager **_streamTransportManager** = null
- Photon.Voice.Unity.Logger **_logger** = new Photon.Voice.Unity.Logger(LogLevel.Info)

Properties

- GameObject StreamTransportGameObject `[get, set]`
- IStreamTransportManager StreamTransportManager `[get, set]`
- StreamTransport **Transport** `[get]`

Properties inherited from IStreamTransportManagerListener

- GameObject **gameObject** `[get]`

3.151.1 Detailed Description

Base class for stream senders/receivers which rely on a IRegisterableStreamTransportManager to provide the underlying LoadBalancingTransport upon which the stream run Automatically tries to find _streamTransportManager IRegisterableStreamTransportManager, through StreamTransportObserver.FindStreamTransportManager.

OnStreamTransportManagerFound is called when the IRegisterableStreamTransportManager is found

3.151.2 Member Function Documentation

3.151.2.1 FindStreamTransportManagerCoroutine()

```
virtual IEnumerator FindStreamTransportManagerCoroutine ()  [protected], [virtual]
```

Try to find a IRegisterableStreamTransportManager, using StreamTransportObserver.FindStreamTransportManager, that will fill the _streamTransportManager field if found Retries a few times (around 10s) if not found, before giving up.

Returns

3.151.3 Property Documentation

3.151.3.1 StreamTransportGameObject

```
GameObject StreamTransportGameObject  [get], [set]
```

Implements IStreamTransportManagerListener.

3.151.3.2 StreamTransportManager

```
IStreamTransportManager StreamTransportManager  [get], [set]
```

Implements IStreamTransportManagerListener.

3.152 StreamTransportObserver Class Reference

Wrapper handling LoadBalancingTransport callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the IRegisterableStreamTransportManager interface.

Inherits IStreamTransportManager, IDisposable, IConnectionCallbacks, and IMatchmakingCallbacks.

Public Member Functions

- LogLevel StreamLogLevel ()
- **StreamTransportObserver** (StreamTransport transport, LogLevel logLevel=LogLevel.Info)
- void **RegisterOnTransport** ()
- void RegisterStreamSender (IStreamSender streamSender)
- void RegisterStreamReceiver (IStreamReceiver streamReceiver)
- void **Dispose** ()
- void **OnDisconnected** (DisconnectCause cause)
- void **OnConnectedToMaster** ()
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)

Static Public Member Functions

- static IEnumerator FindStreamTransportManager (IStreamTransportManagerListener listener, bool change↩ StreamTransportGameObjectWithResult=true, bool logFailure=true)

  Stream Transport Manager finder function, that will, for a given game object:

Public Attributes

- bool **isRoomJoined** = false

Protected Member Functions

- virtual void **OnRemoteVoiceAdd** (int channelId, int playerId, byte voiceId, VoiceInfo voiceInfo, ref RemoteVoiceOptions options)
- void **NotifiyJoinedRoom** ()
- void **NotifyLeftRoom** ()
- virtual void **NotifyRemoteVoiceAdd** (int channelId, int playerId, byte voiceId, VoiceInfo i, ref RemoteVoiceOptions options)
- virtual void **CheckJoinStatus** ()

Protected Attributes

- List< IStreamSender > **_registeredStreamSenders** = new List<IStreamSender>()
- List< IStreamReceiver > **_registeredStreamReceivers** = new List<IStreamReceiver>()
- LogLevel **_logLevel** = LogLevel.Info

Properties

- StreamTransport Transport = null `[get, set]`

Properties inherited from IStreamTransportManager

- long **RoundTripTime** `[get]`
- long **BytesOut** `[get]`
- long **PacketsOut** `[get]`
- long **BytesIn** `[get]`
- long **PacketsIn** `[get]`

3.152.1 Detailed Description

Wrapper handling LoadBalancingTransport callbacks for a given transport, to be used as a wrapper around component handling a transport but not compatible with the IRegisterableStreamTransportManager interface.

Provide the StreamTransportObserver.FindStreamTransportManager helper, that will, for a given game object:

1. find a IRegisterableStreamTransportManager
2. or wrap a IStreamTransportManager in its children to enhance it with callback handling
3. or, wrap a VoiceConnection (like FusionVoiceClient), if any in the children

3.152.2 Member Function Documentation

3.152.2.1 FindStreamTransportManager()

```
IEnumerator FindStreamTransportManager (
            IStreamTransportManagerListener listener,
            bool changeStreamTransportGameObjectWithResult = true,
            bool logFailure = true)  [static]
```

Stream Transport Manager finder function, that will, for a given game object:

1. find a IRegisterableStreamTransportManager
2. or wrap a IStreamTransportManager in its children to enhance it with callback handling
3. or, wrap a VoiceConnection (like FusionVoiceClient), if any in the children or in the scene
4. will look for a StreamTransportClient in the scene

For the found manager, if the Transport is not yet set, wait for it to be set before returning

3.152.2.2 RegisterStreamReceiver()

```
void RegisterStreamReceiver (
            IStreamReceiver streamReceiver)
```

Implements IStreamTransportManager.

3.152.2.3 RegisterStreamSender()

```
void RegisterStreamSender (
            IStreamSender streamSender)
```

Implements IStreamTransportManager.

3.152.2.4 StreamLogLevel()

```
LogLevel StreamLogLevel ()
```

Implements IStreamTransportManager.

3.152.3 Property Documentation

3.152.3.1 Transport

```
StreamTransport Transport = null  [get], [set]
```

Implements IStreamTransportManager.

3.153 ImageBufferInfo.StrideSet Struct Reference

Public Member Functions

- **StrideSet** (int length, int s0=0, int s1=0, int s2=0, int s3=0)

Properties

- int **this[int key]** `[get, set]`
- int **Length** `[get]`

3.154 StringSessionProperty Struct Reference

Public Attributes

- string **propertyName**
- string **value**

3.155 AudioUtil.TempoUp< T > Class Template Reference

Public Member Functions

- void **Begin** (int channels, int changePerc, int skipGroup)
- int **Process** (T[ ] s, T[ ] d)
- int **End** (T[ ] s)
- int **endFloat** (float[ ] s)
- int **endShort** (short[ ] s)

3.156 TestTone Class Reference

Inherits MonoBehaviour.

3.157 TimerWorker Class Reference

Public Member Functions

- **TimerWorker** (Action callback, int interval)
- void **Stop** ()

Static Public Member Functions

- static void **CallbackStatic** (int handle)

3.158 AudioUtil.ToneAudioPusher< T > Class Template Reference

IAudioPusher that provides a constant tone signal.

Inherits AudioUtil.GeneratorPusher< T >.

Public Member Functions

- ToneAudioPusher (int frequency=440, int bufSizeMs=100, int samplingRate=48000, int channels=1)
  Create a new ToneAudioReader instance.

Public Member Functions inherited from AudioUtil.GeneratorPusher< T >

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)
  Set the callback function used for pushing data.
- void **Dispose** ()

Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

Protected Member Functions inherited from AudioUtil.GeneratorPusher< T >

- int **Gen** (T[] buf, long timeSamples)

Additional Inherited Members

Protected Attributes inherited from AudioUtil.GeneratorPusher< T >

- long **timeSamples**

Properties inherited from AudioUtil.GeneratorPusher< T >

- int Channels `[get]`
  Number of channels in the audio signal.
- int SamplingRate `[get]`
  Sampling rate of the audio signal (in Hz).
- string Error `[get]`
  If not null, audio object is in invalid state.

3.158.1 Detailed Description

IAudioPusher that provides a constant tone signal.

3.158.2 Member Function Documentation

3.158.2.1 ToneAudioPusher()

```
ToneAudioPusher (
            int frequency = 440,
            int bufSizeMs = 100,
            int samplingRate = 48000,
            int channels = 1)
```

Create a new ToneAudioReader instance.

Parameters

frequency	Frequency of the generated tone (in Hz).
bufSizeMs	Size of buffers to push (in milliseconds).
samplingRate	Sampling rate of the audio signal (in Hz).
channels	Number of channels in the audio signal.

3.159 AudioUtil.ToneAudioReader< T > Class Template Reference

IAudioReader that provides a constant tone signal.

Inherits AudioUtil.GeneratorReader< T >.

Public Member Functions

- ToneAudioReader (Func< double > clockSec=null, double frequency=440, int samplingRate=48000, int channels=1)

  Create a new ToneAudioReader instance.

Public Member Functions inherited from AudioUtil.GeneratorReader< T >

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool Read (T[ ] buf)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Protected Member Functions

- override int **Gen** (T[ ] buf, long timeSamples)

Protected Member Functions inherited from AudioUtil.GeneratorReader< T >

- int **Gen** (T[ ] buf, long timeSamples)

Additional Inherited Members

Properties inherited from AudioUtil.GeneratorReader< T >

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.159.1 Detailed Description

IAudioReader that provides a constant tone signal.

Because of current resampling algorithm, the tone is distorted if SamplingRate does not equal encoder sampling rate.

3.159.2 Member Function Documentation

3.159.2.1 ToneAudioReader()

```
ToneAudioReader (
            Func< double > clockSec = null,
            double frequency = 440,
            int samplingRate = 48000,
            int channels = 1)
```

Create a new ToneAudioReader instance.

Parameters

	clockSec	Function to get current time in seconds. In Unity, pass in '() => AudioSettings.dspTime' for better results.
	frequency	Frequency of the generated tone (in Hz).
	samplingRate	Sampling rate of the audio signal (in Hz).
	channels	Number of channels in the audio signal.

3.160 UnityAudioOut Class Reference

Inherits AudioOutDelayControl< T >.

Public Member Functions

- **UnityAudioOut** (AudioSource audioSource, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (float[ ] data, int offsetSamples)
- override void Stop ()

Public Member Functions inherited from AudioOutDelayControl< T >

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()

Protected Attributes

- readonly AudioSource **source**
- AudioClip **clip**

Protected Attributes inherited from AudioOutDelayControl< T >

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

Properties

- override long **OutPos** `[get]`

Properties inherited from AudioOutDelayControl< T >

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

Additional Inherited Members

Static Public Attributes inherited from AudioOutDelayControl< T >

- const int **FRAME_POOL_CAPACITY** = 50

3.160.1 Member Function Documentation

3.160.1.1 Stop()

```
override void Stop ()  [virtual]
```

Reimplemented from AudioOutDelayControl< T >.

3.161 UnityLogger Class Reference

Static Public Member Functions

- static void **Log** (LogLevel level, Object obj, string tag, string objName, string msg)
- static void **Log** (LogLevel level, Object obj, string tag, string objName, string fmt, params object[ ] args)

3.162 UnityMicrophone Class Reference

A wrapper around UnityEngine.Microphone to be able to safely use Microphone and compile for WebGL.

Static Public Member Functions

- static void **End** (string deviceName)
- static void **GetDeviceCaps** (string deviceName, out int minFreq, out int maxFreq)
- static int **GetPosition** (string deviceName)
- static bool **IsRecording** (string deviceName)
- static AudioClip **Start** (string deviceName, bool loop, int lengthSec, int frequency)
- static string **CheckDevice** (Voice.ILogger logger, string logPref, string device, int suggestedFrequency, out int frequency)

Properties

- static string[] **devices** `[get]`

3.162.1 Detailed Description

A wrapper around UnityEngine.Microphone to be able to safely use Microphone and compile for WebGL.

3.163 UnityVoiceClient Class Reference

Component that represents a Voice client and manages a simple Unity integration: a single Recorder and multiple remote speakers.

Inherits VoiceConnection.

Public Member Functions

- override bool ConnectUsingSettings (AppSettings overwriteSettings=null)

  Connect to Photon server using Settings

Public Member Functions inherited from VoiceConnection

- bool AddSpeaker (Speaker speaker, object userData)

  Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.
- Speaker InstantiateSpeakerPrefab (GameObject parent, bool destroyOnRemove)

  Instantiates SpeakerPrefab, optionally attaches it to the provided parent.
- bool **AddRecorder** (Recorder rec)
- void **RemoveRecorder** (Recorder rec)

Public Attributes

- bool **UseVoiceAppSettings** = false

  Whether or not to use the Voice AppId and all the other AppSettings from Fusion's RealtimeAppSettings Scriptable↩ Object singleton in the Voice client/app.

Public Attributes inherited from VoiceConnection

- AppSettings **Settings**

  Settings to be used by this Voice Client.

Protected Member Functions

- virtual void **Start** ()
- override Speaker InstantiateSpeakerForRemoteVoice (int playerId, byte voiceId, object userData)

Protected Member Functions inherited from VoiceConnection

- override void **Awake** ()
- virtual void **Update** ()
- virtual void **OnDestroy** ()
- virtual void **OnVoiceStateChanged** (ClientState fromState, ClientState toState)
- void **CalcStatistics** ()
- virtual void **OnOperationResponseReceived** (OperationResponse operationResponse)

Properties

- override bool **AlwaysUsePrimaryRecorder** `[get]`

Properties inherited from VoiceConnection

- virtual bool **AlwaysUsePrimaryRecorder** `[get]`
- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- new LoadBalancingTransport **Client** `[get]`
- VoiceClient **VoiceClient** `[get]`

  Returns underlying Photon Voice client.
- ClientState **ClientState** `[get]`

  Returns Photon Voice client state.
- float **FramesReceivedPerSecond** `[get]`

  Number of frames received per second.
- float **FramesLostPerSecond** `[get]`

  Number of frames lost per second.
- float **FramesLostPercent** `[get]`

  Percentage of lost frames.
- GameObject **SpeakerPrefab** `[get, set]`

  Prefab that contains Speaker component to be instantiated when receiving a new remote audio source info.
- Recorder **PrimaryRecorder** `[get, set]`

  Primary Recorder to be used by VoiceConnection implementations directly or via integration objects.
- bool **UsePrimaryRecorder** `[get]`

  Use VoiceConnection.PrimaryRecorder directly.
- string **BestRegionSummaryInPreferences** `[get, set]`

  Used to store and access the "Best Region Summary" in the Player Preferences.

Additional Inherited Members

Static Public Attributes inherited from VoiceConnection

- const int **ChannelAudio** = 1

  Recommended Photon Transport channel for audio. Chosen not to interfere with video and default channel.
- const int **ChannelVideo** = 2

  Recommended Photon Transport channel for video. Chosen not to interfere with audio and default channel.

Events inherited from VoiceConnection

- Action< Speaker > **SpeakerLinked**

  Fires when a speaker has been linked to a remote audio stream.
- Action< RemoteVoiceLink > **RemoteVoiceAdded**

  Fires when a remote voice stream is added.

3.163.1 Detailed Description

Component that represents a Voice client and manages a simple Unity integration: a single Recorder and multiple remote speakers.

3.163.2 Member Function Documentation

3.163.2.1 ConnectUsingSettings()

```
override bool ConnectUsingSettings (
            AppSettings overwriteSettings = null)  [virtual]
```

Connect to Photon server using Settings

Parameters

	overwriteSettings	Overwrites Settings before connecting

Returns

If true voice connection command was sent from client

Reimplemented from VoiceConnection.

3.163.2.2 InstantiateSpeakerForRemoteVoice()

```
override Speaker InstantiateSpeakerForRemoteVoice (
            int playerId,
            byte voiceId,
            object userData)  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.164 UnsupportedCodecException Class Reference

Exception thrown if an unsupported codec is encountered.

Inherits Exception.

Public Member Functions

- UnsupportedCodecException (string info, Codec codec)
  Create a new UnsupportedCodecException.

3.164.1 Detailed Description

Exception thrown if an unsupported codec is encountered.

3.164.2 Constructor & Destructor Documentation

3.164.2.1 UnsupportedCodecException()

```
UnsupportedCodecException (
            string info,
            Codec codec)
```

Create a new UnsupportedCodecException.

Parameters

	info	The info prepending standard message.
	codec	The codec actually encountered.

3.165 UnsupportedPlatformException Class Reference

Exception thrown if an unsupported platform is encountered.

Inherits Exception.

Public Member Functions

- UnsupportedPlatformException (string subject, string platform=null)
  Create a new UnsupportedPlatformException.

3.165.1 Detailed Description

Exception thrown if an unsupported platform is encountered.

3.165.2 Constructor & Destructor Documentation

3.165.2.1 UnsupportedPlatformException()

```
UnsupportedPlatformException (
            string subject,
            string platform = null)
```

Create a new UnsupportedPlatformException.

Parameters

	subject	The info prepending standard message.

///

Parameters

	platform	Optional platform name.

3.166 UnsupportedSampleTypeException Class Reference

Exception thrown if an unsupported audio sample type is encountered.

Inherits Exception.

Public Member Functions

- UnsupportedSampleTypeException (Type t)
  Create a new UnsupportedSampleTypeException.

3.166.1 Detailed Description

Exception thrown if an unsupported audio sample type is encountered.

PhotonVoice generally supports 32-bit floating point ("float") or 16-bit signed integer ("short") audio, but it usually won't be converted automatically due to the high CPU overhead (and potential loss of precision) involved.

3.166.2 Constructor & Destructor Documentation

3.166.2.1 UnsupportedSampleTypeException()

```
UnsupportedSampleTypeException (
            Type t)
```

Create a new UnsupportedSampleTypeException.

Parameters

	t	The sample type actually encountered.

3.167 OpusCodec.Util Class Reference

3.168 Utility Class Reference

Static Public Member Functions

- static bool **IsAudio** (this Codec c)
- static bool **IsVideo** (this Codec c)

3.169 VideoInEnumerator Class Reference

Inherits DeviceEnumeratorBase.

Public Member Functions

- **VideoInEnumerator** (ILogger logger)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Properties

- override string Error `[get]`

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

3.169.1 Member Function Documentation

3.169.1.1 Refresh()

```
override void Refresh ()
```

Implements IDeviceEnumerator.

3.169.2 Property Documentation

3.169.2.1 Error

```
override string Error  [get]
```

Implements IDeviceEnumerator.

3.170 VideoInEnumerator Class Reference

Inherits DeviceEnumerator.

Public Member Functions

- **VideoInEnumerator** (ILogger logger)

Public Member Functions inherited from DeviceEnumerator

- **DeviceEnumerator** (ILogger logger, Windows.Devices.Enumeration.DeviceClass deviceClass)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.171 VoiceClient Class Reference

Voice client interact with other clients on network via IVoiceTransport.

Inherits IDisposable.

Classes

- struct CreateOptions

Public Member Functions

- delegate void **RemoteVoiceInfoDelegate** (int channelId, int playerId, byte voiceId, VoiceInfo voiceInfo, ref RemoteVoiceOptions options)

  Remote voice info event delegate.
- IEnumerable< LocalVoice > **LocalVoicesInChannel** (int channelId)

  Iterates through copy of all local voices list of given channel.
- void **LogSpacingProfiles** ()
- void **LogStats** ()
- void **SetRemoteVoiceDelayFrames** (Codec codec, int delayFrames)
- void **SetRemoteVoiceDelayFrames** (Func< Codec, bool > predicate, int delayFrames)
- **VoiceClient** (IVoiceTransport transport, ILogger logger, CreateOptions opt=default(CreateOptions))

  Creates VoiceClient instance.
- void **Service** ()

  This method dispatches all available incoming commands and then sends this client's outgoing commands. Call this method regularly (2..20 times a second).
- LocalVoice CreateLocalVoice (VoiceInfo voiceInfo, int channelId, VoiceCreateOptions options=default(VoiceCreateOptions))

  Creates basic outgoing stream w/o data processing support. Provided encoder should generate output data stream.
- LocalVoiceAudio< T > **CreateLocalVoiceAudio**< **T** > (VoiceInfo voiceInfo, IAudioDesc audioSourceDesc, int channelId, VoiceCreateOptions options=default(VoiceCreateOptions))
- LocalVoice CreateLocalVoiceAudioFromSource (VoiceInfo voiceInfo, IAudioDesc source, AudioSampleType sampleType, int channelId, VoiceCreateOptions options=default(VoiceCreateOptions))

  Creates outgoing audio stream of type automatically assigned and adds procedures (callback or serviceable) for consuming given audio source data. Adds audio specific features (e.g. resampling, level meter) to processing pipeline and to returning stream handler.
- LocalVoiceVideo CreateLocalVoiceVideo (VoiceInfo voiceInfo, IVideoRecorder recorder, int channelId, VoiceCreateOptions options=default(VoiceCreateOptions))

  Creates outgoing video stream consuming sequence of image buffers.
- void **RemoveLocalVoice** (LocalVoice voice)

  Removes local voice (outgoing data stream).

Parameters

	voice	*Handler of outgoing stream to be removed.*

- void **onJoinChannel** (int channelId)
- void **onJoinAllChannels** ()
- void **onLeaveChannel** (int channel)
- void **onLeaveAllChannels** ()
- void **onPlayerJoin** (int channelId, int playerId)
- void **onPlayerJoin** (int playerId)
- void **onPlayerLeave** (int channelId, int playerId)
- void **onPlayerLeave** (int playerId)
- void **onVoiceInfo** (int channelId, int playerId, byte voiceId, int eventBufferSize, VoiceInfo info)
- void **onVoiceRemove** (int playerId, byte[] voiceIds)
- void **onFrame** (int playerId, byte voiceId, ushort evNumber, ref FrameBuffer receivedBytes, bool isLocal↩ Player)
- void **Dispose** ()

Properties

- bool **ThreadingEnabled** `[get, set]`
- int **EventsLost** `[get, set]`

  Lost events counter (the number of empty frames sent to the deocder).
- int **FramesLost** `[get, set]`

  Lost frames counter (the number of empty frames sent to the deocder).
- int **FramesFragPart** `[get, set]`

  The counter of assembled frames, fragments of which are partially missing.
- int **FramesRecovered** `[get, set]`

  Recovered frames counter.
- int **FramesLate** `[get, set]`

  Counter of late (incorrectly ordered) frames.
- int **FramesReceived** `[get]`

  Received frames counter.
- int **FramesReceivedFEC** `[get, set]`

  Received FEC events counter.
- int **FramesTryFEC** `[get, set]`

  FEC recorery attempts counter.
- int **FramesReceivedFragments** `[get, set]`

  Received events for fragmented frames counter.
- int **FramesReceivedFragmented** `[get, set]`

  Assembled fragmented frames counter.
- int **FramesSent** `[get]`

  Sent frames counter.
- int **FramesSentBytes** `[get]`

  Sent frames bytes counter.
- int **RoundTripTime** `[get]`

  Average time required voice packet to return to sender.
- int **RoundTripTimeVariance** `[get]`

  Average round trip time variation.
- bool **SuppressInfoDuplicateWarning** `[get, set]`

  Do not log warning when duplicate info received.
- RemoteVoiceInfoDelegate **OnRemoteVoiceInfoAction** `[get, set]`

  Register a method to be called when remote voice info arrived (after join or new new remote voice creation). Metod parameters: (int channelId, int playerId, byte voiceId, VoiceInfo voiceInfo, ref RemoteVoiceOptions options);.
- int **DebugLostPercent** `[get, set]`

Lost frames simulation ratio.

- IEnumerable< LocalVoice > **LocalVoices** `[get]`

  Iterates through copy of all local voices list.

- IEnumerable< RemoteVoiceInfo > **RemoteVoiceInfos** `[get]`

  Iterates through all remote voices infos.

3.171.1 Detailed Description

Voice client interact with other clients on network via IVoiceTransport.

3.171.2 Member Function Documentation

3.171.2.1 CreateLocalVoice()

```
LocalVoice CreateLocalVoice (
            VoiceInfo voiceInfo,
            int channelId,
            VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates basic outgoing stream w/o data processing support. Provided encoder should generate output data stream.

Parameters

voiceInfo	Outgoing stream parameters.
channelId	Transport channel specific to transport.
options	Voice creation options.

Returns

Outgoing stream handler.

3.171.2.2 CreateLocalVoiceAudioFromSource()

```
LocalVoice CreateLocalVoiceAudioFromSource (
            VoiceInfo voiceInfo,
            IAudioDesc source,
            AudioSampleType sampleType,
            int channelId,
            VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates outgoing audio stream of type automatically assigned and adds procedures (callback or serviceable) for consuming given audio source data. Adds audio specific features (e.g. resampling, level meter) to processing pipeline and to returning stream handler.

Parameters

voiceInfo	Outgoing stream parameters.
source	Streaming audio source.
sampleType	Voice's audio sample type. If does not match source audio sample type, conversion will occur.
channelId	Transport channel specific to transport.
options	Voice creation options.

Returns

Outgoing stream handler.

audioSourceDesc.SamplingRate and voiceInfo.SamplingRate may do not match. Automatic resampling will occur in this case.

3.171.2.3 CreateLocalVoiceVideo()

```
LocalVoiceVideo CreateLocalVoiceVideo (
            VoiceInfo voiceInfo,
            IVideoRecorder recorder,
            int channelId,
            VoiceCreateOptions options = default(VoiceCreateOptions))
```

Creates outgoing video stream consuming sequence of image buffers.

Parameters

voiceInfo	Outgoing stream parameters.
recorder	Video recorder.
channelId	Transport channel specific to transport.
options	Voice creation options.

Returns

Outgoing stream handler.

3.172 VoiceComponent Class Reference

Inherits MonoBehaviour.

Inherited by PhotonVoiceView, AudioChangesHandler, FMODRecorderSetup, Recorder, RecorderPreset, Speaker, MicAmplifier, MicrophonePermission, SaveIncomingStreamToFile, SaveOutgoingStreamToFile, and WebRtcAudioDsp.

Protected Member Functions

- virtual void **Awake** ()

Properties

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.173 VoiceComponentImpl Class Reference

Public Member Functions

- void **Awake** (MonoBehaviour mb)

Properties

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.174 VoiceConnection Class Reference

Component that represents a Voice client.

Inherits ConnectionHandler.

Inherited by UnityVoiceClient, and VoiceFollowClient.

Public Member Functions

- virtual bool ConnectUsingSettings (AppSettings overwriteSettings=null)

  Connect to Photon server using Settings
- bool AddSpeaker (Speaker speaker, object userData)

  Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.
- Speaker InstantiateSpeakerPrefab (GameObject parent, bool destroyOnRemove)

  Instantiates SpeakerPrefab, optionally attaches it to the provided parent.
- bool **AddRecorder** (Recorder rec)
- void **RemoveRecorder** (Recorder rec)

Public Attributes

- AppSettings **Settings**

  Settings to be used by this Voice Client.

Static Public Attributes

- const int **ChannelAudio** = 1

  Recommended Photon Transport channel for audio. Chosen not to interfere with video and default channel.
- const int **ChannelVideo** = 2

  Recommended Photon Transport channel for video. Chosen not to interfere with audio and default channel.

Protected Member Functions

- override void **Awake** ()
- virtual void **Update** ()
- virtual void **OnDestroy** ()
- virtual Speaker **InstantiateSpeakerForRemoteVoice** (int playerId, byte voiceId, object userData)
- virtual void **OnVoiceStateChanged** (ClientState fromState, ClientState toState)
- void **CalcStatistics** ()
- virtual void **OnOperationResponseReceived** (OperationResponse operationResponse)

Properties

- virtual bool **AlwaysUsePrimaryRecorder** `[get]`
- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- new LoadBalancingTransport **Client** `[get]`
- VoiceClient **VoiceClient** `[get]`

  Returns underlying Photon Voice client.
- ClientState **ClientState** `[get]`

  Returns Photon Voice client state.
- float **FramesReceivedPerSecond** `[get]`

  Number of frames received per second.
- float **FramesLostPerSecond** `[get]`

  Number of frames lost per second.
- float **FramesLostPercent** `[get]`

  Percentage of lost frames.
- GameObject **SpeakerPrefab** `[get, set]`

  Prefab that contains Speaker component to be instantiated when receiving a new remote audio source info.
- Recorder **PrimaryRecorder** `[get, set]`

  Primary Recorder to be used by VoiceConnection implementations directly or via integration objects.
- bool **UsePrimaryRecorder** `[get]`

  Use VoiceConnection.PrimaryRecorder directly.
- string **BestRegionSummaryInPreferences** `[get, set]`

  Used to store and access the "Best Region Summary" in the Player Preferences.

Events

- Action< Speaker > **SpeakerLinked**

  Fires when a speaker has been linked to a remote audio stream.
- Action< RemoteVoiceLink > **RemoteVoiceAdded**

  Fires when a remote voice stream is added.

3.174.1 Detailed Description

Component that represents a Voice client.

3.174.2 Member Function Documentation

3.174.2.1 AddSpeaker()

```
bool AddSpeaker (
            Speaker speaker,
            object userData)
```

Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.

Parameters

	speaker	Speaker ot try linking.
	userData	UserData object used to bind local Speaker with remote voice stream.

Returns

3.174.2.2 ConnectUsingSettings()

```
virtual bool ConnectUsingSettings (
            AppSettings overwriteSettings = null)  [virtual]
```

Connect to Photon server using Settings

Parameters

	overwriteSettings	Overwrites Settings before connecting

Returns

If true voice connection command was sent from client

Reimplemented in UnityVoiceClient.

3.174.2.3 InstantiateSpeakerPrefab()

```
Speaker InstantiateSpeakerPrefab (
            GameObject parent,
            bool destroyOnRemove)
```

Instantiates SpeakerPrefab, optionally attaches it to the provided parent.

VoiceConnection manages the instantiated object (destroys on OnRemoteVoiceRemoveAction).

Parameters

	parent	The object to attach S
	destroyOnRemove	Automatically destroy instantiated prefab when remote voice is removed (the caller does not manages the i

Returns

Instantiated Speaker or null.

3.175 VoiceConnectionMatchmaking Class Reference

Use matchmaking to connect and join a VoiceConnection.

Inherits MonoBehaviour, IConnectionCallbacks, and IMatchmakingCallbacks.

Public Member Functions

- void **ConnectNow** ()
- virtual void **JoinRoom** ()

  Join the room once connected to the server. The default implementation relies on random matchmaking, using EnterRoomParams parameters.
- virtual void **OnConnectedToMaster** ()
- virtual void **OnConnected** ()
- virtual void **OnRegionListReceived** (RegionHandler regionHandler)
- virtual void **OnCustomAuthenticationFailed** (string debugMessage)
- virtual void **OnCustomAuthenticationResponse** (Dictionary< string, object > data)
- virtual void **OnJoinedRoom** ()
- virtual void **OnLeftRoom** ()
- virtual void **OnFriendListUpdate** (List< FriendInfo > friendList)
- virtual void **OnCreatedRoom** ()
- virtual void **OnCreateRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRoomFailed** (short returnCode, string message)
- virtual void **OnJoinRandomFailed** (short returnCode, string message)
- void **OnDisconnected** (DisconnectCause cause)

Public Attributes

- bool **autoConnect** = true
- StreamMatchmakingSettings **matchmakingSettings** = new StreamMatchmakingSettings()
- List< StringSessionProperty > **roomCustomPropertiesPreview** = new List<StringSessionProperty>()

Protected Member Functions

- virtual void **Start** ()

Protected Attributes

- VoiceConnection **voiceConnection**

Properties

- virtual AuthenticationValues **AuthValues** `[get]`

3.175.1 Detailed Description

Use matchmaking to connect and join a VoiceConnection.

3.176 VoiceConnectionPermissionWaiter Class Reference

Wait for Microphone access permissions to be available before letting VoiceConnection connect Note: record↩ WhenJoined should be disabled on the recorder.

Inherits PermissionWaiter.

Public Member Functions

- override void OnPermissionGranted ()

  Should be called manually (for instance in a PermissionsRequester callback) if checkPermisisonGrantAutomatically is not checked.

Static Public Attributes

- const int **LOWPRIORITY_EXECUTION_ORDER** = PermissionWaiter.EXECUTION_ORDER + 50

Static Public Attributes inherited from PermissionWaiter

- const int **EXECUTION_ORDER** = PermissionsRequester.EXECUTION_ORDER + 1

Protected Member Functions

- override void Awake ()
- override void OnPermissionRequired ()

  Called when the permission is required.
- override void Update ()

Protected Member Functions inherited from PermissionWaiter

- virtual void **OnPermissionChanged** (bool isGranted)

Properties

- override string **PermissionName** `[get]`

Properties inherited from PermissionWaiter

- bool **IsWaitingForPermission** `[get]`
- virtual string **PermissionName** `[get]`
- bool **IsPermissionGranted** `[get]`

Additional Inherited Members

Public Attributes inherited from PermissionWaiter

- bool **checkPermisisonGrantAutomatically** = true
- UnityEvent< bool > **onPermissionChanged** = new UnityEvent<bool>()
- UnityEvent **onPermissionGranted** = new UnityEvent()

Protected Attributes inherited from PermissionWaiter

- bool **waitingForPermission** = false

3.176.1 Detailed Description

Wait for Microphone access permissions to be available before letting VoiceConnection connect Note: record↩ WhenJoined should be disabled on the recorder.

Should be used alongside a PermissionsRequester, to align permissions on the desired timing (to avoid conflicting requests)

3.176.2 Member Function Documentation

3.176.2.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from PermissionWaiter.

3.176.2.2 OnPermissionGranted()

```
override void OnPermissionGranted ()  [virtual]
```

Should be called manually (for instance in a PermissionsRequester callback) if checkPermisisonGrantAutomatically is not checked.

Reimplemented from PermissionWaiter.

3.176.2.3 OnPermissionRequired()

```
override void OnPermissionRequired ()  [protected], [virtual]
```

Called when the permission is required.

Reimplemented from PermissionWaiter.

3.176.2.4 Update()

```
override void Update ()  [protected], [virtual]
```

Reimplemented from PermissionWaiter.

3.177 VoiceCreateOptions Struct Reference

Used to initialize optional properties of the LocalVoice instance at creation time.

Public Attributes

- IEncoder **Encoder**

  Encoder.
- byte **InterestGroup**

  See LocalVoice.InterestGroup.
- int[ ] **TargetPlayers**

  See LocalVoice.TargetPlayersSet to [] to make sure that LocalVoice does not create Remote voices during creation.
- bool **DebugEchoMode**

  See LocalVoice.DebugEchoMode.
- bool **Reliable**

  See LocalVoice.Reliable.
- bool **Encrypt**

  See LocalVoice.Encrypt.
- bool **Fragment**

  See LocalVoice.Fragment.
- int **FEC**

  See LocalVoice.FEC.
- int **EventBufSize**

  Receiver's ring buffer size (default is 256)

3.177.1 Detailed Description

Used to initialize optional properties of the LocalVoice instance at creation time.

3.178 VoiceDebugScript Class Reference

Utility script to be attached next to PhotonVoiceView & PhotonView on the player prefab to be network instantiated. Call voiceDebugScript.CantHearYou() on the networked object of the remote (or local) player if you can't hear the corresponding player.

Inherits MonoBehaviourPun.

Public Member Functions

- void **CantHearYou** ()

Public Attributes

- bool **ForceRecordingAndTransmission**

  Make sure recorder.TransmitEnabled and recorder.RecordingEnabled are true.
- AudioClip **TestAudioClip**

  Audio file to be broadcast when TestUsingAudioClip is enabled.
- bool **TestUsingAudioClip**

  Broadcast Audio file to make sure transmission over network works if microphone (audio input device/hardware) is not reliable. Requires setting AudioClip in TestAudioClip.
- bool **DisableVad**

  Disable recorder.VoiceDetection for easier testing.
- bool **IncreaseLogLevels**

  Set main voice component's log level to ALL (max).
- bool **LocalDebug**

  Debug DebugEcho mode (Can't Hear My Self?!).

3.178.1 Detailed Description

Utility script to be attached next to PhotonVoiceView & PhotonView on the player prefab to be network instantiated. Call voiceDebugScript.CantHearYou() on the networked object of the remote (or local) player if you can't hear the corresponding player.

3.179 AudioUtil.VoiceDetector< T > Class Template Reference

Simple voice activity detector triggered by signal level.

Inherits IProcessor< T >, and AudioUtil.IVoiceDetector.

Inherited by AudioUtil.VoiceDetectorFloat, and AudioUtil.VoiceDetectorShort.

Public Member Functions

- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Protected Attributes

- float **norm**
- float **threshold**
- int **activityDelay**
- int **autoSilenceCounter** = 0
- int **valuesCountPerSec**
- int **activityDelayValuesCount**

Properties

- bool On `[get, set]`

  If true, voice detection enabled.
- float Threshold `[get, set]`

  Voice detected as soon as signal level exceeds threshold.
- bool Detected `[get, protected set]`

  If true, voice detected.
- DateTime DetectedTime `[get]`

  Last time when switched to detected state.
- int ActivityDelayMs `[get, set]`

  Keep detected state during this time after signal level dropped below threshold.

Events

- Action **OnDetected**

  Called when switched to detected state.

Events inherited from AudioUtil.IVoiceDetector

- Action **OnDetected**

  Called when switched to detected state.

3.179.1 Detailed Description

Simple voice activity detector triggered by signal level.

3.179.2 Member Function Documentation

3.179.2.1 Process()

```
T[] Process (
            T[] buf)  [abstract]
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements IProcessor< T >.

3.179.3 Property Documentation

3.179.3.1 ActivityDelayMs

```
int ActivityDelayMs  [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implements AudioUtil.IVoiceDetector.

3.179.3.2 Detected

```
bool Detected  [get], [protected set]
```

If true, voice detected.

Implements AudioUtil.IVoiceDetector.

3.179.3.3 DetectedTime

```
DateTime DetectedTime  [get]
```

Last time when switched to detected state.

Implements AudioUtil.IVoiceDetector.

3.179.3.4 On

```
bool On  [get], [set]
```

If true, voice detection enabled.

Implements AudioUtil.IVoiceDetector.

3.179.3.5 Threshold

```
float Threshold  [get], [set]
```

Voice detected as soon as signal level exceeds threshold.

Implements AudioUtil.IVoiceDetector.

3.180 AudioUtil.VoiceDetectorCalibration< T > Class Template Reference

Calibration Utility for Voice Detector.

Inherits IProcessor< T >.

Public Member Functions

- VoiceDetectorCalibration (IVoiceDetector voiceDetector, ILevelMeter levelMeter, int samplingRate, int channels)

  Create new VoiceDetectorCalibration instance.
- void Calibrate (int durationMs, Action< float > onCalibrated=null)

  Start calibration.
- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Protected Attributes

- int **calibrateCount**

Properties

- bool **IsCalibrating** [get]

3.180.1 Detailed Description

Calibration Utility for Voice Detector.

Using this audio processor, you can calibrate the IVoiceDetector.Threshold.

3.180.2 Member Function Documentation

3.180.2.1 Calibrate()

```
void Calibrate (
            int durationMs,
            Action< float > onCalibrated = null)
```

Start calibration.

Parameters

	durationMs	Duration of the calibration procedure (in milliseconds).
	onCalibrated	Optional callback that is called after calibration is complete.

This activates the Calibration process. It will reset the given LevelMeter's AccumAvgPeakAmp (accumulated average peak amplitude), and when the duration has passed, use it for the VoiceDetector's detection threshold.

3.180.2.2 Process()

```
T[] Process (
            T[] buf)
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements IProcessor< T >.

3.180.2.3 VoiceDetectorCalibration()

```
VoiceDetectorCalibration (
            IVoiceDetector voiceDetector,
            ILevelMeter levelMeter,
            int samplingRate,
            int channels)
```

Create new VoiceDetectorCalibration instance.

Parameters

	voiceDetector	Voice Detector to calibrate.
	levelMeter	Level Meter to look at for calibration.
	samplingRate	Sampling rate of the audio signal (in Hz).
	channels	Number of channels in the audio signal.

3.181 AudioUtil.VoiceDetectorDummy Class Reference

Dummy VoiceDetector that doesn't actually do anything.

Inherits AudioUtil.IVoiceDetector.

Properties

- bool On `[get, set]`
  If true, voice detection enabled.
- float Threshold `[get, set]`
  Voice detected as soon as signal level exceeds threshold.
- bool Detected `[get]`
  If true, voice detected.
- int ActivityDelayMs `[get, set]`
  Keep detected state during this time after signal level dropped below threshold.
- DateTime DetectedTime `[get]`
  Last time when switched to detected state.

Events

- Action **OnDetected** `[add, remove]`

Events inherited from AudioUtil.IVoiceDetector

- Action **OnDetected**
  Called when switched to detected state.

3.181.1 Detailed Description

Dummy VoiceDetector that doesn't actually do anything.

3.181.2 Property Documentation

3.181.2.1 ActivityDelayMs

```
int ActivityDelayMs [get], [set]
```

Keep detected state during this time after signal level dropped below threshold.

Implements AudioUtil.IVoiceDetector.

3.181.2.2 Detected

```
bool Detected  [get]
```

If true, voice detected.

Implements AudioUtil.IVoiceDetector.

3.181.2.3 DetectedTime

```
DateTime DetectedTime  [get]
```

Last time when switched to detected state.

Implements AudioUtil.IVoiceDetector.

3.181.2.4 On

```
bool On  [get], [set]
```

If true, voice detection enabled.

Implements AudioUtil.IVoiceDetector.

3.181.2.5 Threshold

```
float Threshold  [get], [set]
```

Voice detected as soon as signal level exceeds threshold.

Implements AudioUtil.IVoiceDetector.

3.182 AudioUtil.VoiceDetectorFloat Class Reference

VoiceDetector specialization for float audio.

Inherits AudioUtil.VoiceDetector< T >.

Public Member Functions

- VoiceDetectorFloat (int samplingRate, int numChannels)
  Create a new VoiceDetectorFloat instance.
- override float[ ] **Process** (float[ ] buffer)

Public Member Functions inherited from AudioUtil.VoiceDetector< T >

- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from AudioUtil.VoiceDetector< T >

- float **norm**
- float **threshold**
- int **activityDelay**
- int **autoSilenceCounter** = 0
- int **valuesCountPerSec**
- int **activityDelayValuesCount**

Properties inherited from AudioUtil.VoiceDetector< T >

- bool On `[get, set]`

  If true, voice detection enabled.
- float Threshold `[get, set]`

  Voice detected as soon as signal level exceeds threshold.
- bool Detected `[get, protected set]`

  If true, voice detected.
- DateTime DetectedTime `[get]`

  Last time when switched to detected state.
- int ActivityDelayMs `[get, set]`

  Keep detected state during this time after signal level dropped below threshold.

Events inherited from AudioUtil.VoiceDetector< T >

- Action **OnDetected**

  Called when switched to detected state.

Events inherited from AudioUtil.IVoiceDetector

- Action **OnDetected**

  Called when switched to detected state.

3.182.1 Detailed Description

VoiceDetector specialization for float audio.

3.182.2 Constructor & Destructor Documentation

3.182.2.1 VoiceDetectorFloat()

```
VoiceDetectorFloat (
            int samplingRate,
            int numChannels)
```

Create a new VoiceDetectorFloat instance.

Parameters

	samplingRate	Sampling rate of the audio signal (in Hz).
	numChannels	Number of channels in the audio signal.

3.183 AudioUtil.VoiceDetectorShort Class Reference

VoiceDetector specialization for float audio.

Inherits AudioUtil.VoiceDetector< T >.

Public Member Functions

- VoiceDetectorShort (int samplingRate, int numChannels)

  Create a new VoiceDetectorFloat instance.
- override short[ ] **Process** (short[ ] buffer)

Public Member Functions inherited from AudioUtil.VoiceDetector< T >

- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from AudioUtil.VoiceDetector< T >

- float **norm**
- float **threshold**
- int **activityDelay**
- int **autoSilenceCounter** = 0
- int **valuesCountPerSec**
- int **activityDelayValuesCount**

Properties inherited from AudioUtil.VoiceDetector< T >

- bool On `[get, set]`

  If true, voice detection enabled.
- float Threshold `[get, set]`

  Voice detected as soon as signal level exceeds threshold.
- bool Detected `[get, protected set]`

  If true, voice detected.
- DateTime DetectedTime `[get]`

  Last time when switched to detected state.
- int ActivityDelayMs `[get, set]`

  Keep detected state during this time after signal level dropped below threshold.

Events inherited from AudioUtil.VoiceDetector< T >

- Action **OnDetected**

  Called when switched to detected state.

Events inherited from AudioUtil.IVoiceDetector

- Action **OnDetected**

  Called when switched to detected state.

3.183.1 Detailed Description

VoiceDetector specialization for float audio.

3.183.2 Constructor & Destructor Documentation

3.183.2.1 VoiceDetectorShort()

```
VoiceDetectorShort (
            int samplingRate,
            int numChannels)
```

Create a new VoiceDetectorFloat instance.

Parameters

	samplingRate	Sampling rate of the audio signal (in Hz).
	numChannels	Number of channels in the audio signal.

3.184 VoiceEvent Class Reference

Static Public Attributes

- const byte Code = 202

  Single event used for voice communications.
- const byte **FrameCode** = 203

3.184.1 Member Data Documentation

3.184.1.1 Code

```
const byte Code = 202  [static]
```

Single event used for voice communications.

Change if it conflicts with other event codes used in the same Photon room.

3.185 VoiceFollowClient Class Reference

This class can be used to automatically sync client states between Leader and Voice clients.

Inherits VoiceConnection.

Inherited by FusionVoiceClient, and PunVoiceClient.

Public Member Functions

- bool ConnectAndJoinRoom ()

  Connect voice client to Photon servers and join a Voice room.
- void **Disconnect** ()

  Disconnect voice client from all Photon servers.

Public Member Functions inherited from VoiceConnection

- virtual bool ConnectUsingSettings (AppSettings overwriteSettings=null)

  Connect to Photon server using Settings
- bool AddSpeaker (Speaker speaker, object userData)

  Tries to link local Speaker with remote voice stream using UserData. Useful if Speaker created after stream is started.
- Speaker InstantiateSpeakerPrefab (GameObject parent, bool destroyOnRemove)

  Instantiates SpeakerPrefab, optionally attaches it to the provided parent.
- bool **AddRecorder** (Recorder rec)
- void **RemoveRecorder** (Recorder rec)

Public Attributes

- bool **AutoConnectAndJoin** = true

  Auto connect voice client and join a voice room when Leader client is joined to a Leader room.

Public Attributes inherited from VoiceConnection

- AppSettings **Settings**

  Settings to be used by this Voice Client.

Protected Member Functions

- string **GetVoiceRoomName** ()
- bool **ConnectVoice** ()
- virtual void **Start** ()
- override void OnDestroy ()
- override void OnOperationResponseReceived (OperationResponse operationResponse)
- void **LeaderStateChanged** (ClientState toState)
- override void OnVoiceStateChanged (ClientState fromState, ClientState toState)
- virtual bool **JoinVoiceRoom** (string voiceRoomName)

Protected Member Functions inherited from VoiceConnection

- override void **Awake** ()
- virtual void **Update** ()
- virtual Speaker **InstantiateSpeakerForRemoteVoice** (int playerId, byte voiceId, object userData)
- void **CalcStatistics** ()

Properties

- bool **LeaderInRoom** `[get]`
- bool **LeaderOfflineMode** `[get]`

Properties inherited from VoiceConnection

- virtual bool **AlwaysUsePrimaryRecorder** `[get]`
- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- new LoadBalancingTransport **Client** `[get]`
- VoiceClient **VoiceClient** `[get]`

  Returns underlying Photon Voice client.
- ClientState **ClientState** `[get]`

  Returns Photon Voice client state.
- float **FramesReceivedPerSecond** `[get]`

  Number of frames received per second.
- float **FramesLostPerSecond** `[get]`

  Number of frames lost per second.
- float **FramesLostPercent** `[get]`

  Percentage of lost frames.
- GameObject **SpeakerPrefab** `[get, set]`

  Prefab that contains Speaker component to be instantiated when receiving a new remote audio source info.
- Recorder **PrimaryRecorder** `[get, set]`

  Primary Recorder to be used by VoiceConnection implementations directly or via integration objects.
- bool **UsePrimaryRecorder** `[get]`

  Use VoiceConnection.PrimaryRecorder directly.
- string **BestRegionSummaryInPreferences** `[get, set]`

  Used to store and access the "Best Region Summary" in the Player Preferences.

Additional Inherited Members

Static Public Attributes inherited from VoiceConnection

- const int **ChannelAudio** = 1

  Recommended Photon Transport channel for audio. Chosen not to interfere with video and default channel.
- const int **ChannelVideo** = 2

  Recommended Photon Transport channel for video. Chosen not to interfere with audio and default channel.

Events inherited from VoiceConnection

- Action< Speaker > **SpeakerLinked**

  Fires when a speaker has been linked to a remote audio stream.
- Action< RemoteVoiceLink > **RemoteVoiceAdded**

  Fires when a remote voice stream is added.

3.185.1 Detailed Description

This class can be used to automatically sync client states between Leader and Voice clients.

3.185.2 Member Function Documentation

3.185.2.1 ConnectAndJoinRoom()

```
bool ConnectAndJoinRoom ()
```

Connect voice client to Photon servers and join a Voice room.

Returns

If true, connection command send from client

3.185.2.2 OnDestroy()

```
override void OnDestroy ()  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.185.2.3 OnOperationResponseReceived()

```
override void OnOperationResponseReceived (
            OperationResponse operationResponse)  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.185.2.4 OnVoiceStateChanged()

```
override void OnVoiceStateChanged (
            ClientState fromState,
            ClientState toState)  [protected], [virtual]
```

Reimplemented from VoiceConnection.

3.186 VoiceInfo Struct Reference

Describes stream properties.

Public Member Functions

- override string **ToString** ()

Static Public Member Functions

- static VoiceInfo CreateAudioOpus (POpusCodec.Enums.SamplingRate samplingRate, int channels, Opus↩ Codec.FrameDuration frameDurationUs, int bitrate, object userdata=null)

  Create stream info for an Opus audio stream.
- static VoiceInfo CreateAudio (Codec codec, int samplingRate, int channels, int frameDurationUs, object user-data=null)

  Create stream info for an audio stream.
- static VoiceInfo CreateVideo (Codec codec, int bitrate, int width, int height, int fps, int keyFrameInt, object userdata=null)

  Create stream info for a video stream.

Properties

- Codec **Codec** `[get, set]`
- int **SamplingRate** `[get, set]`

  Audio sampling rate (frequency, in Hz).
- int **Channels** `[get, set]`

  Number of channels.
- int **FrameDurationUs** `[get, set]`

  Uncompressed frame (audio packet) size in microseconds.
- int **Bitrate** `[get, set]`

  Target bitrate (in bits/second).
- int **Width** `[get, set]`

  Video width.
- int **Height** `[get, set]`

  Video height.
- int **FPS** `[get, set]`

  Video frames per second.
- int **KeyFrameInt** `[get, set]`

  Video keyframe interval in frames.
- object **UserData** `[get, set]`

  Optional user data. Should be serializable by Photon.
- int **FrameDurationSamples** `[get]`

  Uncompressed frame (data packet) size in samples.
- int **FrameSize** `[get]`

  Uncompressed frame (data packet) array size.

3.186.1 Detailed Description

Describes stream properties.

3.186.2 Member Function Documentation

3.186.2.1 CreateAudio()

```
VoiceInfo CreateAudio (
            Codec codec,
            int samplingRate,
            int channels,
            int frameDurationUs,
            object userdata = null)  [static]
```

Create stream info for an audio stream.

Parameters

	codec	Audio codec.
	samplingRate	Audio sampling rate.
	channels	Number of channels.
	frameDurationUs	Uncompressed frame (audio packet) size in microseconds.
	userdata	Optional user data. Should be serializable by Photon.

Returns

VoiceInfo instance.

3.186.2.2 CreateAudioOpus()

```
VoiceInfo CreateAudioOpus (
            POpusCodec.Enums.SamplingRate samplingRate,
            int channels,
            OpusCodec.FrameDuration frameDurationUs,
            int bitrate,
            object userdata = null)  [static]
```

Create stream info for an Opus audio stream.

Parameters

	samplingRate	Audio sampling rate.
	channels	Number of channels.
	frameDurationUs	Uncompressed frame (audio packet) size in microseconds.
	bitrate	Stream bitrate (in bits/second).
	userdata	Optional user data. Should be serializable by Photon.

Returns

VoiceInfo instance.

3.186.2.3 CreateVideo()

```
VoiceInfo CreateVideo (
            Codec codec,
            int bitrate,
            int width,
            int height,
            int fps,
            int keyFrameInt,
            object userdata = null)  [static]
```

Create stream info for a video stream.

Parameters

	codec	Video codec.
	bitrate	Stream bitrate.
	width	Streamed video width. If 0, width and height of video source used (no rescaling).
	height	Streamed video height. If -1, aspect ratio preserved during rescaling.
	fps	Streamed video frames per second.
	keyFrameInt	Keyframes interval in frames.

///

Parameters

	userdata	Optional user data. Should be serializable by Photon.

Returns

VoiceInfo instance.

3.187 AudioUtil.VoiceLevelDetectCalibrate< T > Class Template Reference

Utility Audio Processor Voice Detection Calibration.

Inherits IProcessor< T >.

Public Member Functions

- VoiceLevelDetectCalibrate (int samplingRate, int channels)

  Create new VoiceLevelDetectCalibrate instance.
- void Calibrate (int durationMs, Action< float > onCalibrated=null)

  Start calibration.
- T[ ] Process (T[ ] buf)

  Process a frame of data.
- void **Dispose** ()

Properties

- ILevelMeter **LevelMeter** `[get]`

  The LevelMeter in use.
- IVoiceDetector **VoiceDetector** `[get]`

  The VoiceDetector in use.
- bool **IsCalibrating** `[get]`

3.187.1 Detailed Description

Utility Audio Processor Voice Detection Calibration.

Encapsulates level meter, voice detector and voice detector calibrator in single instance.

3.187.2 Member Function Documentation

3.187.2.1 Calibrate()

```
void Calibrate (
            int durationMs,
            Action< float > onCalibrated = null)
```

Start calibration.

Parameters

	durationMs	Duration of the calibration procedure (in milliseconds).
	onCalibrated	Called when calibration is complete. Parameter is new threshold value.

This activates the Calibration process. It will reset the given LevelMeter's AccumAvgPeakAmp (accumulated average peak amplitude), and when the duration has passed, use it for the VoiceDetector's detection threshold.

3.187.2.2 Process()

```
T[] Process (
            T[] buf)
```

Process a frame of data.

Parameters

	buf	Buffer containing input data

Returns

Buffer containing output data or null if frame has been discarded (VAD)

Implements IProcessor< T >.

3.187.2.3 VoiceLevelDetectCalibrate()

```
VoiceLevelDetectCalibrate (
            int samplingRate,
            int channels)
```

Create new VoiceLevelDetectCalibrate instance.

Parameters

	samplingRate	Sampling rate of the audio signal (in Hz).
	channels	Number of channels in the audio signal.

3.188 VoiceLogger Class Reference

Inherits MonoBehaviour.

Static Public Member Functions

- static VoiceLogger **FindLogger** (GameObject gameObject)
- static VoiceLogger **CreateRootLogger** ()

Public Attributes

- LogLevel **LogLevel** = LogLevel.Warning

3.189 VoiceNetworkObject Class Reference

Inherits NetworkBehaviour.

Public Member Functions

- override void **Spawned** ()
- override void **Despawned** (NetworkRunner runner, bool hasState)

Properties

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- Recorder **RecorderInUse** `[get]`

  The Recorder component currently used by this VoiceNetworkObject.
- Speaker **SpeakerInUse** `[get]`

  The Speaker component currently used by this VoiceNetworkObject.
- bool **IsSpeaking** `[get]`

  If true, this VoiceNetworkObject has a Speaker that is currently playing received audio frames from remote audio source.
- bool **IsRecording** `[get]`

  If true, this VoiceNetworkObject has a Recorder that is currently transmitting audio stream from local audio source.
- bool **IsLocal** `[get]`

3.190 AudioUtil.WaveformAudioPusher< T > Class Template Reference

IAudioPusher that provides the given waveform.

Inherits AudioUtil.GeneratorPusher< T >.

Public Member Functions

- **WaveformAudioPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)

Public Member Functions inherited from AudioUtil.GeneratorPusher< T >

- **GeneratorPusher** (int bufSizeMs=100, int samplingRate=48000, int channels=1)
- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.
- void **Dispose** ()

Protected Member Functions

- override int **Gen** (T[] buf, long timeSamples)

Protected Member Functions inherited from AudioUtil.GeneratorPusher< T >

- int **Gen** (T[] buf, long timeSamples)

Properties

- T[] **Waveform** `[set]`

Properties inherited from AudioUtil.GeneratorPusher< T >

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

Additional Inherited Members

Protected Attributes inherited from AudioUtil.GeneratorPusher< T >

- long **timeSamples**

3.190.1 Detailed Description

IAudioPusher that provides the given waveform.

3.191 AudioUtil.WaveformAudioReader< T > Class Template Reference

IAudioReader that provides the given waveform.

Inherits AudioUtil.GeneratorReader< T >.

Public Member Functions

- **WaveformAudioReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)

Public Member Functions inherited from AudioUtil.GeneratorReader< T >

- **GeneratorReader** (Func< double > clockSec=null, int samplingRate=48000, int channels=1)
- void **Dispose** ()
- bool Read (T[ ] buf)

  Fill full given frame buffer with source uncompressed data or return false if not enough such data.

Protected Member Functions

- override int **Gen** (T[ ] buf, long timeSamples)

Protected Member Functions inherited from AudioUtil.GeneratorReader< T >

- int **Gen** (T[ ] buf, long timeSamples)

Properties

- T[ ] **Waveform** `[set]`

Properties inherited from AudioUtil.GeneratorReader< T >

- int Channels `[get]`

  Number of channels in the audio signal.
- int SamplingRate `[get]`

  Sampling rate of the audio signal (in Hz).
- string Error `[get]`

  If not null, audio object is in invalid state.

3.191.1 Detailed Description

IAudioReader that provides the given waveform.

3.192 WaveWriter Class Reference

Inherits IDisposable.

Public Member Functions

- **WaveWriter** (string fileName, int sampleRate, int bits, int channels)
- **WaveWriter** (Stream stream, int sampleRate, int bitsPerSample, int channels)
- void **Dispose** ()
- void **WriteSample** (float sample)
- void **WriteSamples** (float[ ] samples, int offset, int count)
- void **Write** (byte[ ] buffer, int offset, int count)
- void **Write** (byte value)
- void **Write** (short value)
- void **Write** (int value)
- void **Write** (float value)

Protected Member Functions

- virtual void **Dispose** (bool disposing)

3.193 WebAudioAudioOut Class Reference

Inherits AudioOutDelayControl< T >.

Public Member Functions

- **WebAudioAudioOut** (PlayDelayConfig playDelayConfig, double spatialBlend, ILogger logger, string log↩ Prefix, bool debugInfo)
- override void **OutCreate** (int frequency, int channels, int bufferSamples)
- override void **OutStart** ()
- override void **OutWrite** (float[ ] data, int offsetSamples)
- override void Stop ()
- void **SetVolume** (double x)
- void **SetSpatialBlend** (double x)
- void **SetRefDistance** (double x)
- void **SetMaxDistance** (double x)
- void **SetListenerPosition** (double x, double y, double z)
- void **SetListenerOrientation** (double fx, double fy, double fz, double ux, double uy, double uz)
- void **SetPosition** (double x, double y, double z)

Public Member Functions inherited from AudioOutDelayControl< T >

- void **OutCreate** (int frequency, int channels, int bufferSamples)
- void **OutStart** ()
- void **OutWrite** (T[ ] data, int offsetSamples)
- bool **IsZeroFrame** (T[ ] f)
- **AudioOutDelayControl** (bool processInService, PlayDelayConfig playDelayConfig, ILogger logger, string logPrefix, bool debugInfo)
- void Start (int frequency, int channels, int frameSamples)
- void Service ()
- void Push (T[ ] frame)
- void Flush ()

Static Public Member Functions

- static void **ResumeAudioContext** ()

Protected Attributes

- int **frequency**
- int **channels**
- int **bufferSamples**

Protected Attributes inherited from AudioOutDelayControl< T >

- readonly int **sizeofT** = System.Runtime.InteropServices.Marshal.SizeOf(default(T))
- readonly ILogger **logger**
- readonly string **logPrefix**

Properties

- string **Error** `[get]`
- override long **OutPos** `[get]`

Properties inherited from AudioOutDelayControl< T >

- long **OutPos** `[get]`
- int Lag `[get]`
- bool **IsFlushed** `[get]`
- bool IsPlaying `[get]`

Additional Inherited Members

Static Public Attributes inherited from AudioOutDelayControl< T >

- const int **FRAME_POOL_CAPACITY** = 50

3.193.1 Member Function Documentation

3.193.1.1 Stop()

```
override void Stop () [virtual]
```

Reimplemented from AudioOutDelayControl< T >.

3.194 WebAudioInEnumerator Class Reference

Inherits WebDeviceEnumerator.

Public Member Functions

- **WebAudioInEnumerator** (ILogger logger)

Public Member Functions inherited from WebDeviceEnumerator

- **WebDeviceEnumerator** (ILogger logger, string filter)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.195 WebAudioMicIn Class Reference

Inherits IAudioPusher< T >.

Public Member Functions

- **WebAudioMicIn** (string deviceId, int suggestedFrequency, int suggestedChannels, ILogger logger)
- void **SetCallback** (Action< float[ ]> callback, ObjectFactory< float[ ], int > bufferFactory, int optimalFrame↩ Size)
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[ ]> callback, ObjectFactory< T[ ], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- string Error [get]

  If not null, audio object is in invalid state.
- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- int Channels [get]

  Number of channels in the audio signal.

3.195.1 Property Documentation

3.195.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.195.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.195.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.

3.196 WebDeviceEnumerator Class Reference

Inherits DeviceEnumeratorBase.

Inherited by WebAudioInEnumerator, and WebVideoInEnumerator.

Public Member Functions

- **WebDeviceEnumerator** (ILogger logger, string filter)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.196.1 Member Function Documentation

3.196.1.1 Refresh()

```
override void Refresh ()
```

Implements IDeviceEnumerator.

3.197 WebRtcAudioDsp Class Reference

Inherits VoiceComponent.

Public Member Functions

- void **AdjustVoiceInfo** (ref VoiceInfo voiceInfo, ref AudioSampleType st)

Protected Member Functions

- override void Awake ()

Properties

- bool **AEC** `[get, set]`
- bool **AecHighPass** `[get, set]`
- int **ReverseStreamDelayMs** `[get, set]`
- WebRTCAudioProcessor.NSLevel **NoiseSuppressionLevel** `[get, set]`

  Noise suppression aggressiveness. Default: Moderate.
- bool **AgcLimiter** `[get, set]`

  Hard-limit the AGC output at the target level. Default: true.
- WebRTCAudioProcessor.VADLikelihood **VadLikelihood** `[get, set]`

  How readily the voice detector calls a frame speech. VeryLow (default) yields the fewest false positives.
- bool **NoiseSuppression** `[get, set]`
- bool **HighPass** `[get, set]`
- bool **Bypass** `[get, set]`
- bool **AGC** `[get, set]`
- int **AgcCompressionGain** `[get, set]`
- int **AgcTargetLevel** `[get, set]`
- bool **VAD** `[get, set]`
- bool **IsSupported** `[get]`

Properties inherited from VoiceComponent

- Voice.ILogger **Logger** `[get]`
- VoiceLogger **VoiceLogger** `[get]`
- string **Name** `[set]`

3.197.1 Member Function Documentation

3.197.1.1 Awake()

```
override void Awake ()  [protected], [virtual]
```

Reimplemented from VoiceComponent.

3.198 WebRTCAudioLib Class Reference

Inherited by WebRTCAudioProcessor.

Public Types

- enum **Error**
- enum **Param**
- enum NSLevel

  webrtc AudioProcessing::Config::NoiseSuppression::Level.
- enum VADLikelihood

  Likelihood that the voice detector reports a frame as speech. Maps onto webrtc::Vad::Aggressiveness inversely: VeryLow is the most aggressive setting and yields the fewest false positives.

Public Member Functions

- static IntPtr **webrtc_audio_processor_create** (int samplingRate, int channels, int frameSize, int rev↩ SamplingRate, int revChannels)
- static int **webrtc_audio_processor_init** (IntPtr proc)
- static int **webrtc_audio_processor_set_param** (IntPtr proc, int param, int v)
- static int **webrtc_audio_processor_process** (IntPtr proc, short[ ] buffer, int offset, out bool voiceDetected)
- static int **webrtc_audio_processor_process_reverse** (IntPtr proc, short[ ] buffer, int bufferSize)
- static void **webrtc_audio_processor_destroy** (IntPtr proc)

Properties

- static string **Version** `[get]`

  Which WebRTC revision the native library was built from, e.g. "WebRTC source stamp 2026-09-03T04:09:34, commit 9ab2b29b...". Returns a diagnostic string instead of throwing if the native library is missing or predates this export.

3.199 WebRTCAudioProcessor Class Reference

Inherits WebRTCAudioLib, and IProcessor< T >.

Public Member Functions

- **WebRTCAudioProcessor** (ILogger logger, int frameSize, int samplingRate, int channels, int reverse↩ SamplingRate, int reverseChannels)
- short[ ] **Process** (short[ ] buf)
- void **OnAudioOutFrameFloat** (float[ ] data)
- void **Dispose** ()

Public Member Functions inherited from WebRTCAudioLib

- static IntPtr **webrtc_audio_processor_create** (int samplingRate, int channels, int frameSize, int rev↩ SamplingRate, int revChannels)
- static int **webrtc_audio_processor_init** (IntPtr proc)
- static int **webrtc_audio_processor_set_param** (IntPtr proc, int param, int v)
- static int **webrtc_audio_processor_process** (IntPtr proc, short[ ] buffer, int offset, out bool voiceDetected)
- static int **webrtc_audio_processor_process_reverse** (IntPtr proc, short[ ] buffer, int bufferSize)
- static void **webrtc_audio_processor_destroy** (IntPtr proc)

Public Member Functions inherited from IProcessor< T >

- T[ ] Process (T[ ] buf)

  Process a frame of data.

Static Public Attributes

- static readonly int[ ] **SupportedSamplingRates** = { 8000, 16000, 32000, 48000 }

Properties

- int **AECStreamDelayMs** `[set]`
- bool **AEC** `[set]`
- bool **AECHighPass** `[set]`
- bool **HighPass** `[set]`
- bool **NoiseSuppression** `[set]`
- NSLevel **NoiseSuppressionLevel** `[set]`

  Noise suppression aggressiveness. Default: Moderate.
- bool **AGC** `[set]`
- int **AGCCompressionGain** `[set]`
- int **AGCTargetLevel** `[set]`
- bool **AGCLimiter** `[set]`

  Hard-limit the AGC output at the target level instead of only compressing above it. Default: true.
- bool **AGC2** `[set]`
- bool **VAD** `[set]`
- VADLikelihood **VoiceDetectionLikelihood** `[set]`

  How readily the voice detector calls a frame speech. Default: VeryLow, which reproduces the behaviour of the webrtc voice_detection submodule this wrapper used before it was removed upstream.
- bool **Bypass** `[set]`

Properties inherited from WebRTCAudioLib

- static string **Version** `[get]`

  Which WebRTC revision the native library was built from, e.g. "WebRTC source stamp 2026-09-03T04:09:34, commit 9ab2b29b...". Returns a diagnostic string instead of throwing if the native library is missing or predates this export.

Additional Inherited Members

Public Types inherited from WebRTCAudioLib

- enum **Error**
- enum **Param**
- enum NSLevel

  webrtc AudioProcessing::Config::NoiseSuppression::Level.
- enum VADLikelihood

  Likelihood that the voice detector reports a frame as speech. Maps onto webrtc::Vad::Aggressiveness inversely: VeryLow is the most aggressive setting and yields the fewest false positives.

3.200 WebVideoInEnumerator Class Reference

Inherits WebDeviceEnumerator.

Public Member Functions

- **WebVideoInEnumerator** (ILogger logger)

Public Member Functions inherited from WebDeviceEnumerator

- **WebDeviceEnumerator** (ILogger logger, string filter)
- override void Refresh ()
- override void **Dispose** ()

Public Member Functions inherited from DeviceEnumeratorBase

- **DeviceEnumeratorBase** (ILogger logger)
- IEnumerator< DeviceInfo > **GetEnumerator** ()
- void Refresh ()
- void **Dispose** ()

Additional Inherited Members

Protected Attributes inherited from DeviceEnumeratorBase

- List< DeviceInfo > **devices** = new List<DeviceInfo>()
- ILogger **logger**

Properties inherited from DeviceEnumeratorBase

- virtual bool IsSupported `[get]`
- virtual string Error `[get, protected set]`
- Action OnReady `[protected get, set]`

3.201 WindowsAudioInPusher Class Reference

Inherits IAudioPusher< T >.

Public Member Functions

- **WindowsAudioInPusher** (int deviceID, ILogger logger)
- void **SetCallback** (Action< short[]> callback, ObjectFactory< short[], int > bufferFactory, int optimal↩ FrameSize)
- void **Dispose** ()

Public Member Functions inherited from IAudioPusher< T >

- void SetCallback (Action< T[]> callback, ObjectFactory< T[], int > bufferFactory, int optimalFrameSize)

  Set the callback function used for pushing data.

Properties

- int Channels [get]

  Number of channels in the audio signal.
- int SamplingRate [get]

  Sampling rate of the audio signal (in Hz).
- string Error [get]

  If not null, audio object is in invalid state.

3.201.1 Property Documentation

3.201.1.1 Channels

```
int Channels [get]
```

Number of channels in the audio signal.

Implements IAudioDesc.

3.201.1.2 Error

```
string Error [get]
```

If not null, audio object is in invalid state.

Implements IAudioDesc.

3.201.1.3 SamplingRate

```
int SamplingRate [get]
```

Sampling rate of the audio signal (in Hz).

Implements IAudioDesc.